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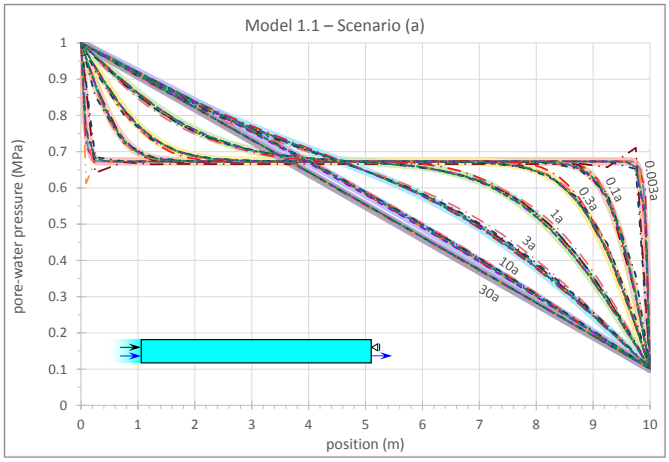
BenVaSim

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International Benchmarking for Verification and Validation of TH²M Simulators with Special Consideration of Fluid Dynamical Processes in Radioactive Waste Repository Systems

Official project title:

Verbundprojekt: Internationales Benchmarking zur Verifizierung und Validierung von TH²M-Simulatoren insbesondere im Hinblick auf fluiddynamische Prozesse in Endlagersystemen (BenVaSim)



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Synthesis report for the BMWi joint research project with grant numbers 02E11567A-B

Reporting period: 01.05.2017 - 31.03.2021

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for the BMWi-funded joint research project

BenVaSim—International Benchmarking for Verification and Validation of TH^{2}M Simulators with Special Consideration of Fluid Dynamical Processes in Radioactive Waste Repository Systems

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Summary

Many countries worldwide are or will be facing the task of finding a suitable geological rock formation for the design and construction of a stable and tight final repository for heat-generating high-level radioactive waste. To reach this aim, numerical predictions of the multiphysical processes taking place during the period of several 100,000 years, in which the decaying radionuclides pose a potential hazard to life and health, form an indispensable contribution. Consequently, research is conducted in many of these countries to properly capture the properties of potential host rocks and to implement software able to simulate the behaviour of a final repository in such a host rock, appropriately extrapolated into the future.

To build confidence in numerical predictions of the behaviour of an underground repository, the implemented simulators have to be verified by some means. Mathematical solutions of the underlying differential equations can however only be derived under strong simplifications, and therefore they provide a very limited basis for verification. Hence, the six teams from the author list gathered to create the international collaboration project "BenVaSim", a simulator benchmarking where systematic comparative simulations on coupled hydraulic-mechanical and thermal-hydraulic-mechanical processes are conducted to confirm the correct workings of the involved simulators and to reveal potential shortcomings of them, thus helping to find and fix potential bugs and add missing features. Generic model examples with framework conditions as well-defined as possible and reasonable were designed and simulated by the project partners. The simulation results were then contrasted to each other and to analytical solutions where available, and the partners had the possibility to double-check their simulation settings and make corrections to settings or simulators. All of the involved simulators or parts of coupled simulators have already been applied in the research for radioactive waste disposal and verified to a certain extent, while particularly modified or updated versions or—as it has been the case with the initiator of the project—a recent simulator coupling require more comprehensive verification work.

The report at hand deals primarily with the juxtaposition of all partners' final results. Preparatory work for the BenVaSim project by its initiator like literature research and conferring with the project-partners-to-be had been undertaken and documented before in a pre-project, and the work, contributions, and benefits of the project partner and coordinator TUC as well as those of the project partners GRS BS and GRS K to/from the simulator benchmarking are summarised in separate individual final reports. The final report by the coordinator additionally contains

further information about the project's background and motivations as well as a comprehensive compilation of process and coupling equations and some mathematical solutions of them.

Even if some discrepancies remained in some of the results of the project partners in mutual comparison, the majority of results for the first three models show a great deal of qualitative and quantitative agreement, especially in the presence of mathematical solutions. Some of the original discrepancies could be attributed to missing physical relations or predefined simulator-intrinsic settings, and in most cases—depending on the possibilities to modify the corresponding simulator's source code—these issues could be resolved. Causes for other discrepancies could be figured out, but have not been or could not be remedied for various (organisational or technical) reasons. Concerning the last model, time for contrasting and analysing of the partners' results became short due to restrictions on assemblies and travelling and due to an increased effort in defining scenarios with framework conditions as similar as possible.

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1 Introduction and motivation

1.1 Framework conditions and objectives of the BenVaSim project

Comprehensive research activities have been taking place worldwide for many decades looking into the final disposal of heat-generating high-level radioactive waste in deep underground geological rock formations. As part of the overall geologic repository planning, long-term safety cases and, by association, predictions of the repository behaviour will have to be provided for the post-closure phase usually covering periods of several 100,000 years up to a million years. In addition to physical and chemical analyses of the properties of the materials used for the specific geologic repository system in question (host rock, backfill and sealing materials, containers) under loads typical for geologic repositories, numerical simulation programs are also required to predict the development of repository systems. These numerical simulators need to be developed, implemented, adjusted, extended, and tested to prove their suitability. Checking the suitability of some of these simulators and especially their computational accuracy is the overall objective to which the project "BenVaSim"—an acronym for

International Benchmarking for Verification and Validation of TH^2M Simulators
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in Radioactive Waste Repository Systems

—hopes to make an important contribution.

Thermo-hydro-mechanical processes (THM processes) play a key role in the long-term evolution of a geologic repository for high-level radioactive waste. The mechanical load-bearing and deformation behaviour of the various repository system components, the hydraulic flow behaviour of the pre-existing and the technically generated fluids and of those penetrating the repository from outside, as well as the development and propagation of heat induced by the radioactive decay of the waste affect one another in intense, diverse, and complex ways. Particularly with respect to the hydraulic processes, interactions between the liquid phase, mainly consisting of formation water, and the gas phase, comprising various gas components—primarily air in the first instance—have to be considered. These hydraulic two-phase-flow processes (H^2) usually have a significant influence on fluid transport and mechanical loads and therefore on the overall TH^2M-coupled repository behaviour. For an assessment of the long-term safety of a geologic repository system as reliable as possible, it is hence unavoidable to analyse the physical-technical as well as the numerical-prognostic aspects of TH^2M processes taking place within the geological and geotechnical materials involved in the system.

The research required for such a long-term safety case therefore also includes confirming the basic reliability of numerical predictive data and thus verification, validation, and where appropriate, the code adaption of simulators used for this purpose. Methodological verification approaches include comparing the numerical simulation results with analytical differential equation solutions, or, in the absence of precise analytical solutions, undertaking comparisons of the results generated by various different simulators for reference examples.

Benchmarking is an internationally proven approach for the quality assurance of simulators. In research done on the final disposal of radioactive wastes, various projects either exist or have existed with the aim of comparing the simulation experience and skills of various organisations. However, the numerical accuracy of the individual simulators used in such projects is usually assumed in advance. Generally, experience has shown that such cooperation projects support the exchange of experience between organisations and particularly their scientists, and that they also give rise to new scientific understanding.

Against this background, the motivation and primary objective of the BenVaSim project were to provide a quality assurance platform for simulators which can be applied for the long-term safety analysis of the TH^2M behaviour of geologic repository systems and their components, with a focus on verification. Validation aspects arise in this project stage especially where underlying process equations implemented in the simulators show differences in their details. Unlike the relatively common type of conventional benchmarking which aims to virtually reproduce realistic material or subsystem behaviour on the basis of measurement data from laboratory or field tests, the BenVaSim simulator benchmarking project primarily aims to check the numerical accuracy of the simulators involved. Besides gaining greater familiarity in handling the applied simulators and expanding the horizon of experience and thus the reliability of predictions on the side of the partners, both particularly regarding their junior scientists, another benefit of the project is the provision of the numerically backed model examples for subsequent work of the partners and of third parties by forming benchmarks for own simulator verifications and for training of junior scientists.

This synthesis report summarises the simulation results obtained by the partners. For more information on background and motivations of the BenVaSim project, the reader is referred to the final report of the initiator of this project, Clausthal University of Technology, *Lux et al. (2021)*, as well as to their German final report for a pre-project covering literature research on

benchmarking projects related to the final disposal of radioactive waste, *Lux & Rutenberg (2018)*.

1.2 Participants, simulators, and structure of the project

Six partner organisations or working groups from Germany (D), Switzerland (CH), and the United States of America (USA) have been involved in the BenVaSim project. The following Table 1.1 lists the partners along with their acronyms as used in this report, as well as references to their final reports or other publications covering their work undertaken as part of the BenVa-Sim project, if applicable.

Technische Universität Clausthal Lehrstuhl für Deponietechnik und Geomechanik Clausthal University of Technology Chair for Waste Disposal Technologies and Geomechanics	**TUC**	Clausthal-Zellerfeld (D) *Lux et al. (2021)*
Gesellschaft für Anlagen- und Reaktorsicherheit gGmbH Abteilung Endlagerforschung / Repository Research Department	**GRS BS**	Brunswick (D) *Czaikowski & Friedenberg (2020)*
Bundesanstalt für Geowissenschaften und Rohstoffe Fachbereich Geotechnische Sicherheitsnachweise Federal Institute for Geosciences and Natural Resources Sub-Department Geotechnical Safety Analyses	**BGR**	Hanover (D) *Pitz & Maßmann (2020)*
Gesellschaft für Anlagen- und Reaktorsicherheit gGmbH Abt. Entsorgungssicherheit / Waste Management Safety Department	**GRS K**	Cologne (D) *Seher et al. (2019)*
Eidgenössisches Nuklearsicherheitsinspektorat Sektion Tiefenlagerung & Sicherheitsanalysen Swiss Federal Nuclear Safety Inspectorate Disposal & Safety Analysis Section	**ENSI**	Brugg, AG (CH)
Lawrence Berkeley National Laboratory Hydrogeology Department	**LBNL**	Berkeley, CA (USA)

Table 1.1: BenVaSim project participants

Moreover, the Swiss Seismological Service at Swiss Federal Institute of Technology in Zurich (ETHZ) has supported ENSI in performing simulations with the simulator TOUGH-FLAC (see list of applied simulators below) for the BenVaSim project.

The simulators listed alphabetically in Table 1.2 participated in the simulator benchmarking project, and Table 1.3 shows which partners used which simulators.

CODE_BRIGHT	versions 7.4, 8.4
	UPC (2017/2018)
COMSOL Multiphysics	versions 5.3, 5.4, 5.5
	COMSOL (2019)
FTK simulator	version 3.00a (linking FLAC^{3D} 4.0 to a slightly modified TOUGH2)
	Lux et al. (2015, 2018), Lux & Rutenberg (2018); *Itasca (2009), Pruess et al. (1999)*
OpenGeoSys	versions 5.7.1, 5.8, 6.3.2 (BGR), version 5.8 (ENSI)
	Kolditz et al. (2012a)
TOUGH-FLAC; TFC *(a variation further developed in parallel by GRS K)*	version TOUGH2 (GRS K, LBNL) / TOUGH3 (ENSI) and version FLAC^{3D} 4.0 (LBNL) / 6.0 (ENSI, GRS K)
	Rutqvist et al. (2002), Rutqvist (2011, 2017), Rinaldi et al. (2018), Hotzel (2014); *Pruess et al. (1999), Navarro (2018), Jung et al. (2018), Itasca (2009/2017)*

Table 1.2: Simulators applied in the BenVaSim project

Simulator		TUC	GRS BS	BGR	GRS K	ENSI	LBNL
CODE_BRIGHT	(CB)		X				
COMSOL Multiphysics	(CM)					X	
FTK simulator	(FTK)	X					
OpenGeoSys	(OGS)			X		X	
TOUGH-FLAC	(TF)					X	X
	(TFC)				X		

Table 1.3: Attribution of simulators to project partners

TUC managed and coordinated the BenVaSim project. On internal meetings, the project partners jointly discussed and agreed on the generic model examples suggested by TUC and adapted them or added others to them later where necessary depending on first simulation results. TUC contrasted the simulation results of all partners and provided assistance in analysing and explaining discrepancies within the scope of possibilities.

The spectrum of the generic model examples agreed for simulation in the BenVaSim project has been divided up into *stages*, *models*, and *scenarios*, with the numbering convention

⟨stage no.⟩ . ⟨model no.⟩ (⟨scenario letter⟩)

being applied to the model examples. A first, coarse subdivision into stages is based on the complexity of the models. For simplicity, a model's complexity has been associated with its dimension. The model examples in stage 1 are therefore one-dimensional (i.e., e.g., without *Poisson* number, transverse stresses and deformations, as well as without any transverse flow components) and thus strongly abstracted; examples for stages 2 and 3 possibly following this research project are to be designed as two- and three-dimensional, respectively. Different models were conceived for each of the different stages, each being patterned on a system component or on a sub-system of a generic geologic repository design but at the same time being abstracted to a certain degree because of the necessary limitations involved. Limiting factors here are the use of those physical-mathematical relations (e.g., constitutive and process models) supported by as many as possible of the participating simulators, the comparability to analytical solutions, as well as the model dimensions themselves. Finally, within each of the models, a distinction into scenarios is drawn where mostly individual parameters or characteristics of model subareas, initial values, or boundary conditions are varied, or the influence of a type of process is completely ignored. This means that it is possible to make comparisons between different scenarios within a model, while the simulation results of different models cannot be directly compared with one another.

More details about the structure and the course of the BenVaSim project can be found in the final report of the coordinator, *Lux et al. (2021)*.

2 Physical modelling equations

2.1 General considerations

Based on literature research about processes and equations supported by the different simulators applied in the BenVaSim project, *Lux & Rutenberg (2018)*, models and scenarios have been suggested and discussed for stage 1 of this project. The emerged final set of mathematical process equations for one-dimensional models are summarised in this section 2. For a more general and more comprehensive description and derivation of analytical and semi-analytical solutions, please refer to *Lux et al. (2021)* and *Rutenberg (in prep.)*.

Process quantities are to be taken as functions of time t and space. Since deformations are considered small compared to the sizes of models simulated in this project stage, a distinction between the initial position X and the current position x of a material point (an infinitesimal volume element) is not deemed necessary except for the definition of displacement and strains. Thus, in respect of a simpler depiction of profiles, the initial reference position X will be used in this report.

The following sign conventions are used in this report: First, the models from this stage are one-dimensional in order to facilitate error analysis of the project partners' simulation results, and therefore, physical quantities possessing a direction (like fluxes and deformations) degenerate from vectors to scalars. Since all models simulated in this stage are furthermore oriented horizontally, positive values of the aforementioned quantities are defined to point in the right direction. Second, the sign convention of geomechanics applies, meaning positive stresses are compressive and positive strains represent material compactions. Third, capillary pressure is defined as the nonnegative quantity many literature sources refer to as "suction".

Also, quantities defined by the terms "generation", "influx", "source", and "supply" are in general to be understood in a non-specific way—the terms are used only to indicate the sign convention of the corresponding quantities. Negative values for them describe consumption, efflux, sink, and removal. Accordingly, negative pressure-volume work is exerted by the fluids instead of on them.

2.2 Mechanical processes

From the basic mechanical processes describing elastic, plastic, and viscous deformation behaviour, it has been agreed by the project partners in the course of the project to use only

Hooke's linear elastic constitutive model for stage 1, although one scenario including *Mohr/ Coulomb* plasticity had been suggested by the coordinator and shown to be working with fixed normal stresses. The models' horizontal one-dimensionality means that *Poisson*'s ratio ν and gravity g both vanish. In addition, mechanical processes are considered to occur fast enough to neglect dynamic behaviour and to focus on static, i.e., balanced mechanical states with a vanishing acceleration $\dot{v}$. Therefore, the mechanical process equations comprise *Cauchy*'s equation of motion as the conservation law for linear momentum and *Hooke*'s stress-strain relationship in the form of Eqs. (2.1) and (2.2), respectively, as well as a definition of strain given further below.

$$\sigma'_{tot} = 0 \frac{MPa}{m} \tag{2.1}$$

$$\dot{\sigma}_{eff} = E\dot{\varepsilon}^m \tag{2.2}$$

where

σ_{tot} total stress in MPa,

σ_{eff} effective stress in MPa,

E drained *Young*'s modulus in MPa,

ε^m mechanical strain, dimensionless.

The quantity ε^m defines strains induced by mechanical effective stresses (thus considering also hydraulic impacts), as opposed to ε denoting the sum of all strains (including thermal strains) which is responsible for spatial displacements. Note that the distinctions between total and effective stresses and between strains and mechanical strains are irrelevant for the sole consideration of uncoupled mechanical processes in this subsection (i.e., $\sigma_{tot} = \sigma_{eff}$ and $\varepsilon^m = \varepsilon$).

Strains can be defined via spatial displacements in two ways: as technical or linearised strains according to Eq. (2.3), relating accumulated displacements to initial model dimensions, and as true or logarithmic strains according to Eq. (2.4), relating every increment of displacement to current model dimensions obtained during the deformation process so far:

$$\varepsilon_{lin} = -u' \tag{2.3}$$

$$\varepsilon_{log} = -\ln x' \tag{2.4}$$

where

ε_{lin} linearised strain, dimensionless,

u displacement in m,

ε_{log} logarithmic strain, dimensionless,

x trajectory of a material point X over time in m, $x = X + u$, $x' = 1 + u'$.

Both types of strains have been used to derive analytical and semi-analytical solutions for the BenVaSim model examples. This section will not focus on this distinction of strains any further, and the generalised designator ε will be used in place of ε_{lin} and ε_{log}.

Volumetric strains can be introduced for three-dimensional models where they quantify the relative change in volume of infinitesimal volume elements. Usually defined by $\varepsilon_{vol} = \mathrm{tr}\,\varepsilon$, one-dimensionality reduces them to $\varepsilon_{vol} = \varepsilon$ and, by analogy for the mechanical case, to:

$$\varepsilon_{vol}^{m} = \varepsilon^{m} \tag{2.5}$$

where

ε_{vol}^{m} mechanical volumetric strain, dimensionless.

2.3 Hydraulic processes for single- and two-phase flow

When dealing with radioactive waste disposal in deep geological formations, the pore space in these formations and in other involved geomaterials is usually occupied by a liquid and a gaseous fluid. In the simplest case, these fluid phases are composed of water and air, appearing as air with water vapour in the gas phase and as water with dissolved air in the liquid phase. This assumption was chosen to form the general framework for hydraulic two-phase-flow processes occurring in the models from this project stage, although no special emphasis has been put on phase-extraneous components (vapour and dissolved air) as their influence was deemed less meaningful. Thus, project partners or simulators may have neglected phase-extraneous components, which has also been done for deriving semi-analytical solutions. In addition, model 1.1 entirely focuses on single-phase flow. Concerning the physical quantities related to the two fluid phases, an index $\varphi \in \{l; g\}$ will be appended to the quantity designators (sometimes complemented by $\varphi = s$ for the grains as the solid phase) while for the two components, $\kappa \in \{w; a\}$ will be used. Sums over φ will not encompass $\varphi = s$.

Molecular diffusion processes have been neglected in this project stage.

Formulation of the process equations in terms of masses

The conserved quantities are masses of components, and Eq. (2.6) represents the corresponding balance equations for both components:

$$-\sum_{\varphi} q'_{m;\kappa\varphi} + \hat{q}_{m;\kappa} = \dot{\zeta}_{m/V;\,\kappa} = \dot{\Theta}_{m/V;\,\kappa} = \frac{d}{dt}\left(\phi \sum_{\varphi} S_{\varphi}\rho_{\varphi}X_{\kappa\varphi}\right) \tag{2.6}$$

where

$q_{m;\kappa\varphi}$ mass flux of component κ in phase φ in $kg/(m^2 s)$,

$\hat{q}_{m;\kappa}$ mass generation rate for component κ in both phases per unit volume of porous medium in $kg/(m^3 s)$,

$\zeta_{m/V;\,\kappa}$ accumulated mass influx of component κ in both phases per unit volume of porous medium in kg/m^3,

$\Theta_{m/V;\,\kappa}$ mass content of component κ in both phases per unit volume of porous medium in kg/m^3,

ϕ material porosity, dimensionless,

S_{φ} saturation of phase φ, dimensionless,

ρ_{φ} mass density of phase φ in kg/m^3,

$X_{\kappa\varphi}$ mass fraction of component κ within phase φ, dimensionless.

Equation (2.6) applies to uncoupled hydraulic processes only. Bulk deformations could be included by extending the term $\dot{\Theta}_{m/V;\,\kappa}$ appropriately, however, since mechanical processes deal with strains and are therefore usually formulated in terms of volumes instead of masses, volumetric formulations of the hydraulic and the H↔M coupling equations will be introduced further down and in section 2.5, respectively.

Pressure-driven convection of fluid phases is modelled by *Darcy*'s law, and the components move accordingly:

$$q_{m;\kappa\varphi} = -\rho_{\varphi}X_{\kappa\varphi} \cdot K_{rel;\varphi}(S_l) \cdot \frac{K}{\eta_{\varphi}} \cdot p'_{\varphi} \tag{2.7}$$

where

$K_{rel;\varphi}$ relative permeability of phase φ, dimensionless,

K intrinsic permeability of the porous medium in m^2,

η_φ dynamic viscosity of phase φ in $MPa \cdot s$,

p_φ pore pressure of phase φ in MPa.

Pressures and densities of the phases are connected by their corresponding bulk moduli:

$$-\breve{K}_\varphi \frac{\dot{V}_\varphi}{V_\varphi} = \dot{p}_\varphi = \breve{K}_\varphi \frac{\dot{\rho}_\varphi}{\rho_\varphi} \tag{2.8}$$

where

$\breve{K}_\varphi$ bulk modulus of phase φ in MPa,

V_φ volume of a representative volume element of phase φ in m^3.

In a simple approach for the liquid phase, the bulk modulus can be taken as a pressure- and temperature-independent constant. For an ideal gas, an expression for the bulk modulus can be derived from the ideal gas law:

$$\breve{K}_g = p_g \tag{2.9}$$

It should be noted that some simulators contain density functions which are more complex than the one defined by Eq. (2.8).

For single-phase flow of water, Eqs. (2.6) and (2.7) restricted to $\varphi = l$ and neglecting dissolved air ($X_{al} = 0$ and, according to Eq. (2.11), $X_{wl} = 1$) are merged under a few additional pre-conditions—no source term, constant values for gas pressure and liquid viscosity—to form the mass formulation of *Richards*' equation:

$$\frac{d}{dt}(\phi S_l \rho_l) = \left(\rho_l \cdot K_{rel;l}(S_l) \cdot \frac{K}{\eta_l} \cdot p_l'\right)' \tag{2.10}$$

Special considerations for two-phase flow

While the connection between both phases' saturations as per Eq. (2.11) is straightforward (as it is the case with mass fractions within each single phase φ), capillary forces introduced by Eq. (2.12) have to be taken into account when comparing both phases' pressures.

$$\sum_{\varphi} S_{\varphi} = 1 = \sum_{\kappa} X_{\kappa\varphi} \tag{2.11}$$

$$p_g - p_l = p_{cap}(S_l) \tag{2.12}$$

where

p_{cap} capillary pressure in MPa.

In many models and scenarios of the BenVaSim project, the capillary-pressure function (or retention curve) by *van Genuchten (1980)*, Eq. (2.13), and the relative-permeability functions from Eqs. (2.14a-b) derived in *Mualem (1976)* and *Parker et al. (1987)* (compiled in *Lux et al., 2015*) were used:

$$p_{cap}(S_l) = \frac{\gamma_l}{\tilde{\alpha}}\left(S_{l;eff}^{-\frac{1}{m}} - 1\right)^{1-m} \tag{2.13}$$

$$K_{rel;l}(S_l) = S_{l;eff}^{\tilde{\varepsilon}}\left(1 - \left(1 - S_{l;eff}^{\frac{1}{m}}\right)^{m}\right)^{2} \tag{2.14a}$$

$$K_{rel;g}(S_l) = \left(1 - S_{l;eff}\right)^{\tilde{\gamma}}\left(1 - S_{l;eff}^{\frac{1}{m}}\right)^{2m} \tag{2.14b}$$

where

$\tilde{\alpha}$ curve-shape parameter from *van Genuchten*'s approach in m^{-1},

$S_{\varphi;eff}$ effective saturation of phase φ, dimensionless,

m curve-shape parameter from *van Genuchten*'s approach, dimensionless,

$\tilde{\varepsilon}$ pore-connectivity parameter for the liquid phase from *Mualem*'s approach, dim'less,

$\tilde{\gamma}$ pore-connectivity parameter for the gas phase from *Mualem*'s approach, dim'less.

In Equation (2.13), the pre-factor $\gamma_l/\tilde{\alpha}$ itself is sometimes treated as the curve-shape parameter (hereafter denoted by p_{cap0}) in place of $\tilde{\alpha}$.

The effective saturations used to determine relative permeabilities and capillary pressure are defined by:

$$S_{\varphi;eff} = \frac{S_{\varphi} - S_{\varphi r}}{S_{\varphi s} - S_{\varphi r}} \tag{2.15}$$

where

$S_{\varphi r}$ residual saturation of phase φ, dimensionless,

$S_{\varphi s}$ maximum saturation of phase φ, dimensionless, $S_{\varphi s} = 1 - S_{\psi r}$ for phase $\psi \neq \varphi$.

Some of the benchmarks include simplified retention curves and/or relative-permeability functions instead. Functions used in these cases are trivial constant functions ($p_{cap} \equiv 0\ MPa$, $K_{rel;\varphi} \equiv 1$) or linear identities ($K_{rel;\varphi} = S_\varphi$).

Volumetric analogues

As an alternative to the mass formulation introduced above, balance and transport equations are sometimes expressed in terms of phase volumes. Particularly, hydraulic processes in the simulator FLAC^{3D} are built around volumetric analogues to Eqs. (2.6) and (2.7). The advantage of the following volumetric formulation of hydraulic equations is an easier way of coupling them to mechanical equations at the cost of an accurate conservation of masses and, in case of two-phase flow, at the cost of considering phase-extraneous components. Volume balance and transport equations therefore read:

$$-q'_{V;\varphi} + \hat{q}_{V;\varphi} = \dot{\zeta}_{V/V;\,\varphi} \tag{2.16}$$

$$q_{V;\varphi} = -K_{rel;\varphi}(S_l) \cdot \frac{K}{\eta_\varphi} \cdot p'_\varphi \tag{2.17}$$

where

$q_{V;\varphi}$ volumetric flux of phase φ in m/s,

$\hat{q}_{V;\varphi}$ volumetric generation rate for phase φ per unit volume of porous medium in s^{-1},

$\zeta_{V/V;\,\varphi}$ accumulated volumetric influx of phase φ per unit volume of porous medium, dim'less.

If solid grains are assumed incompressible and, as a consequence, porosity remains constant, a connection from volumetric influx to the hydraulic state variables p_φ and S_φ can be established as follows:

$$\frac{1}{S_\varphi}\dot{\zeta}_{V/V;\,\varphi} = \frac{\phi}{K_\varphi}\dot{p}_\varphi + \frac{\phi}{S_\varphi}\dot{S}_\varphi \tag{2.18}$$

2.4 Thermal processes

The conservative quantity relevant for thermal processes is internal energy, which changes solely upon heat flow as long as fluid flow is not yet considered. Radiative heat flow was neglected in this project, so heat conduction is the only considered mode for transfer of internal energy in this subsection.

Equations (2.19a-b) show the internal-energy balance and, alternatively implemented in some of the simulators, the enthalpy balance. The conductive heat flux modelled by *Fourier*'s law is given in Eq. (2.20):

$$-q'_Q + \hat{q}_Q = \dot{\Theta}_{U/V} = \frac{d}{dt}\left((1-\phi)\rho_s\, u_s + \phi \sum_{\varphi} S_\varphi \rho_\varphi u_\varphi \right) \tag{2.19a}$$

$$-q'_Q + \hat{q}_Q = \dot{\Theta}_{H/V} = \frac{d}{dt}\left((1-\phi)\rho_s\, h_s + \phi \sum_{\varphi} S_\varphi \rho_\varphi h_\varphi \right) \tag{2.19b}$$

$$q_Q = -\lambda T' \tag{2.20}$$

where for $\varphi \in \{l; g; s\}$

q_Q heat flux in $J/(m^2 s)$,

$\hat{q}_Q$ heat generation rate per unit volume of porous medium in $J/(m^3 s)$,

$\Theta_{U/V}$ content of internal energy in J/m^3,

ρ_φ mass density of phase φ in kg/m^3,

u_φ specific internal energy of phase φ in J/kg, $\dot{u}_\varphi = c_\varphi \dot{T}$,

c_φ specific heat capacity of phase φ in $J/(kg\ K)$,

T temperature in K,

$\Theta_{H/V}$ content of enthalpy in J/m^3,

h_φ specific enthalpy of phase φ in J/kg, $\dot{h}_\varphi = c_\varphi \dot{T}$,

λ thermal conductivity in $W/(m \cdot K)$.

2.5 Process couplings

Couplings between M, H, and T processes for which equations have been specified in this benchmarking comprise pressure-induced changes of mechanical stresses (H→M), temperature-induced strains of the bulk material (T→M), adaptions of phase pressures and saturations to pore-size changes (M→H), and advective transport of energy by fluid motion as well as saturation-dependent thermal parameters (H→T). The internal-energy balance equation has additionally to be expanded to take into account pressure-volume work exerted by moving fluids and by internal sources. Compressibility of solid grains according to *Biot*'s theory of poroelasticity is considered for M↔H effects, *Biot (1941), Detournay & Cheng (1993).* An M→T coupling has been considered insignificant in its effect and hence omitted while T→H coupling has been reduced to temperature dependencies of hydraulic properties. On this aspect, every applied simulator includes its own non-trivial functions for calculation of hydraulic properties from temperature and pressure (assumed to reflect reality in a good manner), and a concrete specification of functions to be used by all partners does not seem feasible. The only T→H specifications given for some scenarios are constant parameter values.

For the purpose of distinguishing between contributions from different physical processes to the same effect, a few quantities defined in the preceding subsections have to be subdivided or generalised as follows, where the symbols $f(...)$ denote as of yet unspecified functional correlations:

$$\sigma_{eff} = \sigma_{tot} + \sigma_{hyd} \tag{2.21}$$

$$\varepsilon = \varepsilon^{m} + \varepsilon^{th} \tag{2.22}$$

$$q_E = q_Q + q_E^{adv} \tag{2.23}$$

$$\hat{q}_E = \hat{q}_Q + \hat{q}_E^{adv} \tag{2.24}$$

where

σ_{hyd} hydrostatic stress induced by the pore pressure in MPa, $\sigma_{hyd} = f(p_{eq})$,

p_{eq} equivalent pore pressure composed of phase pore pressures in MPa, $p_{eq} = f(p_l; p_g)$,

ε strain, dimensionless,

ε^{th} thermal strain, dimensionless,

q_E total flux of energy in $J/(m^2 s)$,

q_E^{adv} advective flux of energy (enthalpy or internal energy) by mass flow in W/m^2,

$\hat{q}_E$ total internal supply rate of energy per unit volume of porous medium in $J/(m^3 s)$,

$\hat{q}_E^{adv}$ supply rate of energy (enthalpy or internal energy) by fluid generation in both phases φ per unit volume of porous medium in $J/(m^3 s)$.

Some properties are to be considered for the porous bulk material while depending on its composition of solid and fluid phases. This means in terms of an arbitrary bulk parameter a composed of phase parameters a_φ:

$$a = (1-\phi)a_s + \phi \sum_{\varphi} S_\varphi a_\varphi \tag{2.25}$$

Equation (2.25) is used to determine density ρ, thermal conductivity λ, and volume-specific heat capacity $\rho \cdot c$ of the bulk material.

H↔M coupling: Biot's theory of poroelasticity

The quintessence from *Biot*'s theory of poroelasticity regarding the H↔M process coupling are the definition of *Biot*'s effective stress (H→M) as well as an equation describing the influence of compaction/dilation on fluid pressure (M→H). Extended to two fluid phases, and setting $\sigma_{hyd} = -\alpha p_{eq}$ and $\varepsilon_{vol}^m = \varepsilon^m$ from Eq. (2.5), the following equations (2.26)-(2.28) emerge:

$$\sigma_{eff} = \sigma_{tot} - \alpha p_{eq} \tag{2.26}$$

$$\frac{1}{\breve{M}_\varphi}\dot{p}_\varphi + \frac{\phi}{S_\varphi}\dot{S}_\varphi = \frac{1}{S_\varphi}\dot{\zeta}_{V/V;\,\varphi} + \alpha\dot{\varepsilon}^m \tag{2.27}$$

$$\breve{M}_\varphi = \frac{\breve{K}_\varphi}{\phi + (\alpha-\phi)(1-\alpha)\,\breve{K}_\varphi/\breve{K}} \quad \Leftrightarrow \quad \check{s}_\varphi = \frac{1}{\breve{M}_\varphi} = \frac{\phi}{\breve{K}_\varphi} + \frac{\alpha-\phi}{\breve{K}_s} \tag{2.28}$$

where

α *Biot* coefficient, dimensionless,

$\breve{M}_\varphi$ *Biot* modulus of phase φ in MPa,

$\breve{K}_\varphi$ bulk modulus of phase $\varphi \in \{l; g; s\}$ in MPa,

$\breve{K}$ drained bulk modulus of the porous medium in MPa,

$\check{s}_\varphi$ volume-specific fluid storage in MPa^{-1}.

The equivalent pore pressure is set to be the saturation-weighted average of both phase pressures according to Eq. (2.29a), while one single scenario uses the maximum phase pressure, i.e., the pore-gas pressure, from Eq. (2.29b) instead:

$$p_{eq} = \sum_{\varphi} S_{\varphi} p_{\varphi} \tag{2.29a}$$

$$p_{eq} = p_g \tag{2.29b}$$

T→HM coupling: thermal expansion and thermal strains

By analogy with Eq. (2.8), temperature and volume or density of a phase are connected by its corresponding volumetric thermal expansion coefficient:

$$\frac{\dot{V}_{\varphi}}{V_{\varphi}} = \gamma_{\varphi}^{th} \dot{T} = -\frac{\dot{\rho}_{\varphi}}{\rho_{\varphi}} \tag{2.30}$$

where for $\varphi \in \{l; g; s\}$

γ_{φ}^{th} volumetric thermal expansion coefficient of phase φ in K^{-1}.

As with the bulk modulus, the thermal expansion coefficients for the solid and the liquid phases can be taken as a pressure- and temperature-independent constants in a simple approach. For an ideal gas, an expression for the volumetric thermal expansion coefficient can be derived from the ideal gas law by analogy to Eq. (2.9):

$$\gamma_g^{th} = \frac{1}{T} \tag{2.31}$$

Again, some simulators compute the fluid densities in a more complex manner.

Based on Eq. (2.30), the thermal strain for the solid skeleton can be derived to:

$$\dot{\varepsilon}^{th} = -\alpha^{th} \dot{T} \tag{2.32}$$

where

α^{th} linear thermal expansion coefficient of the solid skeleton in K^{-1}, $3\alpha^{th} = \gamma_s^{th}$.

H→T coupling: advective energy transport and pressure-volume work

Internal energy may also change due to pressure-volume work exerted by moving fluids and by fluid sources, and the fluids in motion carry internal energy along with them as well. Therefore, the internal-energy balance equation (2.19a), $(\lambda T')' + \hat{q}_Q = \dot{\Theta}_{U/V}$, has to be extended by advective and work terms, and rearranging them yields an equation containing the specific fluid enthalpies:

$$\left(\lambda T' - \underbrace{\sum_{\varphi} h_\varphi q_{m;\varphi}}_{=q_H^{adv}}\right)' + \left(\hat{q}_Q + \underbrace{\sum_{\varphi} h_\varphi \hat{q}_{m;\varphi}}_{=\hat{q}_H^{adv}}\right) = \frac{d}{dt}\left((1-\phi)\rho_s u_s + \phi \sum_{\varphi} S_\varphi \rho_\varphi u_\varphi\right) \tag{2.33}$$

where

$q_{m;\varphi}$ mass flux of phase φ in $kg/(m^2 s)$,

q_H^{adv} advective flux of enthalpy by mass flow in W/m^2,

$\hat{q}_{m;\varphi}$ mass generation rate for phase φ per unit volume of porous medium in $kg/(m^3 s)$,

$\hat{q}_H^{adv}$ supply rate of enthalpy by fluid generation in both phases φ per unit volume of porous medium in $J/(m^3 s)$.

In the simulators applied in the framework of the BenVaSim project, the thermal balance equation is formulated in different ways. Pressure-volume work is often omitted. The following equations (2.34)-(2.36) are implemented into the simulators according to *UPC (2017), Olivella et al. (1994), COMSOL (2019), Kolditz et al. (2012a), Pruess et al. (1999/2012), Jung et al. (2018),* and personal communication:

CODE_BRIGHT: $$\dot{\Theta}_{U/V} = \left(\lambda T' - q_U^{adv}\right)' + \hat{q}_Q \tag{2.34}$$

COMSOL Multiphysics and OpenGeoSys: $$\dot{\Theta}_{H/V} = \left(\lambda T' - q_H^{adv}\right)' + \hat{q}_Q \tag{2.35}$$

TOUGH2 and TOUGH3: $$\dot{\Theta}_{U/V} = \left(\lambda T' - q_H^{adv}\right)' + \left(\hat{q}_Q + \hat{q}_H^{adv}\right) \tag{2.36}$$

where

q_U^{adv} advective flux of internal energy by mass flow in W/m^2.

It was assumed that pressure-volume work has a negligible influence on energy distribution and temperature evolution in the benchmark models compared to heat generation and transport, and thus that the differences in the formulations of energy balance carry only minor weight.

2.6 Tables of designators

The designators of all alphabetic symbols used in this report are summarised in the following two tables.

c_φ	$J/(kg\ K)$	specific heat capacity of phase φ
E	MPa	drained *Young*'s modulus
E_u	MPa	undrained *Young*'s modulus
h_φ	J/kg	specific enthalpy of phase φ
K	m^2	intrinsic permeability of the porous medium
$K_{rel;\varphi}$	–	relative permeability of phase φ
$\breve{K}$	MPa	drained bulk modulus of the porous medium
$\breve{K}_\varphi$	MPa	bulk modulus of phase φ
$\breve{M}_\varphi$	MPa	*Biot* modulus of phase φ
m	–	curve-shape parameter from *van Genuchten*'s approach
p_φ	MPa	pore pressure of phase φ
p_{cap}	MPa	capillary pressure
p_{cap0}	MPa	pre-factor from *van Genuchten*'s approach
p_{eq}	MPa	equivalent pore pressure composed of phase pore pressures
q_E	$J/(m^2 s)$	total flux of energy
q_Q	$J/(m^2 s)$	heat flux
q_E^{adv}	W/m^2	advective flux of energy (enthalpy or internal energy)
q_H^{adv}	W/m^2	advective flux of enthalpy
q_U^{adv}	W/m^2	advective flux of internal energy
$q_{m;\varphi}$	$kg/(m^2 s)$	mass flux of phase φ
$q_{m;\kappa\varphi}$	$kg/(m^2 s)$	mass flux of component κ in phase φ
$q_{V;\varphi}$	m/s	volumetric flux of phase φ

Table 2.1a: Designators with a Latin basic letter

$\hat{q}_E$	$J/(m^3 s)$	total internal supply rate of energy per unit volume of porous medium
$\hat{q}_E^{adv}$	$J/(m^3 s)$	supply rate of energy by fluid generation in both phases φ per unit volume of porous medium
$\hat{q}_H^{adv}$	$J/(m^3 s)$	supply rate of enthalpy by fluid generation in both phases φ per unit volume of porous medium
$\hat{q}_Q$	$J/(m^3 s)$	heat generation rate per unit volume of porous medium
$\hat{q}_{m;\varphi}$	$kg/(m^3 s)$	mass generation rate for phase φ per unit volume of porous medium
$\hat{q}_{m;\kappa}$	$kg/(m^3 s)$	mass generation rate for component κ in both phases per unit volume of porous medium
$\hat{q}_{V;\varphi}$	s^{-1}	volumetric generation rate for phase φ per unit volume of porous medium
S_φ	–	saturation of phase φ
$S_{\varphi;eff}$	–	effective saturation of phase φ
$S_{\varphi r}$	–	residual saturation of phase φ
$S_{\varphi s}$	–	maximum saturation of phase φ
$\check{s}_\varphi$	MPa^{-1}	volume-specific fluid storage
T	K	temperature
t	s	time
u	m	displacement
u_φ	J/kg	specific internal energy of phase φ
V_φ	m^3	volume of a representative volume element of phase φ
X	m	initial reference position of a material point
$X_{\kappa\varphi}$	–	mass fraction of component κ within phase φ
x	m	current position of a material point (trajectory over time)

Table 2.1b: Designators with a Latin basic letter (continued)

α	–	*Biot* coefficient
$\tilde{\alpha}$	m^{-1}	curve-shape parameter from *van Genuchten*'s approach
α^{th}	K^{-1}	linear thermal expansion coefficient of the solid skeleton
$\tilde{\gamma}$	–	pore-connectivity parameter for the gas phase from *Mualem*'s approach
γ_{φ}^{th}	K^{-1}	volumetric thermal expansion coefficient of phase φ
ε	–	strain
ε_{lin}	–	linearised strain
ε_{log}	–	logarithmic strain
ε^{m}	–	mechanical strain
ε^{th}	–	thermal strain
ε_{vol}^{m}	–	mechanical volumetric strain
$\tilde{\varepsilon}$	–	pore-connectivity parameter for the liquid phase from *Mualem*'s approach
$\zeta_{m/V;\,\kappa}$	kg/m^3	accumulated mass influx of component κ in both phases per unit volume of porous medium
$\zeta_{V/V;\,\varphi}$	–	accumulated volumetric influx of phase φ per unit volume of porous medium
η_{φ}	$MPa \cdot s$	dynamic viscosity of phase φ
$\Theta_{H/V}$	J/m^3	content of enthalpy
$\Theta_{U/V}$	J/m^3	content of internal energy
$\Theta_{m/V;\,\kappa}$	kg/m^3	mass content of component κ in both phases per unit volume of porous medium
κ		fluid component – w: water; a: air
λ	$W/(m \cdot K)$	thermal conductivity
ρ_{φ}	kg/m^3	mass density of phase φ
σ_{eff}	MPa	effective stress
σ_{hyd}	MPa	hydrostatic stress induced by the pore pressure
σ_{tot}	MPa	total stress
φ		fluid phase – l: liquid phase; g: gas phase; s: solid phase
ϕ	–	material porosity

Table 2.2: Designators with a Greek basic letter

3 Simulators used in this project stage

3.1 CODE_BRIGHT (GRS BS)

The simulations of GRS BS are performed using the finite-element code CODE_BRIGHT which is developed by the Polytechnic University of Catalonia (UPC) in Barcelona. It is evolved for a coupled modelling of thermal-hydraulic-mechanical (THM) processes in geological media. The theoretical approach is based on a set of governing equations, a set of constitutive laws, and a special computational approach. The problem formulations are established on a multi-phase, multi-species approach. The computational language of CODE_BRIGHT is FORTRAN, *UPC (2017)*.

For pre- and post-processing, CODE_BRIGHT uses GiD system, developed by the International Center for Numerical Methods in Engineering (CIMNE) in Barcelona, *UPC (2017)*.

3.2 COMSOL Multiphysics (ENSI)

The simulator COMSOL Multiphysics has been applied in its versions 5.3, 5.4, and 5.5, *COMSOL (2019)*.

3.3 FTK simulator (TUC)

The coupled FTK simulator (FLAC^{3D}-TOUGH2-Kopplungs-Simulator) had been developed by TUC in the framework of a research project funded by BMWi, *Lux et al. (2015)*, with the aim of being able to simulate TH^{2}M processes in nuclear waste repository components and other subsurface constructions on the basis of well-established simulators for mechanical and two-phase-flow processes. During subsequent research projects, the simulator could be further developed in some aspects, *Lux et al. (2018)*, *Lux & Rutenberg (2018)*. Works on its verification had been conducted only to a certain extent in the course of these projects, works on its validation yielded publications like *Blanco Martín et al. (2016)* and formed an essential part of a doctoral dissertation, *Zhao (2017)*.

The sequentially coupled simulator has been developed with the geomechanical simulator FLAC^{3D} as the master process using functions in its inherent programming language FISH to repeatedly call TOUGH2 for the thermal-hydraulical process simulation parts and to exchange data between both simulators. A user familiar with FLAC^{3D} can implement a model for an FTK

simulation in a very similar way than for a FLAC^{3D} simulation, giving additional input necessary for TOUGH2 via the predefined FISH variables and functions ("FTK functions") and via external text files with predefined structures.

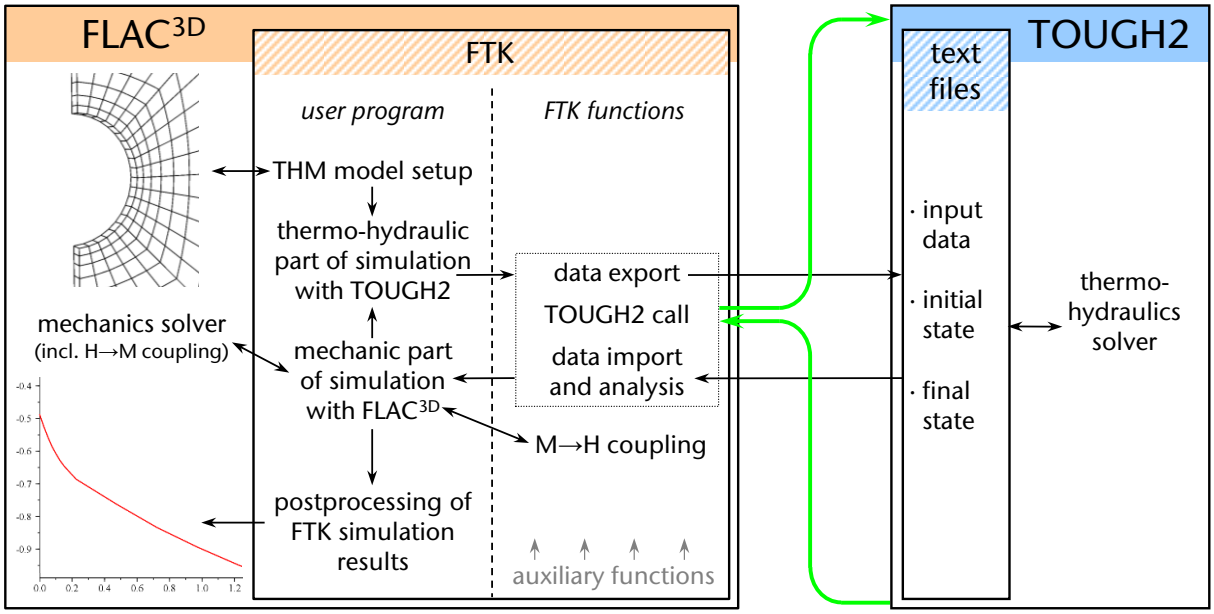


Figure 3.1: Simplified visualisation of the workings of the FTK simulator, translated from Lux & Rutenberg (2018)

Figure 3.1 illustrates in a simplified way how a user program is processed by the FTK simulator. When the user program is started with FLAC^{3D}, all data necessary for the thermo-hydraulic simulation part including the current zone state variables will be written into text files as input for TOUGH2, then, TOUGH2 will be called and perform a TH simulation for the first coupling time step while leaving the model's strains and displacements unchanged. After re-import of the updated TH state from TOUGH2 into FLAC^{3D} again via text files, FLAC^{3D} performs a mechanical balancing of the now perturbed stress state during which pressures and saturations keep being updated by means of FTK functions. The TH and M steps will then be carried out for the next coupling time step, making this solution method a "staggered" method. This procedure is repeated until the simulated time reaches its end point, by when the results of the coupled simulation can be examined by plots and curves from FLAC^{3D} and postprocessed in further ways.

Basically, the equations involved in FLAC^{3D} and TOUGH2 are those given in section 2, with the following clarifications:

– In every mechanical simulation cycle, FLAC^{3D} computes linear strains with respect to current nodal positions and zone dimensions; however, since these dimensions change with time, the resulting strains correspond in total to the logarithmic strains from Eq. (2.4), $\varepsilon = \varepsilon_{log}$.

– The hydraulic simulations are performed by the simulator TOUGH2 whose hydraulic equations are built around mass balance and transport of the fluids. Contrary to this, the M→H coupling in the FTK simulator is realised in an analogous way to FLAC^{3D} on the basis of the fluids' volumes by means of Eq. (2.27) without the volumetric influx rate, i.e., $\dot{p}_\varphi / \breve{M}_\varphi + \phi\, \dot{S}_\varphi / S_\varphi = \alpha \dot{\varepsilon}^m$.

– Equations (2.8) and (2.30) are not applied for computation of densities in TOUGH2; instead, according to *Pruess et al. (1999/2012)*, densities of liquid water and of water vapour are computed based on compilations from *IFC (1967)*, and density of gas is composed of those of vapour and of air, the latter one being treated as an ideal gas. Figure 3.2 shows the densities of liquid water and of vapour-saturated air as functions of the corresponding phase pressure and temperature for the states where the boiling point of water is not reached.

– TOUGH2 considers pressure-volume work according to Eq. (2.36) for balancing internal energy, $\dot{\Theta}_{U/V} = \left(\lambda T' - q_H^{adv}\right)' + \left(\hat{q}_Q + \hat{q}_H^{adv}\right)$.

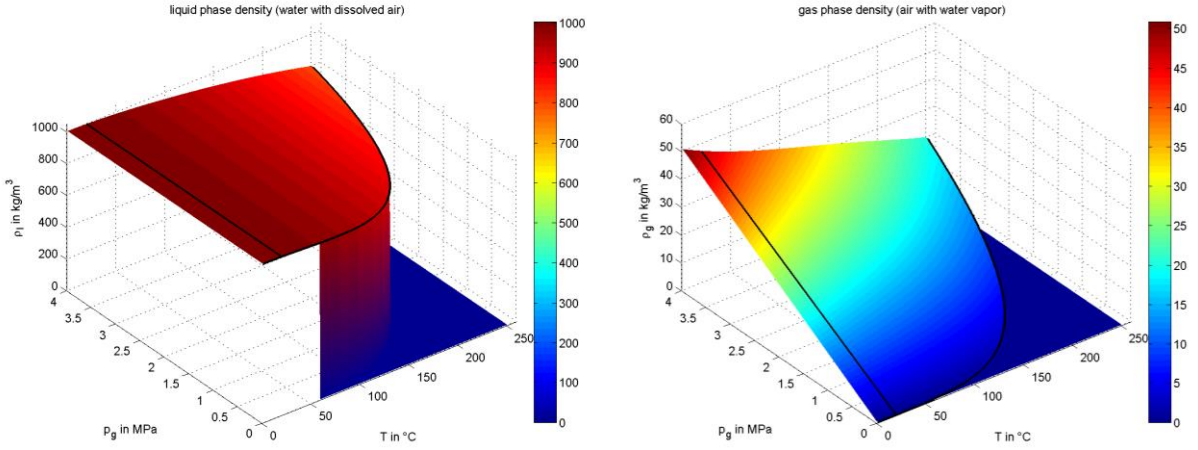


Figure 3.2: Water density (left) and density of water-saturated air (right) versus pressure and temperature for two-phase states (with $p_{cap} = 0\ MPa$ set for computing)

Version 4.0 of FLAC^{3D} (*Itasca, 2009*) and version 2.0 of LBNL's TOUGH2 (*Pruess et al., 1999*), including the bug fixes that were published on their website as well as slight own modifications, have been used for this stage of the BenVaSim project.

3.4 OpenGeoSys (BGR, ENSI)

To model TH^{2}M coupled processes, BGR and ENSI deploy the open source FEM code OpenGeoSys in version 5 (OGS-5) and version 6 (OGS-6) which is currently being developed by various research entities under the lead of the Department of Environmental Informatics at the Helmholtz Centre for Environmental Research (UFZ) in Leipzig. OGS-6 represents an advancement to the previous FEM code OGS-5, *Kolditz et al. (2012a)*, *Bilke et al. (2019)*. The main objective of this implementation is the application of numerical methods to simulate thermo-hydraulic-mechanically and chemically coupled processes in porous and fractured media. Apart from nuclear waste disposal, other typical fields of application of the code are: CO_2 sequestration, shallow and deep geothermal energy systems as well as hydrogeological and water resources management applications. Both OGS-5 and OGS-6 use the same primary variables for the coupled processes (heat transport process: temperature T; hydraulic processes: gas pressure p_g and capillary pressure p_{cap}; mechanical process: displacement vector u). A general difference between the two codes is the numerical solving scheme: In OGS-5, coupled processes are solved generally in a staggered scheme where processes are solved one after another in each time step: updated main variables of one process are then passed on to the next process before solving it. This passing of variables between processes is repeated iteratively in each time step. An exception is the option `DEFORMATION_FLOW`, where the HM coupled process is solved monolithically. In OGS-6, the entire TH^{2}M coupled process is solved monolithically.

For a reliable quality assurance, both OGS-5 and OGS-6 are tested permanently and automatically using benchmark problems (*Kolditz et al., 2012b*; *Kolditz et al., 2015*; *Kolditz et al., 2016*) for which analytical solutions as well as results from other numerical codes are available.

Within the framework of the development of OGS-6, *Naumov et al. (2018)*, *Bilke et al. (2019)*, a new TH^{2}M coupled implementation is currently being developed, *Böttcher et al. (2017)*. This implementation is in a stage of testing and verification. Thus, the BenVaSim project represents the opportunity to test and verify the already well tested implementation of OGS-5 as well as the new implementation of OGS-6.

3.5 TOUGH-FLAC and TFC (LBNL, ENSI, GRS K)

The project partners LBNL, ENSI, and GRS K each applied a coupling of the simulators TOUGH and FLAC^{3D} based on the coupled TOUGH-FLAC simulator implemented by LBNL, *Rutqvist et al. (2002)*. However, different versions of the basic simulators are incorporated in the different TOUGH-FLAC simulators:

- LBNL used TOUGH2 2.0, *Pruess et al. (1999)*, with the bug fixes from their website, together with FLAC^{3D} 4.0, *Itasca (2009)*. A comprehensive overview of this version including examples of application has been given in *Rutqvist (2011)*.
- ENSI used the recently developed TOUGH3, *Jung et al. (2018)*, together with FLAC^{3D} 6.0, *Itasca (2017)*. The correspondingly updated simulator coupling was developed in collaboration of the Swiss Seismological Service at ETHZ with ENSI and LBNL. The approach was first presented at the TOUGH Symposium in 2018, *Rinaldi et al. (2018)* (at that time still with FLAC^{3D} 5.0), and it is now further described in a preprint at ESSOAr, *Rinaldi et al. (2021)*.
- GRS K used an own version of TOUGH-FLAC called TFC, *Hotzel (2014)*, which is based on an own version of LBNL's TOUGH2 called TOUGH2-GRS, coupled with FLAC^{3D}. The simulator TOUGH2-GRS in itself is based on the bug-fixed and further developed TOUGH2 2.0 and was used for BenVaSim in its version 2, *Navarro (2018)*. TFC solely uses a modified version of the two-phase-flow module `EOS7` of TOUGH2-GRS, which features brine as a third component besides water and air, and the simulator has been adapted to the more recent version 6 of FLAC^{3D}, *Seher et al. (2019)*, *Itasca (2017)*.

4 Model examples and simulation results

4.1 Model 1.1

4.1.1 Model description

Model 1.1 is a homogeneous model with a length of 10 metres, fixed at its right end and free to move due to effective stress changes at its left end where a given total stress is applied. The void space of the model's porous material is fully saturated with water, and further amounts of water penetrate into the model on its left-hand side by means of a fixed boundary condition, raising the pore pressure inside the model and eventually leading to an efflux of water on the model's right-hand side. Figure 4.1 visualises the model's setting. Single-phase flow following *Richards*' equation, Eq. (2.10), is assumed for the hydraulic processes, and *Hooke*'s linear-elastic constitutive model is applied for the mechanical processes in every model of this project stage. Several analytical solutions can be derived for various state variables in different stages of the H[1]M process (instantaneous response, transient material behaviour, steady state), *Lux et al. (2021).*

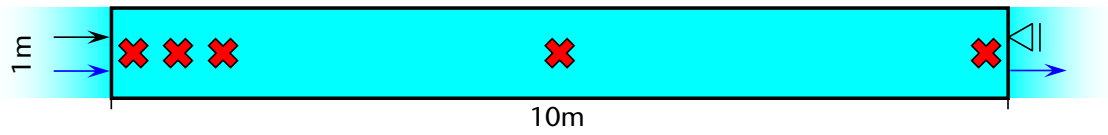


Figure 4.1: Visualisation of model-1.1 geometry and initial and boundary conditions for scenarios 1.1(a-c), and evaluation points

The curves of some state variables plotted against time and/or position have been chosen for the juxtaposition of the project partners' results. For model 1.1, the relevant state variables are pore-water pressure, strain, and displacement. The evaluation points for recording state variables over time were given to be $X = 0.25\ m$, $0.75\ m$, $1.25\ m$, $5\ m$, and $9.75\ m$ (cf. Fig. 4.1), and state profiles in each scenario ought to be taken at predefined points of time, which may differ from each other due to different process speeds in the respective scenarios. While the seven points of time $t = 0.003\ a$, $0.1\ a$, $0.3\ a$, $1\ a$, $3\ a$, $10\ a$, and $30\ a$ were designated for scenarios 1.1(a-b), the points $t = 0.1\ a$, $3\ a$, $10\ a$, $30\ a$, $100\ a$, $300\ a$, and $1{,}000\ a$ are more appropriate for scenario 1.1(c).

Material and fluid parameters as well as initial and boundary conditions for the base scenario 1.1(a), which also serve as the basis for the other model-1.1 scenarios, are listed in Table 4.1. In the other two scenarios, single material parameters are modified as compared with the base

scenario: compressible grains associated with a reduced *Biot* coefficient of $\alpha = 0.75$ are assumed for scenario 1.1(b), and a softer matrix associated with a significantly reduced drained *Young* modulus of $E = 150\ MPa$ is supposed for scenario 1.1(c).

Material and fluid parameters		
Young's modulus (dr.)	E	8,000 MPa
Poisson's ratio	ν	0
porosity	ϕ	0.15
intrinsic permeability	K	10^{-20} m^2
Biot coefficient	α	1
liquid bulk modulus	$\breve{K}_l$	2,100 MPa
liquid viscosity	η_l	10^{-9} $MPa \cdot s$

Initial conditions		
total stress	σ_{tot0}	0.1013 MPa
pore-water pressure	p_{l0}	0.1013 MPa
liquid saturation	S_{l0}	1

Boundary conditions		
lhs total stress	$\sigma_{tot;lhs}$	1 MPa
lhs pore pressure	$p_{l;lhs}$	1 MPa
lhs liquid saturation	$S_{l;lhs}$	1
rhs displacement	u_{rhs}	0 m
rhs pore pressure	$p_{l;rhs}$	0.1013 MPa
rhs liquid saturation	$S_{l;rhs}$	1

Modified material parameter for 1.1(b)		
Biot coefficient	α	0.75

Modified material parameter for 1.1(c)		
Young's modulus (dr.)	E	150 MPa

Table 4.1: Parameters and initial and boundary conditions for scenario 1.1(a) together with parameter adaptions for scenarios 1.1(b-c)

4.1.2 Individual approaches for simulation

4.1.2.1 BGR (simulator OGS)

The numerical simulation was performed with OGS-5 using the monolithic, hydraulic-mechanically (HM) coupled implementation `DEFORMATION_FLOW`. Testing showed that using the staggered solving scheme of the HM coupling leads to numerical difficulties due to the divergence of the numerical solution, *Pitz & Maßmann (2020)*. The finite element type used was quadrangular *Taylor/Hood* elements (quadratic order elements for the mechanic process and linear order elements for the hydraulic process). This results in a linear equation system, which was solved using the SpBiCGSTAB linear solver, an internal OGS-5 implementation of the

stabilised biconjugate gradient method, on a single core within about 1 minute of calculation time.

Initially, a mesh with 20 elements was used based on a suggestion by the project coordinator. With constant element size, this results in an element length of $0.5\,m$ and a node distance of $0.25\,m$ for the quadratic nodes. This discretisation resulted in mesh-dependent oscillations and was subsequently refined.

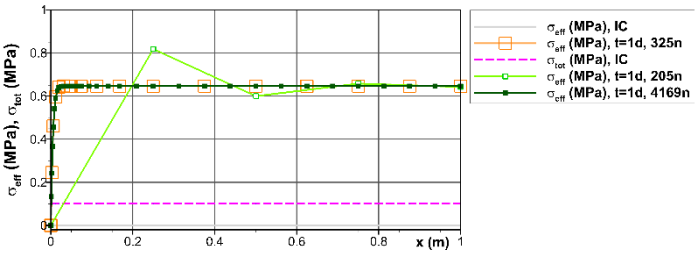


Figure 4.2: Mesh-dependent oscillations of the solution for the effective stress distribution. Finer mesh discretisations do not show such oscillations.

Some mesh convergence testing showed that increasing the number of elements to 256 (325 nodes) and refining elements towards the left and right domain boundaries eliminates the numerical oscillations. This refinement was tested against a simulation with 4,169 nodes which led to similar results and showed that mesh convergence is sufficient with 256 elements:

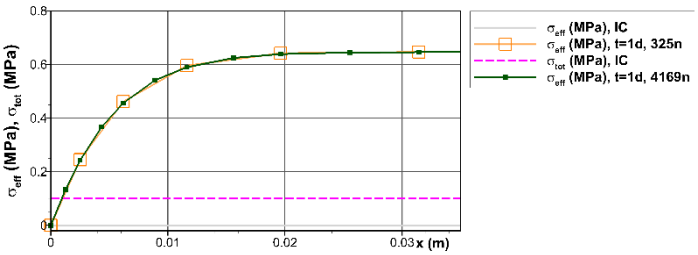


Figure 4.3: Comparison of effective stress results using different refinements. Solutions obtained by using a mesh with 325 nodes and 4,169 nodes show a good agreement. Therefore it can be stated that a mesh with 325 nodes yields good results.

Temporal discretisation was realised with 132 time steps, where time step size was increased by a log-scheme. This allows for an analysis both of the short-term behaviour (such as the immediate response of the system to non-equilibrium initial conditions) and the long-term behaviour of the system.

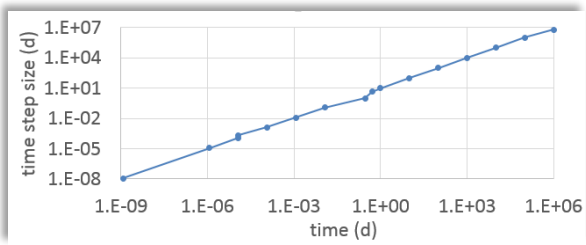


Figure 4.4: Time step size increasing with time in the logarithmic scheme

At the right domain boundary, hydraulic and mechanical boundary conditions (BCs) were implemented as *Dirichlet*-type BCs (zero x-displacement and pore-water pressure kept at the value of the initial condition). At the left domain boundary, the hydraulic BC was implemented as a ramp: Initially, the value of the *Dirichlet*-type BC equalled the initial condition and over the course of one day, it was ramped up to equal the value specified in the model definition. A similar one-day ramp was implemented for the mechanical *Neumann*-type stress source term at the left hand side boundary.

4.1.2.2 ENSI (simulator TOUGH-FLAC)

ENSI used the two-phase-flow module `EOS3` from TOUGH3. A minor modification in the coupling routine between TOUGH3 and FLAC^{3D} needed to be introduced to account for application of boundary conditions at time 0^+.

The mesh accounted for 50 elements, with more fine discretisation near the right- and left-hand-side boundaries.

No issues were reported when running the simulation with non-unity *Biot*'s coefficient or with small *Young*'s modulus.

4.1.2.3 ENSI (simulator OGS)

ENSI used OGS-5 with the monolithic, hydraulic-mechanically (HM) coupled implementation `DEFORMATION_FLOW` to model the fully saturated cases. The mesh used for this case was a 2D mesh with 20 cells along x-axis and 1 cell along z-axis as it was initially suggested by the project coordinator. A simulation time of 31,700 years was split into 138 time steps. The parameters, initial and boundary conditions were assigned as described in the task.

4.1.2.4 GRS BS (simulator CODE_BRIGHT)

The numerical simulation was executed with the FEM code CODE_BRIGHT and is described in *Czaikowski & Friedenberg (2020)*.

For the simulation of the scenario with reduced *Biot* coefficient, GRS BS had some issues. Towards the prediction of a lower liquid pressure level than in scenario 1.1(a) (Fig. 4.10), the submitted initial results of GRS BS showed higher liquid pressure levels (Fig. 4.12). As a result, the displacements were lower (Fig. 4.13).

Within the project, it could be clarified that indeed the mechanical part of the *Biot* theory is implemented in CODE_BRIGHT, the hydraulic part considering the liquid pressure change due to strain changes and liquid fluxes by the use of the *Biot* modulus was not incorporated, *Czaikowski & Friedenberg (2020)*. For the particular case of the considered scenario, the missing hydraulic effect could be reproduced by an adaption of the solid compressibility leading to satisfying results compared to the project partners (curve-fitting procedure).

After the end of the BenVaSim project, a new version of CODE_BRIGHT with a fixed implementation of the *Biot* theory became available and ready to be employed in future simulations.

4.1.2.5 GRS K (simulator TFC)

For the simulations by GRS K, the brine component featured in module `EOS7` of TOUGH2-GRS has been neglected by setting the brine mass fraction to zero.

4.1.2.6 LBNL (simulator TOUGH-FLAC)

The simulations by LBNL have been conducted using the two-phase-flow module `EOS3` from TOUGH2.

4.1.2.7 TUC (FTK simulator)

Due to the saturated conditions specified for model 1.1, the equation-of-state (EOS) module `EOS9` for single-phase flow following *Richards*' equation (2.10) has been applied by TUC for the simulation parts conducted with TOUGH2. In this module, the gas phase (which may well be present) is considered a "passive bystander", *Pruess et al. (1999/2012)*, meaning that

changes in pore-gas pressure are neglected and thus that no mass-balance or flow equations have to be solved for this phase. Besides gas pressure and liquid viscosity, also temperature and liquid bulk modulus are kept constant, whereas liquid density is calculated by means of the simple relation Eq. (2.8).

In scenario 1.1(c), application of the drained *Young* modulus leads to simulation issues that could be overcome by using the undrained *Young* modulus $E_u = E + \alpha^2 \breve{M}_l$ and omitting the M→H feedback routines instead, *Lux et al. (2021)*.

4.1.3 Results and conclusions

Scenario 1.1(a) is the simplest of all scenarios from stage 1 of the BenVaSim project, and thus results have been collected from all project partners showing that all applied simulators are capable of reproducing the system behaviour with only minor differences. Analytical solutions elaborated on in *Lux et al. (2021)* can be brought in as a benchmark and confirm the partners' numerical results. Temporal and spatial discretisations chosen too coarse have posed the main issues in this scenario, leading sometimes at the start of the time-dependent simulation to a slightly delayed flow process as well as to numerical oscillations next to the model boundaries due to the initially large pore-pressure gradients. Consequently, refining the coarse discretisations helped to reduce corresponding discrepancies.

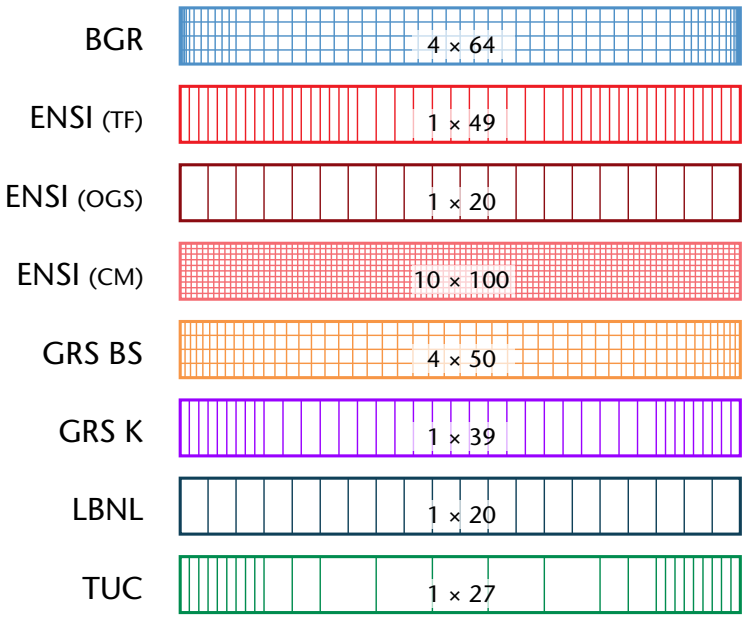


Figure 4.5: Discretisations applied by the partners for simulations of model-1.1 scenarios

Figure 4.5 shows the different model discretisations used by the partners in their final simulations for all three scenarios of model 1.1. A coarse discretisation of 1×20 equally shaped rectangular or cuboid zones initially suggested by the project coordinator had been adopted and kept by two parties, however, most simulations have been performed on refined discretisations with relatively high numbers of zones especially towards the model boundaries where pore-pressure gradients can become very large. Although a finer discretisation usually leads to more accurate simulation results and to smoother curves, partners had to weigh up accuracy against duration of their simulation. For this reason, most uncoupled simulators have managed to simulate the model-1.1 scenarios with discretisations between 200 and 1,000 zones in a reasonable time, whereas the discretisations used for simulator couplings remained at zone numbers below 50.

A standard colour code and dashing pattern is applied to most stage-1 figures defining the curves of the different project partners:

- blue long-dash curves represent the results of BGR,
- red long-dash-dot curves represent the results of ENSI with TOUGH-FLAC,
- dark red long-dash-dot curves represent the results of ENSI with OpenGeoSys,
- light red long-dash-dot curves represent the results of ENSI with COMSOL Multiphysics,
- yellow long-dash-dot-dot curves represent the results of GRS, team Brunswick,
- purple dashed curves represent the results of GRS, team Cologne,
- dark blue dash-dot curves represent the results of LBNL, and
- green short-dash curves represent the results of TUC.

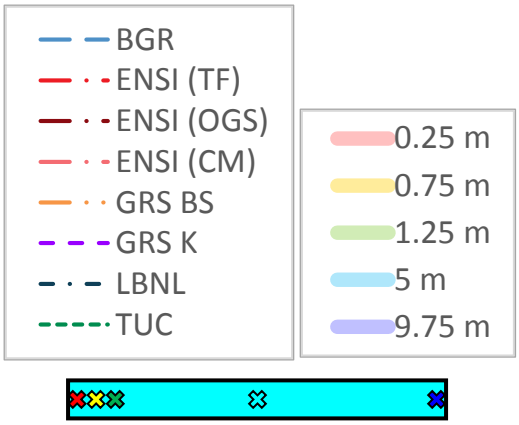


Figure 4.6: General colour coding and dashing pattern for curves in models 1.1 and 1.2

In addition, for models 1.1 and 1.2, temporal evolutions of state variables are recorded at the five model positions $X = 0.25\ m$, $0.75\ m$, $1.25\ m$, $5\ m$, and $9.75\ m$, with a colour coding for corresponding analytical solutions using red, yellow, green, cyan, and blue, respectively. The legends for the curves showing both colour codings are depicted in Fig. 4.6. Analytical solutions for spatial distributions at different points of time are displayed in colours from a specific colour set for the respective model or scenario, with the steady-state solutions always being displayed in grey.

In some of the figures, a distinction of the simulation results on the basis of the different partners is not intended. In those cases, the curves have been given the colour of the corresponding analytical or semi-analytical solutions or, when such solutions are not present, colours of a specific colour set for the respective model or scenario.

Scenario 1.1(a)

Figures 4.7 and 4.8 show the temporal evolution of pore-water pressure for scenario 1.1(a) at the five evaluation positions given above. The computed results match each other and the analytical solution well, where it should be noted that the discrepancy of the dark red curve from the others for position $X = 5\ m$ is owed to an erroneous adoption of the position. A logarithmic scale for the time axis magnifies the short-term reaction of the system, revealing that especially for the two positions near the model boundaries, the evolution of pore-water pressure is a little delayed in some of the simulations. Compared to the first curves that have been submitted by every partner shown in Fig. 4.9 (distinguished only according to the positions), refinements of model discretisations as well as refinements in time-stepping and concomitantly an improved balancing of the instantaneous system response helped to reduce the delay of pressure evolution and also to reduce the numerical oscillations obtained for the instantaneous response. Reduced oscillations near the boundaries can still be identified in the pressure profiles from Fig. 4.10.

Finally, the system's displacement profiles are depicted in Fig. 4.11. The accuracy of the partners' results for steady-state distribution appears to be a little behind that for the steady-state results for pore-water pressure from Fig. 4.10, but on the whole, the simulations produced very satisfactory results. Concurrent displacements have also been obtained by the different simulations where analytical solutions are missing, where the slight mismatch of the dark blue curves can be associated with a coarse time-stepping responsible also for the delayed pressure evolution identified in Fig. 4.8.

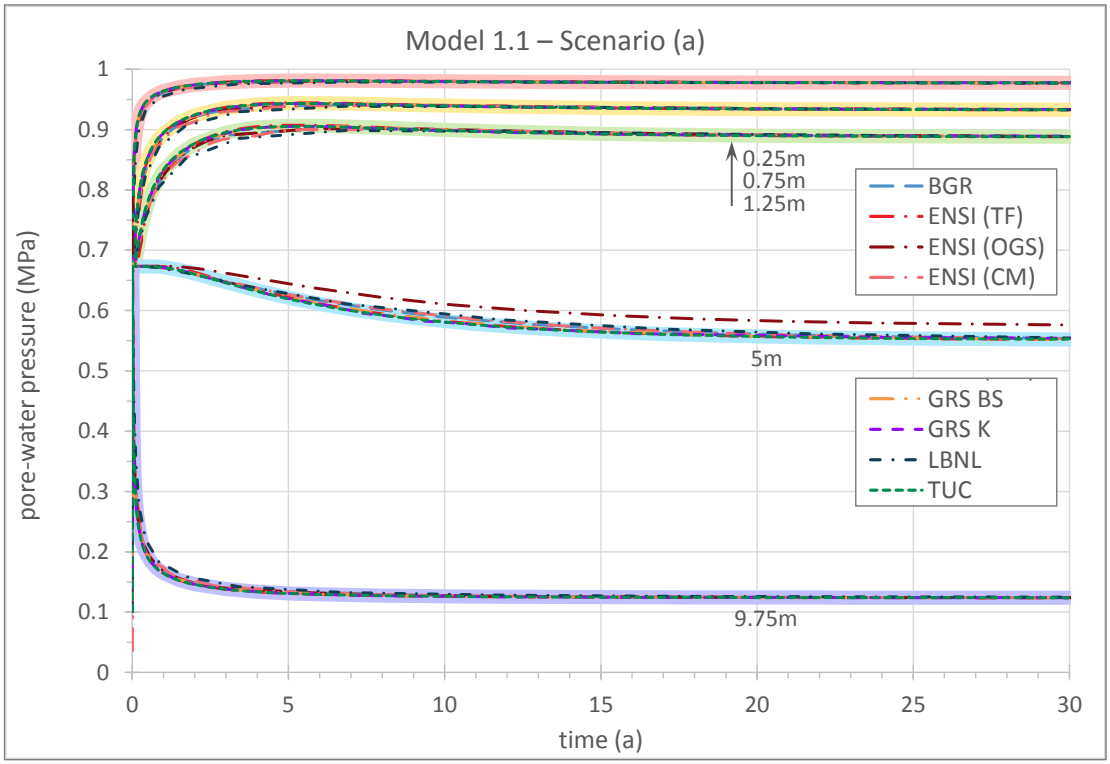


Figure 4.7: Simulation results for scenario 1.1(a) compared to analytical solutions – evolution of pore-water pressure (linear time axis)

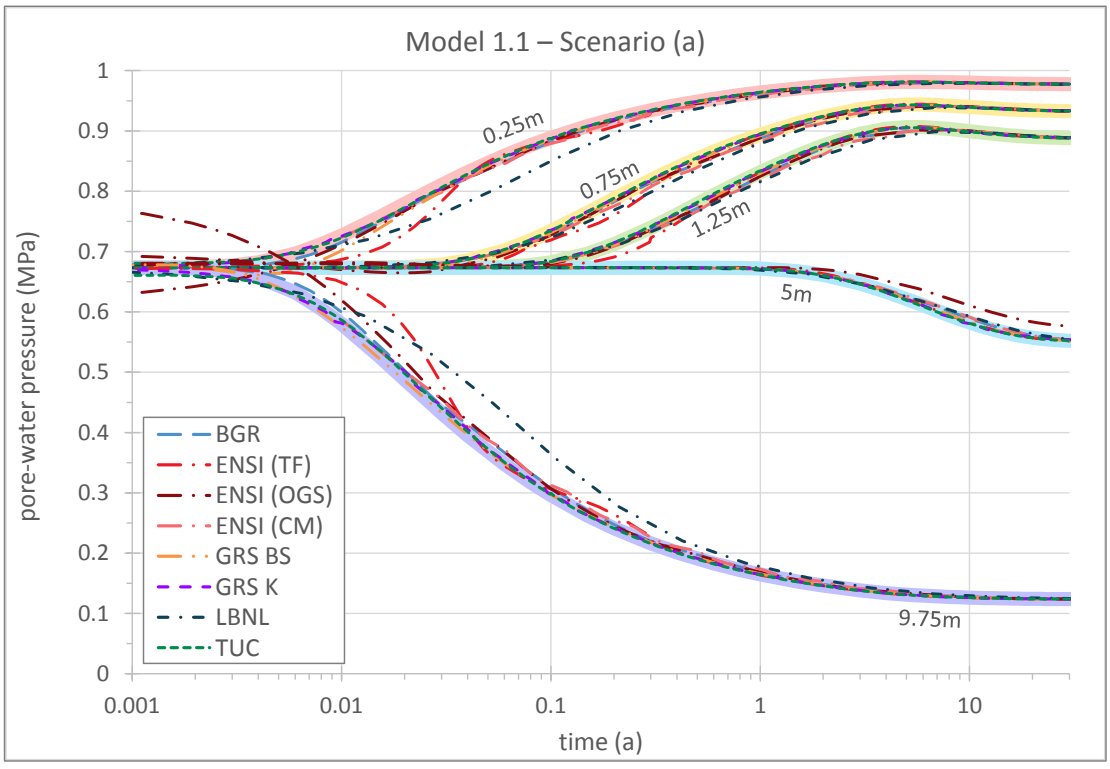


Figure 4.8: Simulation results for scenario 1.1(a) compared to analytical solutions – evolution of pore-water pressure (logarithmic time axis)

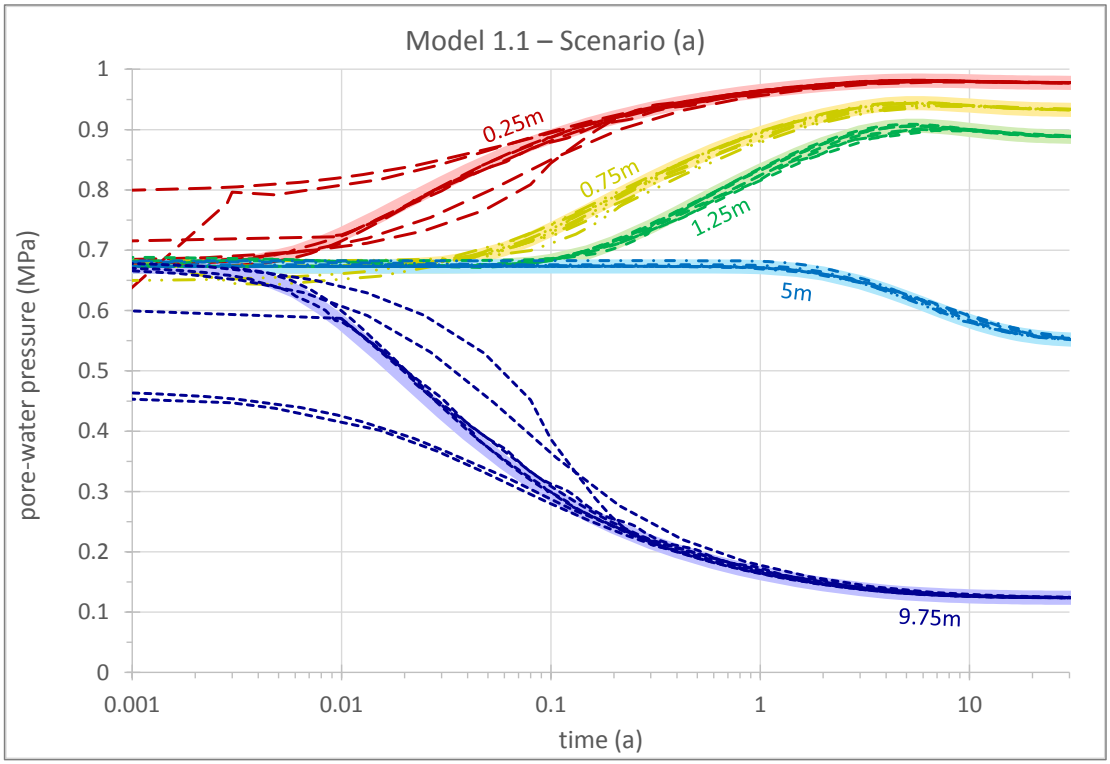


Figure 4.9: First simulation results for scenario 1.1(a) compared to analytical solutions – evolution of pore-water pressure (logarithmic time axis)

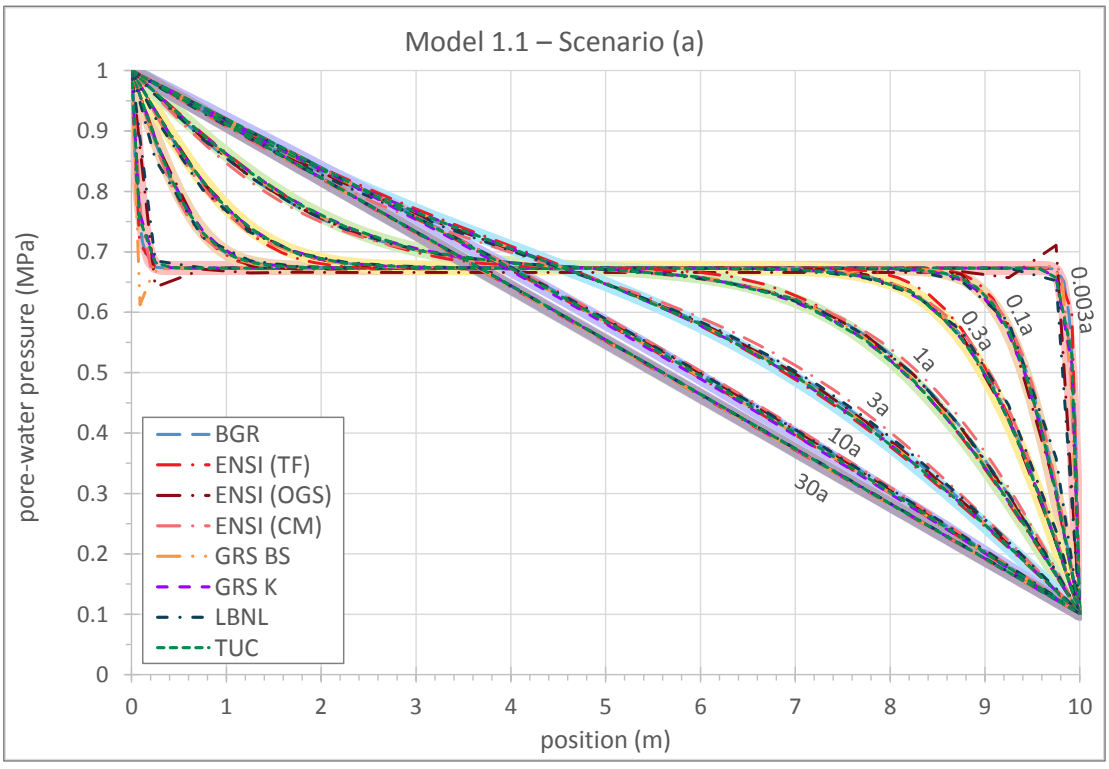


Figure 4.10: Simulation results for scenario 1.1(a) compared to analytical solutions – spatial distribution of pore-water pressure

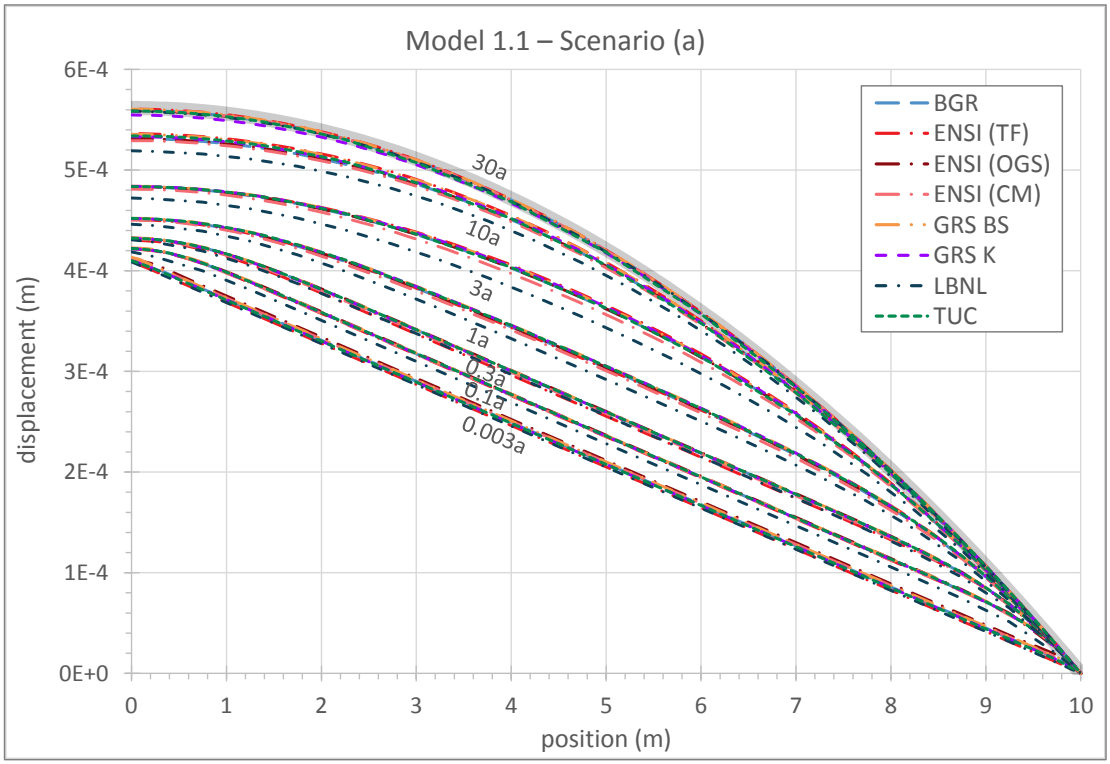


Figure 4.11: Simulation results for scenario 1.1(a) with steady-state analytical solution – spatial distribution of displacement

Scenario 1.1(b)

In scenario 1.1(b), incompressibility of the solid grains has been withdrawn by setting *Biot*'s coefficient to $\alpha = 0.75$. Regarding *Biot*'s effective-stress definition, increasing the interstitial pressure now leads to a reduced expansion of the matrix since the forces exerted by the pore fluids become cushioned by the compressible grains to a certain extent. Similarly, when reducing the ratio of pore-space volume to fluid mass, either by compacting the material or by leading additional fluid mass into the pore space, the resulting pressure build-up turns out to be attenuated compared to a material with incompressible grains.

Considering these stated effects of compressibility of grains, the instantaneous pressure response of the pore water is expected to decrease compared to scenario 1.1(a) in conjunction with an increase of compaction, i.e. in this model's setting, a displacement further to the right side. The hydraulic steady state is independent of *Biot*'s coefficient, which means that the liquid-pressure difference between initial and final state amounts to the same for both scenarios 1.1(a) and 1.1(b) and thus that the final degree of compaction ought to be larger in case of more compressible grains in 1.1(b). The analytical solutions for the profiles for pore-water pressure

and liquid saturation displayed in Figs. 4.12 and 4.13 compared to the corresponding Figs. 4.10 and 4.11 from scenario 1.1(a) meet these expectations, and most partners' results manage to reproduce the HM-coupled behaviour of the model. A slightly elevated instantaneous pressure response in the pink curves seems to lead to a slightly diminished pressure build-up (in order to obtain the same steady-state pressure) in conjunction with slightly less expansive strains and accumulating displacements to the right. As for the yellow curves, the *Biot* theory was found to be implemented incompletely into the simulator CODE_BRIGHT: while the effective-stress equation is considered for a simulation with this theory, the correct adaption of pore pressure upon strain change or water influx/efflux with the help of *Biot*'s modulus and the corresponding coupling equation was found to be missing, *Czaikowski & Friedenberg (2020)*. According to that source, GRS BS was able to obtain a solution matching the others by fitting the grain compressibility in CODE_BRIGHT. Figures 4.12 and 4.13 contain the results GRS BS had obtained normally with the simulator version available during the project period (cf. section 4.1.2.4) to show the identified shortcomings with respect to the partners' results.

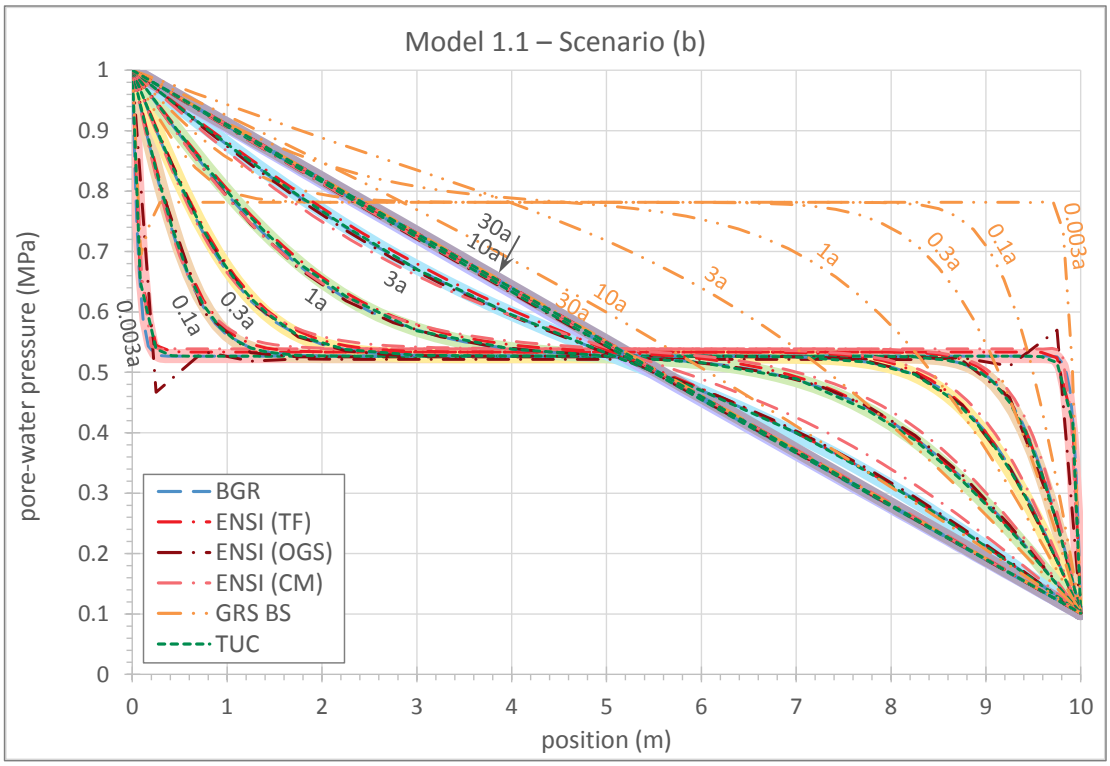


Figure 4.12: Simulation results for scenario 1.1(b) compared to analytical solutions – spatial distribution of pore-water pressure

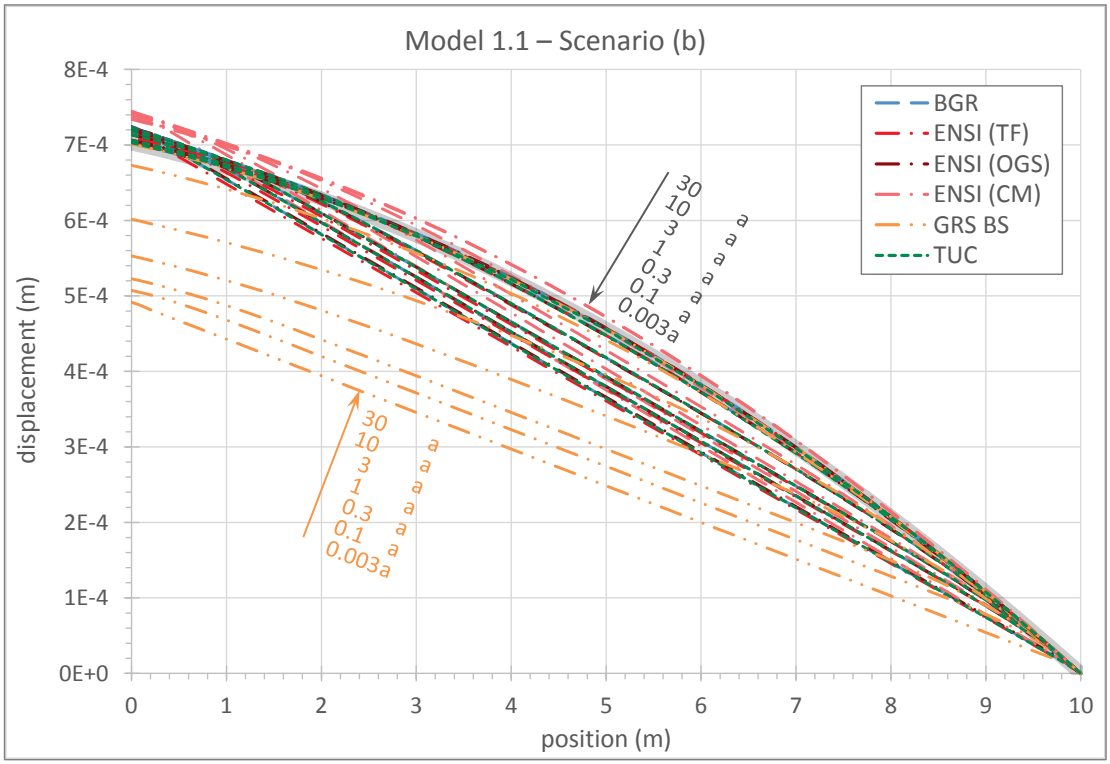


Figure 4.13: Simulation results for scenario 1.1(b) with steady-state analytical solution – spatial distribution of displacement

Scenario 1.1(c)

Finally, Figure 4.14 shows the profiles for pore-water pressure obtained by the partners' simulations for scenario 1.1(c) with a soft matrix possessing a *Young* modulus of $E = 150\ MPa$. Aside from numerical oscillations as already observed before—here especially near the right model boundary since the instantaneous response yields a pore pressure very close to its left boundary value—the approximation of steady state seems to be a little less accurate for some of the simulators than in both preceding scenarios. Nevertheless, the results produced by the different simulators match each other and the analytical solution well.

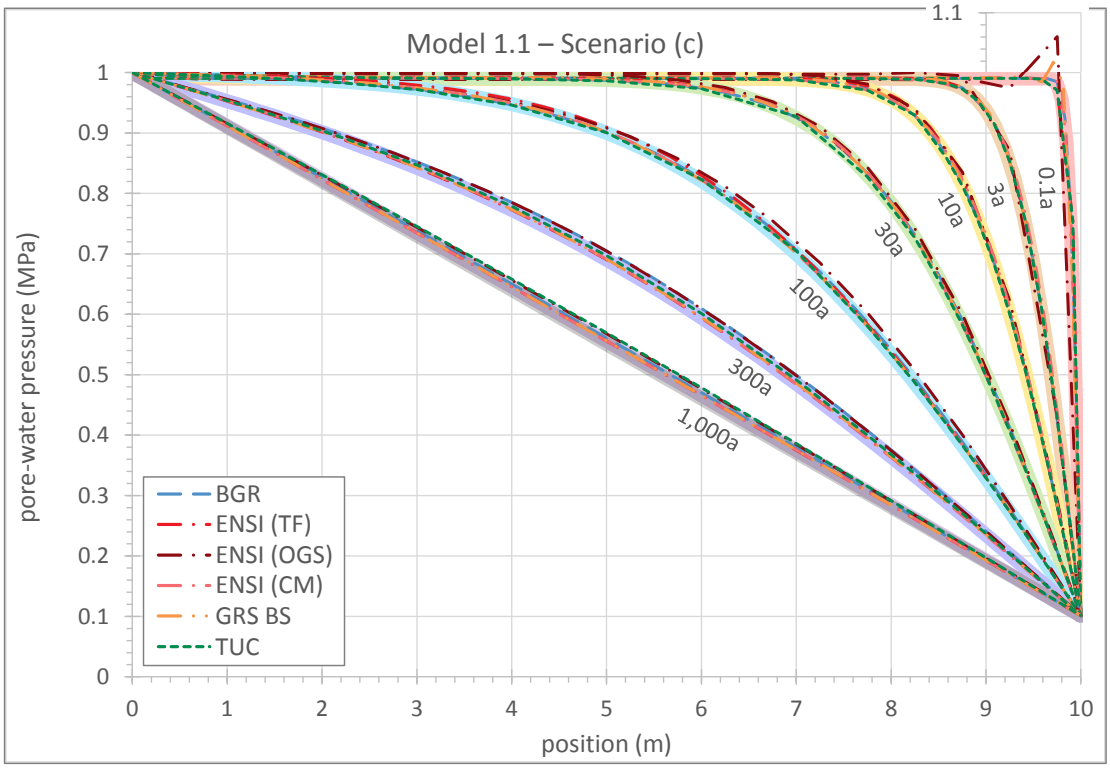


Figure 4.14: Simulation results for scenario 1.1(c) compared to analytical solutions – spatial distribution of pore-water pressure

4.2 Model 1.2

4.2.1 Model description

Model 1.2 looks very similar to model 1.1, with the major difference of taking *Darcy* flow also of the gaseous phase into account. The model has the same dimensions and spatial evaluation points as the previous one, it is clamped at its right boundary while free to move at its left one, and the principal fluid-flow direction is mostly to the right, cf. Fig. 4.15. Note that in Fig. 4.15 and in all similar figures, the separation of blue (fraction of liquid) and white (fraction of gas) areas is just meant to visualise the degree of saturation, not to indicate a vertical distribution of phases.

However—aside from changed parameters and initial and boundary conditions—the addition of two-phase flow including dependency of relative permeabilities on liquid saturation leads to differences in the model's flow behaviour, especially when capillary forces are considered as well. Similarly to model 1.1, an overpressure of the gas phase on the model's left-hand side induces a gas flow towards the right, but when capillary forces are present, an increased suction on the left-hand side coming from a reduced liquid saturation lets the liquid phase move in the opposite direction. In the course of a simulation, fluids at boundaries may initially flow in a certain direction due to the initial pore-pressure distributions, but change direction later; this behaviour is visualised in Fig. 4.15 by means of a double arrow whose larger arrow indicates the steady-state flow direction.



Figure 4.15: Visualisation of model-1.2 geometry and initial and boundary conditions for scenarios 1.2(a-c), and evaluation points

Scenario-dependent points of time for taking profiles are listed in Table 4.2 together with the spatial evaluation points already mentioned.

Evolutions at $X =$	Profiles at $t =$	$\approx$
$0.25\ m$	$8 \cdot 10^{8}\ s$	$25\ a$
$0.75\ m$	$8 \cdot 10^{9}\ s$	$250\ a$
$1.25\ m$	$2.1 \cdot 10^{10}\ s$	$650\ a$
$5\ m$	$8 \cdot 10^{10}\ s$	$2{,}500\ a$
$9.75\ m$	$3 \cdot 10^{11}\ s$	$10{,}000\ a$
	$3 \cdot 10^{12}\ s$	$100{,}000\ a$
	and for 1.2(b-c) also at	
	$1.2 \cdot 10^{13}\ s$	$380{,}000\ a$

Table 4.2: Positions for temporal evolutions and points of time for profiles concerning model 1.2

Material and fluid parameters		
Young's modulus (dr.)	E	$650\ MPa$
Poisson's ratio	ν	0
porosity	ϕ	0.33
intrinsic permeability	K	$2.5 \cdot 10^{-21}\ m^2$
Biot coefficient	α	1
res. liquid saturation	S_{lr}	0.02
res. gas saturation	S_{gr}	0
van Gen. parameter	m	0.5
van Gen. pre-factor	p_{cap0}	$11\ MPa$
pore connectivity par's	$\tilde{\varepsilon}; \tilde{\gamma}$	0.5
liquid bulk modulus	$\breve{K}_l$	$2{,}220\ MPa$
liquid density	ρ_l	$1{,}000\ kg/m^3$
liquid viscosity	η_l	$10^{-9}\ MPa \cdot s$
gas viscosity	η_g	$1.8 \cdot 10^{-11}\ MPa \cdot s$

Initial conditions		
total stress	σ_{tot0}	$4\ MPa$
pore-gas pressure	p_{g0}	$0.2\ MPa$
liquid saturation	S_{l0}	0.63

Boundary conditions		
lhs total stress	$\sigma_{tot;lhs}$	$4\ MPa$
lhs pore pressure	$p_{g;lhs}$	$3\ MPa$
lhs liquid saturation	$S_{l;lhs}$	0.5
rhs displacement	u_{rhs}	$0\ m$
rhs pore pressure	$p_{g;rhs}$	$0.5\ MPa$
rhs liquid saturation	$S_{l;rhs}$	0.9

Table 4.3: Parameters and initial and boundary conditions for scenario 1.2(a)

For the base scenario 1.2(a), a retention curve $p_{cap}(S_l)$ according to *van Genuchten* and relative-permeability functions $K_{rel;\varphi}(S_l)$ according to *Mualem/van Genuchten* are assumed. Table 4.3 lists the parameters, the initial conditions, and the boundary conditions for this scenario.

In the subsequent scenarios 1.2(b-c), the capillary pressure is neglected ($p_{cap}(S_l) \equiv 0\ MPa$) and the relative-permeability functions are simplified ($K_{rel;\varphi} = S_\varphi$, with the residual liquid saturation set to $S_{lr} = 0$). Under these constraints, semi-analytical solutions can be derived for steady-state pore pressures, saturations, strains, and displacements, and for derived quantities like flow rates, *Rutenberg (in prep.)*, *Lux et al. (2021)*.

Neglection of capillary forces implicates that saturation at an outflow boundary will develop independently from its given value at that boundary. More details on this topic are given in the aforementioned references. However, the simulators of some of the BenVaSim project partners were unable to work without an invertible capillary-pressure function for technical reasons, cf. section 4.2.3.1, and when replacing the predefined zero function by a linear one, saturation at the outflow boundary of model 1.2 coincides with the boundary value again. For a better comparison to other simulators, scenario 1.2(c) was therefore designed with an outflow boundary condition for saturation corresponding to the value computed with the semi-analytical solution, $S_{l;rhs} = 1/7 \approx 0.143$.

Table 4.4 shows the modified parameters for scenarios 1.2(b-c) as compared to scenario 1.2(a).

Modified material parameters		
residual liquid saturation	S_{lr}	0
van Genuchten parameter	m	– (1)
van Genuchten pre-factor	p_{cap0}	– ($0\,MPa$)
pore connectivity parameters	$\tilde{\varepsilon}; \tilde{\gamma}$	– (-1)

Modified boundary condition for 1.2(c)		
rhs liquid saturation	$S_{l;rhs}$	$1/7$

Table 4.4: Updated parameters and boundary condition for scenarios 1.2(b-c) in relation to the base scenario

4.2.2 Individual approaches for simulation

4.2.2.1 BGR (simulator OGS)

For the numerical simulation in OGS-5, the `MULTI_PHASE_FLOW` implementation and the `DEFORMATION` implementation were used to simulate hydraulic and mechanical processes. The coupling was realised in a staggered scheme. In contrast to scenario 1.1, the domain in this simulation is not fully saturated initially. As a consequence, the entire system reacts less stiffly. That is to say that for example a change in the medium compaction is compensated by the highly compressible gas and by a change of the pore-water saturation. Therefore the change of pore-water pressure in response to medium compaction or extension is not as stark as is would be in a fully saturated model (vice versa, a change in pore-water pressure does not induce a strong medium deformation in comparison to a fully saturated model). Therefore, the staggered HM-coupling scheme does not lead to numerical challenges like it did in model 1.1. The finite element type used was quadrangular *Taylor/Hood* elements (quadratic order elements for the mechanic process and linear order elements for the hydraulic processes). The resulting non-linear equation was numerically solved by bcgs, the stabilised version of biconjugate gradient method implemented in PETSc (https://www.mcs.anl.gov/petsc/). Hypre (https://hypre.readthedocs.io) gmres was used as a preconditioner and the calculation was performed parallelly on two CPUs, where the parallelisation is based on MPI. The staggered solving scheme was set to perform a minimum of 3 iterations between the hydraulic and mechanic process solvers. Maximum iterations were set to 5. The total calculation time for this coupled problem ended up to be about 6,000 seconds.

In the scenarios 1.2(b) and 1.2(c), the model definition requires that $p_{cap} = 0\ MPa$ constantly. In OGS-5, p_{cap}, p_g, and u (1-dimensional) represent the main variables for the balance equations. Consequently, p_{cap} cannot be set to zero. To overcome this challenge, a linear relation between water saturation and capillary pressure was chosen (see Fig. 4.16).

Scenario 1.2(a) also represents a suitable model to test the new monolithic TH^{2}M implementation in OGS-6, *Pitz et al. (2021)*. A comparison of results obtained using OGS-5 and OGS-6 shows good agreement between the results (see Fig. 4.17).

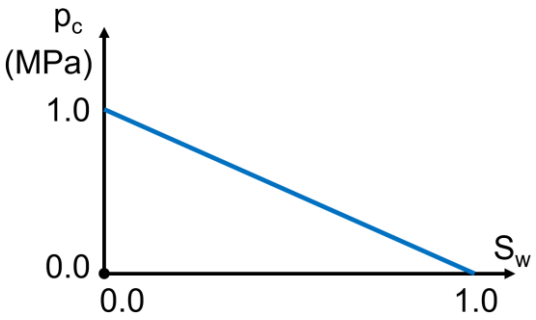


Figure 4.16: Linear relation between capillary pressure and water saturation. Since p_{cap} is a main variable in OGS-5, it cannot be constant zero.

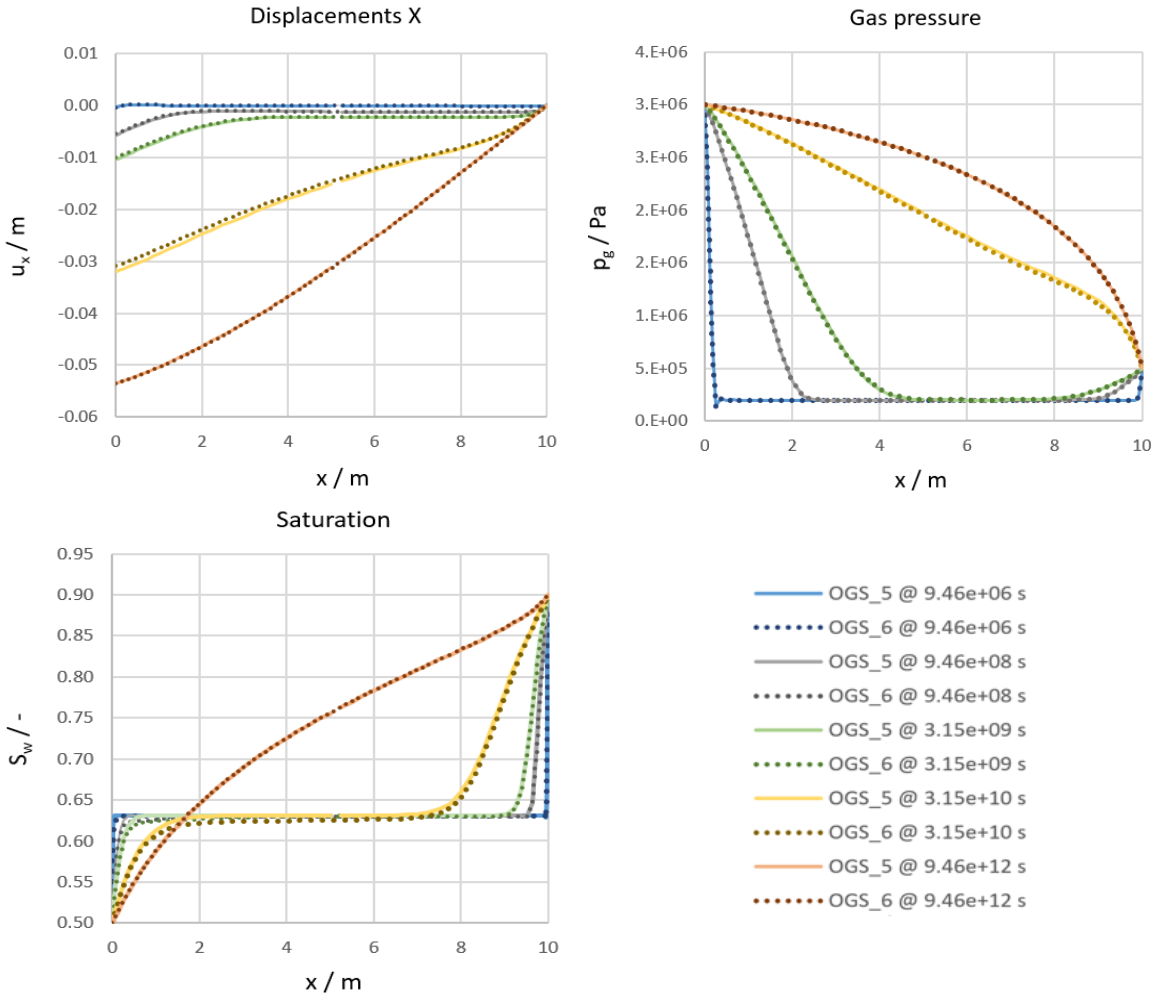


Figure 4.17: Comparison of results obtained using OGS-5 (staggered H^2M) and OGS-6 (monolithic TH^2M) shows good agreement of the solutions. Minor differences may be explained by the different coupling schemes.

4.2.2.2 ENSI (simulator TOUGH-FLAC)

ENSI used the two-phase-flow module `EOS3` from TOUGH3 with no further modification. Instabilities arise at the right-hand-side boundary if the mobilities are averaged between elements. However, when assuming upstream weighting for both the mobilities and the permeabilities, the instabilities are resolved and the simulation is stable, reaching the steady state after 400,000 years.

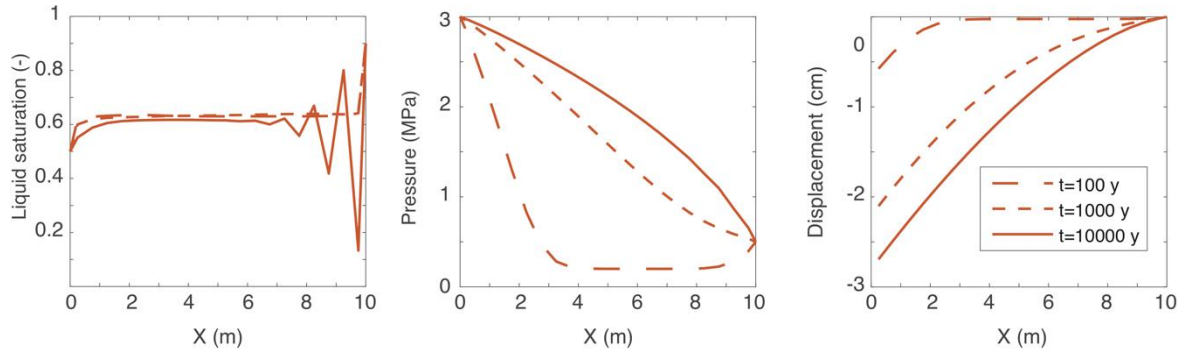


Figure 4.18: Results for TOUGH3-FLAC3D when setting average mobility. The issue is solved by using upstream weighting for both mobilities and permeabilities.

4.2.2.3 ENSI (simulator OGS)

The cases are unsaturated and therefore a staggered simulation of the hydromechanical processes was used in OGS-5. The mesh used for this case was a 2D mesh with 80 cells along x-axis and 2 cells along z-axis. A simulation time of 1.9 million years was split into 834 time steps. The parameters, initial and boundary conditions were assigned as described in the task. The applied stress conditions and the evolving pore-water pressure over time causes a change in effective stress that causes an extension of the model. Due to the unsaturated conditions, the compression of air damps the impact on model geometry. For the scenarios 1.2(b) and 1.2(c), the task definition couldn't be exactly fulfilled because a capillary pressure of zero cannot be defined permanently. To achieve comparable results, a linear relationship for the retention curve was used, starting from $p_{cap} = 1\ MPa$ at unsaturated conditions and decreasing to $p_{cap} = 0\ MPa$ for saturated conditions.

4.2.2.4 GRS BS (simulator CODE_BRIGHT)

In this modelling stage, the neglection of capillary pressure leads to issues in the application of CODE_BRIGHT since the basis of CODE_BRIGHT is a poromechanical approach for the coupled analysis in geological media requiring phase interactions. Thus, GRS BS simulated only the models considering the phase interactions by using the capillary relationship, *Czaikowski & Friedenberg (2020)*.

During the simulation work, a discrepancy in the use of the gas phase relative permeability law was observed since the *Mualem/van Genuchten* approach was not available in CODE_-BRIGHT. Instead, a default law was specified which is defined by one minus the liquid phase relative permeability. This issue could be solved rapidly due to the implementation of the *Mualem/van Genuchten* gas phase relative permeability. More information about the modelling procedure can be found in *Czaikowski & Friedenberg (2020)*.

4.2.2.5 TUC (FTK simulator)

The usual EOS module for two-phase flow with water and air, `EOS3`, has been applied by TUC for the simulation parts conducted with TOUGH2.

4.2.3 Results and conclusions

4.2.3.1 Simulation results obtained under regular conditions

Model 1.2 differs from the preceding model in that it features two-phase-flow processes in association with partial saturation and saturation-dependent phase interactions, i.e., capillary pressures $p_{cap}(S_l)$ and relative fluid permeabilities $K_{rel;\varphi}(S_l)$. Experiences with the fineness of discretisations made with model 1.1 have been incorporated into the present model, and it can be gathered from Figs. 4.19 and 4.20 that some discretisations have been kept for model 1.2 while reworkings have been done with others concerning overall and/or boundary refinement to overcome deficits and to prevent inaccuracies for the more complex H^2M processes. Nevertheless, the issues encountered by the project partners are more numerous than with the simpler preceding model. Besides a few numerical oscillations, the phase interactions and especially the neglection of capillary pressure in some of the scenarios have formed major obstacles to coherent simulation results of the partners. While some simulators work with pore-gas

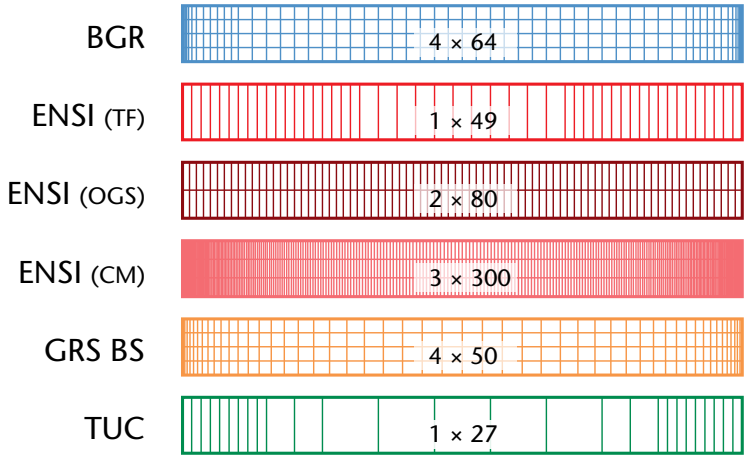


Figure 4.19: Discretisations applied by the partners for simulations of scenario 1.2(a)

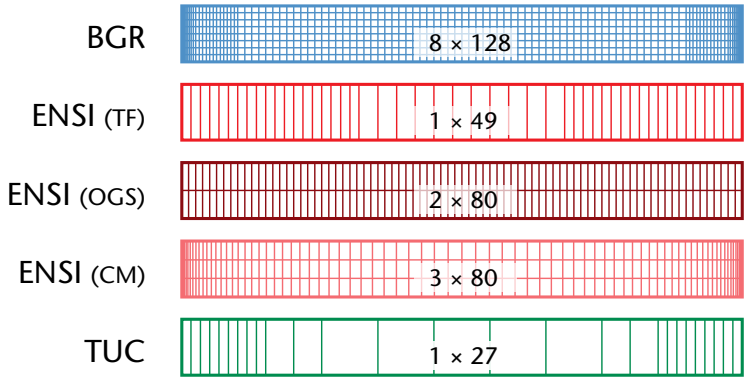


Figure 4.20: Discretisations applied by the partners for simulations of scenario 1.2(b-c)

pressure and liquid saturation as the two independent primary state variables, others use capillary pressure as one of the independent state variables instead. This choice is reasonable for a realistic modelling of subsurface systems and tests since the effect of capillary forces at the fluid interface is usually relevant for determining flux magnitudes and directions and should not be neglected. However, the theoretical approach of setting $p_{cap}(S_l) \equiv 0\ MPa$ in order to be able to obtain semi-analytical solutions for an additional verification hence cannot be put into practice with this kind of simulators. Another issue has been different process speeds arising from preset relative-permeability functions named equally (after *Mualem/van Genuchten*) but defined in different manners, and also from an erroneous simulation setting. In some cases, wrong formulas for an H→M coupling relation or a mechanical quantity lead to significant discrepancies in displacements. Further erroneous inputs concerning initial values and effective saturations have been given, and some outputs have been misdefined or jumbled. For the scenarios with neglected capillary pressure, a semi-analytical solution could be derived, but only

for the steady state. Thus, the partners had only limited possibilities to verify their simulation results, which particularly concerns the issue of different process speeds when they lead to the same steady state. With the exception of capillary-pressure neglection and—due to limited processing time of a partner not particularly funded for this project—also of the erroneous simulator setting, the initial mismatches have been overcome in the course of the project.

Scenario 1.2(a)

Figures 4.21 and 4.22 show the temporal evolution of the pore-gas pressure versus linear and logarithmic time axes, respectively, finally obtained for scenario 1.2(a). Though sometimes accompanied by minor numerical oscillations, the partners' results are consistent with each other, with the exception of the pink curves subject to the aforementioned erroneous simulator setting. This finding is confirmed by the profiles for pore-gas pressure, liquid saturation, pore-water pressure, and displacement following in Figs. 4.23, 4.24, 4.25, and 4.26, respectively, showing that also the mechanical processes are working equivalently in the different simulations.

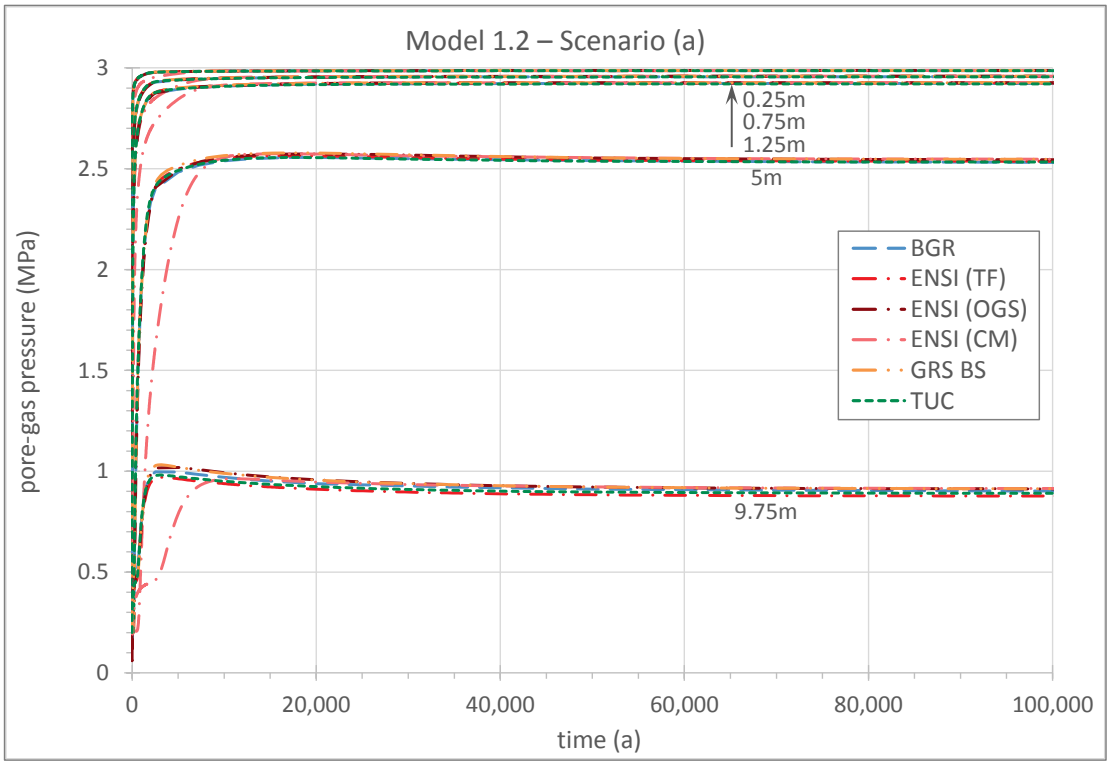


Figure 4.21: Simulation results for scenario 1.2(a) – evolution of pore-gas pressure (linear time axis)

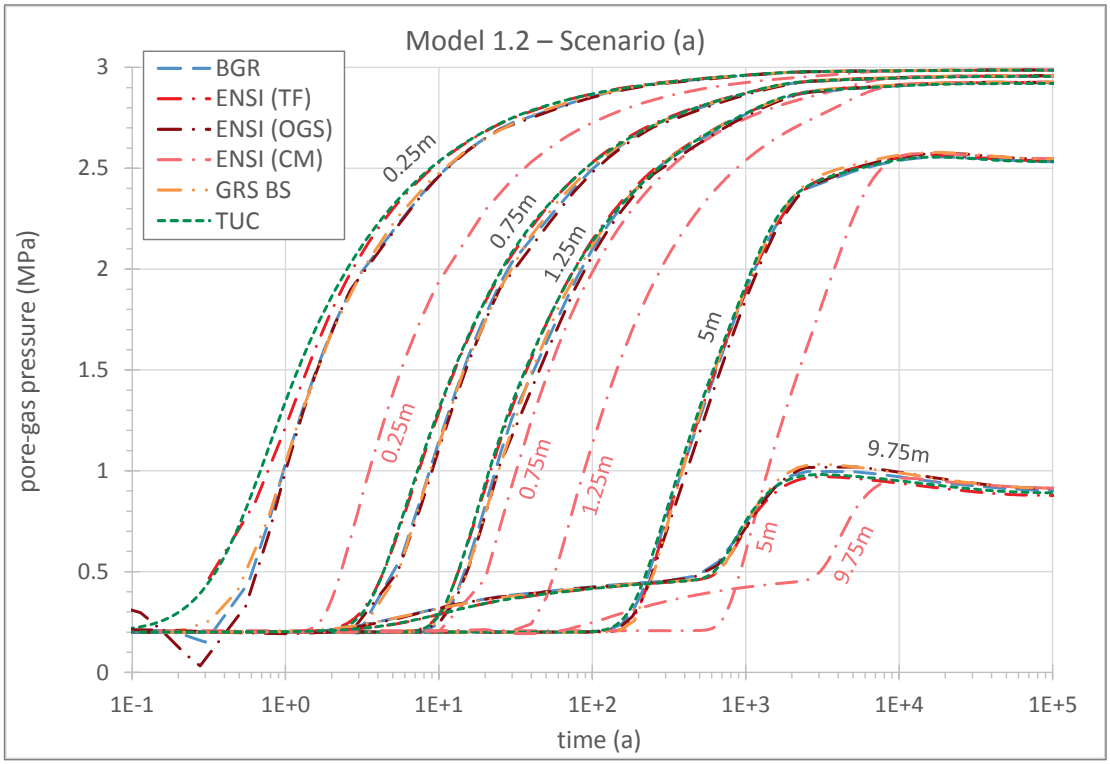


Figure 4.22: Simulation results for scenario 1.2(a) – evolution of pore-gas pressure (logarithmic time axis)

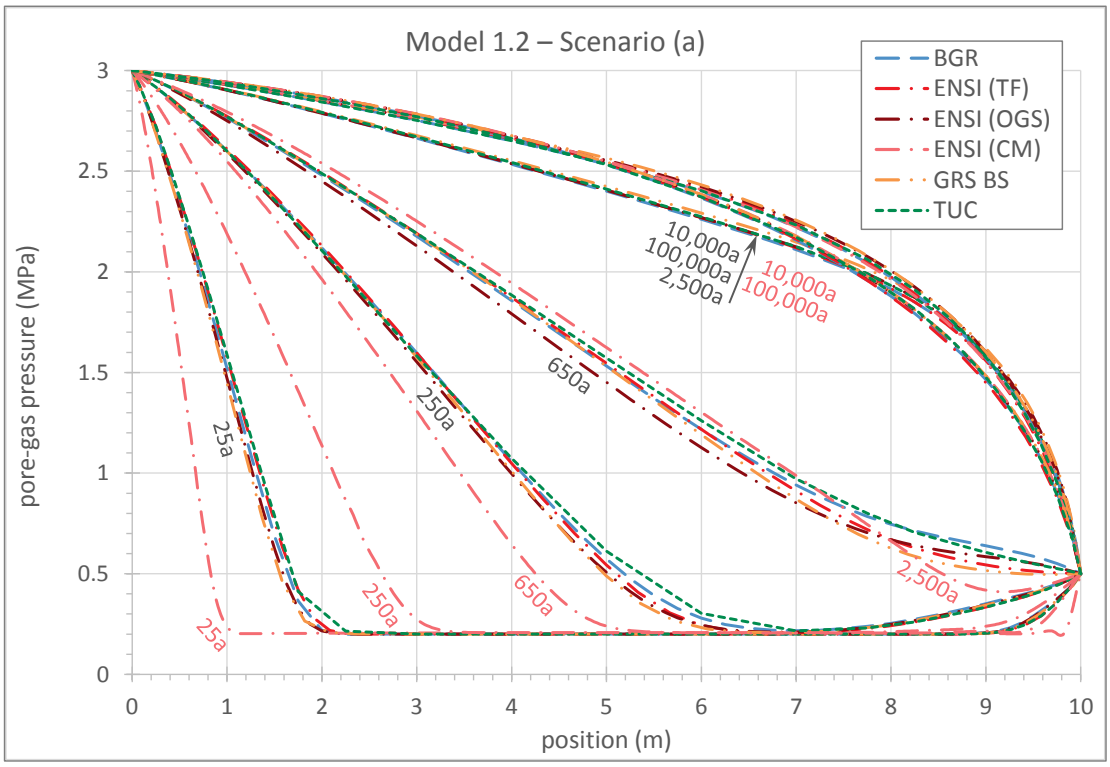


Figure 4.23: Simulation results for scenario 1.2(a) – spatial distribution of pore-gas pressure

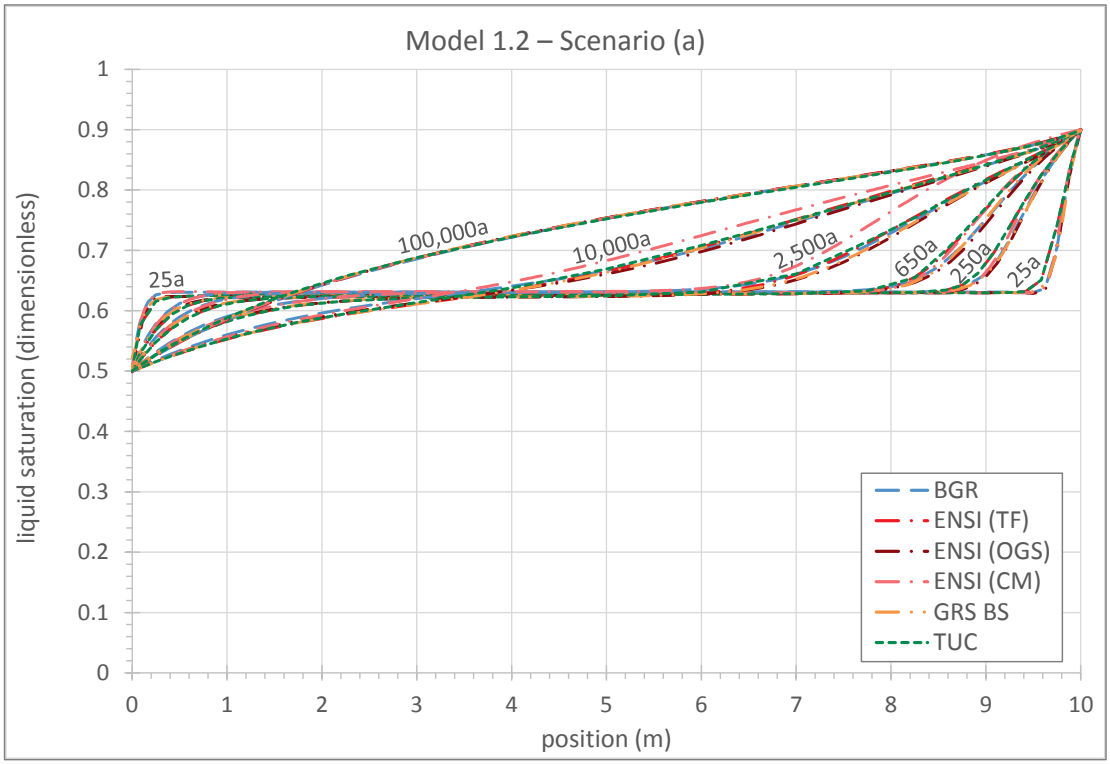


Figure 4.24: Simulation results for scenario 1.2(a) – spatial distribution of liquid saturation

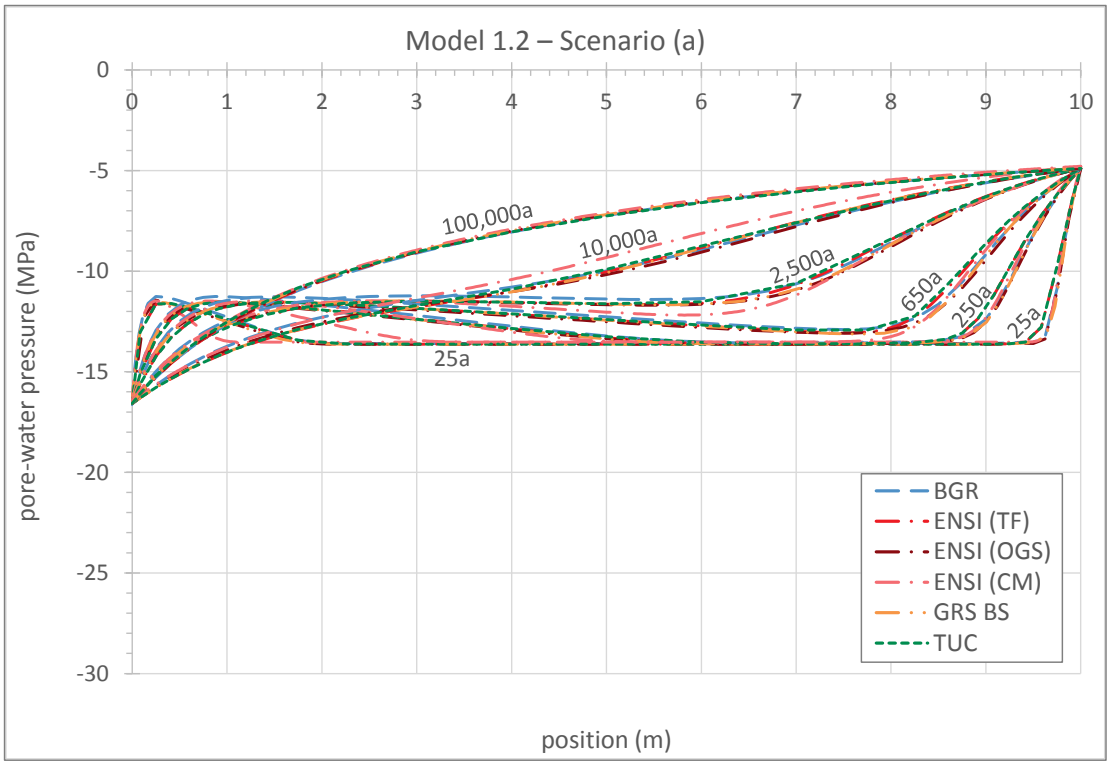


Figure 4.25: Simulation results for scenario 1.2(a) – spatial distribution of pore-water pressure

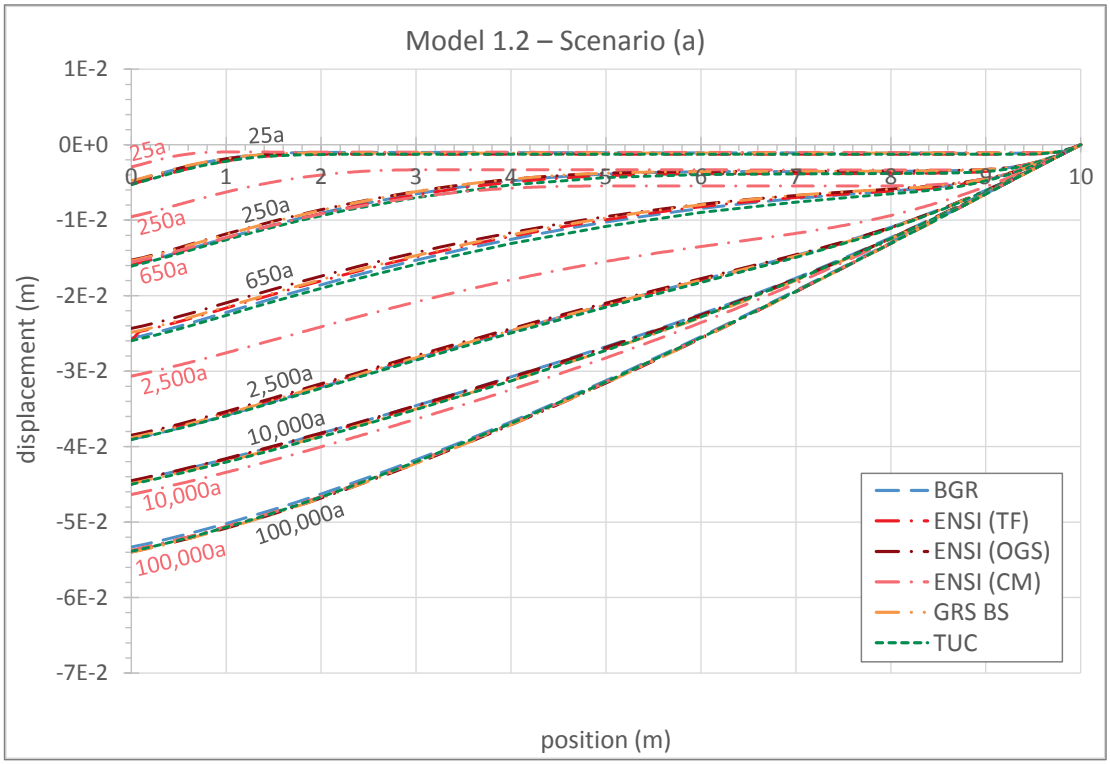


Figure 4.26: Simulation results for scenario 1.2(a) – spatial distribution of displacement

Slightly increased variances in pore pressures and displacement can be observed in the profiles for $t \approx 650\ a$, but they decrease again as soon as the gas phase has reached its preliminary pressure equilibrium. A notably good agreement of results is obtained on the other hand at steady state, especially for saturation and pore-water pressure.

Profiles from first simulations submitted by the partners for pore-gas pressure and displacement can be gathered from Fig. 4.27, without distinction according to the partners. Comparisons with

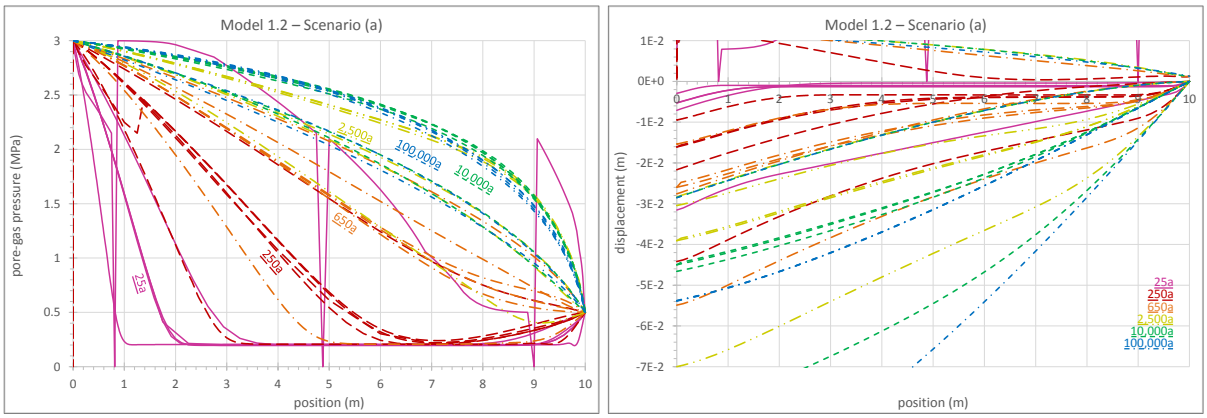


Figure 4.27: First simulation results for scenario 1.2(a) –
spatial distribution of pore-gas pressure (left) and displacement (right)

Figs. 4.23 and 4.26, respectively, convey an impression of the spreading widths in the initial results, revealing many of the issues stated in the first paragraph of this subsection. In particular, displacements differed by a factor of approximately 2 in both directions and in one case also by the sign.

Scenario 1.2(b)

Scenario 1.2(b) differs from its predecessor in that capillary forces are neglected completely. As a consequence, the phase pressures coincide ($p_l \equiv p_g$), and both phases flow into the same direction. The corresponding H^2M-coupled processes have been properly simulated by only three partners—one of which being subject to the simulator setting issue again—since for the simulations with OpenGeoSys and CODE_BRIGHT, setting $p_{cap}(S_l) \equiv 0\ MPa$ was not permitted for the reasons stated above. Nevertheless, the partners applying the simulator OpenGeoSys decided to perform simulations with a linear retention curve $p_{cap}(S_l) = S_g\ MPa$ instead in order to see in how far the results of the original task without capillary forces could be approximated by their simulator. A corresponding comparison will be shown in the following subsection 4.2.3.2.

Figures 4.28 and 4.29 show the profiles for pore-gas pressure and liquid saturation for various points of time from the simulations. In addition, semi-analytical solutions for the steady-state pressure and saturation distributions are displayed as wide, light grey curves put into the background of the figures. It can be gathered from these two figures that the simulation results are in good accordance to each other (except for the pink curves) and also to the steady-state solution (this time also including the pink curve).

One major difference to the results obtained for scenario 1.2(a) (cf. Figs. 4.23 and 4.24) can be found in the saturations not heading to their right-hand-side boundary value of $S_{l;rhs} = 90\ \%$ during the balancing process; instead, saturation at the right model boundary ends up at $S_l(X = 10m) = 1/7 \approx 14.3\ \%$. This is due to the mathematical fact that the semi-analytical solution contains only three degrees of freedom, one fewer than boundary conditions applied, together with the physical explanation that knowledge of pressures is required to determine the predominant and final flow direction and knowledge of saturation at the inflow boundary is required to quantify the volume of each fluid in the pore space. On the other hand, the composition of pore content at the outflow boundary is irrelevant since whatever fluids can be found

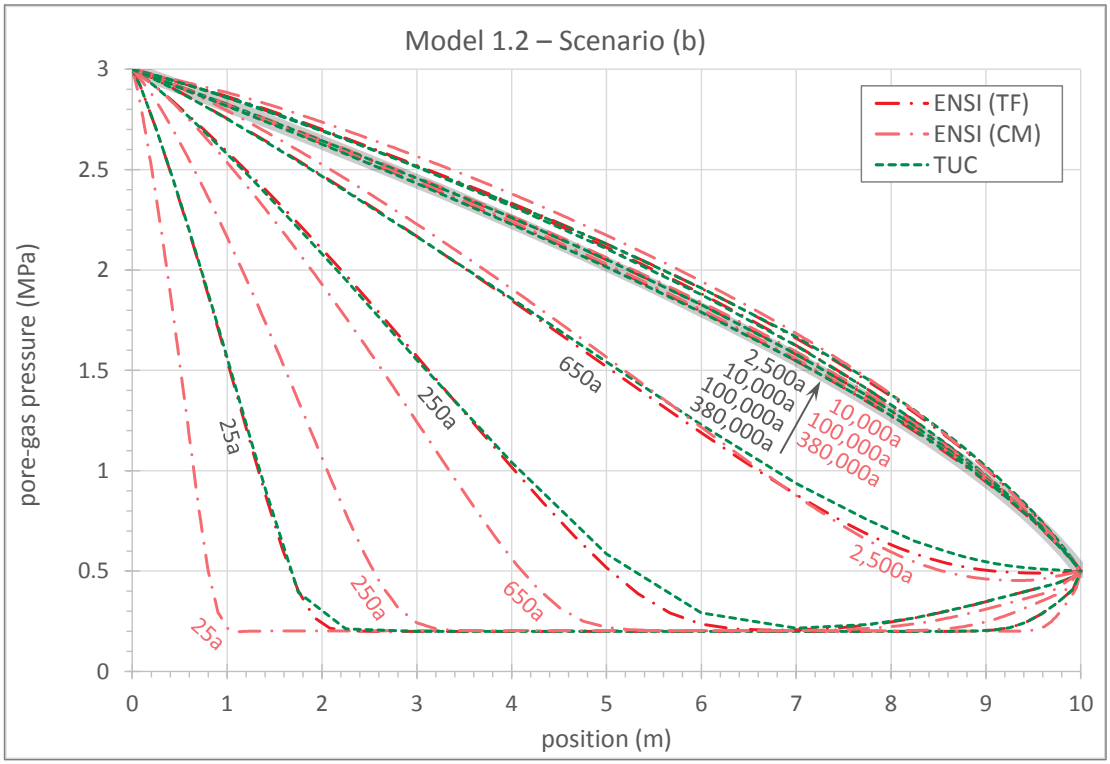


Figure 4.28: Simulation results for scenario 1.2(b) & steady-state semi-analytical solution – spatial distribution of pore-gas pressure

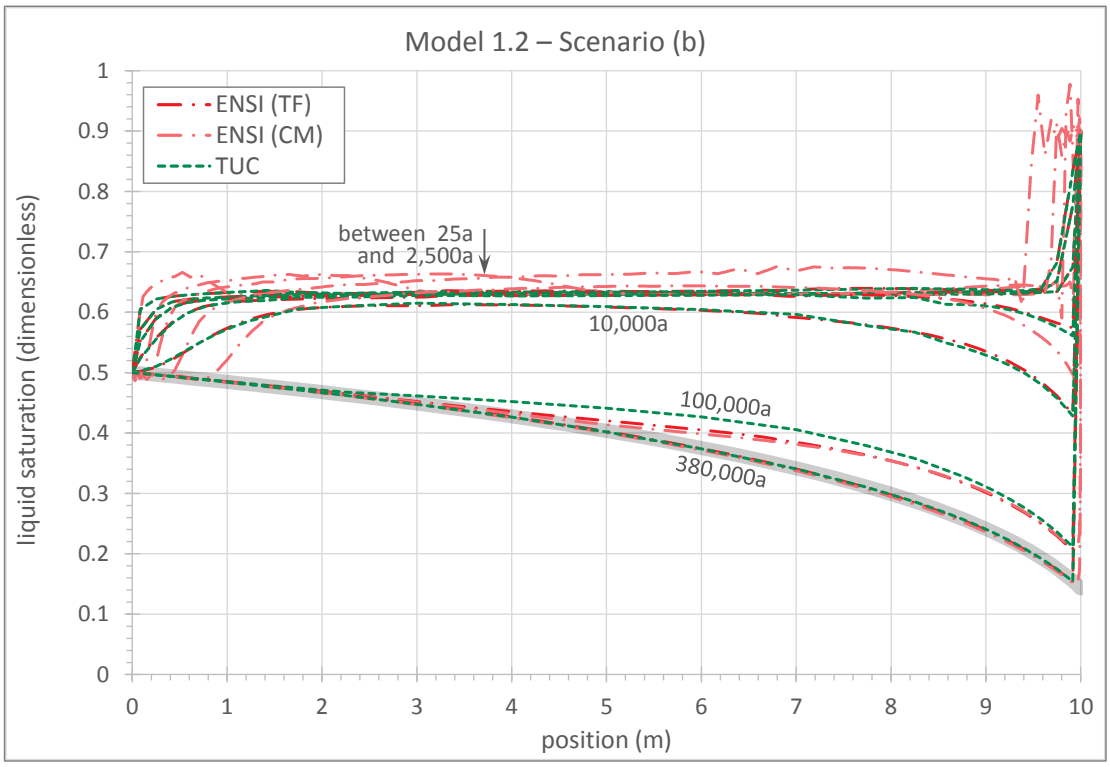


Figure 4.29: Simulation results for scenario 1.2(b) & steady-state semi-analytical solution – spatial distribution of liquid saturation

there will be pushed further away from the model by the leaking fluids, and because of $p_{cap}(S_l) \equiv 0\ MPa$, their volumetric ratio does not have any other impact on the processes inside the model. At the beginning of simulation, both boundaries are inflow boundaries, so all four boundary conditions have to be defined, but in the course of the process, the flow direction globally changes to "from left to right", and from then on, saturation at the right-hand boundary will leave its boundary value and will be controlled by the other three boundary conditions instead. As a further consequence of the reduced liquid saturation and accordingly the elevated gas saturation on the right of the model, the relative permeability of the gas phase rises on that side, and in order for the gas-pressure distribution to end at the same given boundary value as before, the pressure gradient has to decrease on that side to prevent an acceleration of the gas flow. Seen from the opposite perspective, the right-hand-side low gas saturation and gas mobility in scenario 1.2(a) forces the gas to increase significantly in speed in order to be able to escape from the model, meaning its pressure gradient has to increase accordingly which in turn leads to the more pronounced steady-state "hump" in Fig. 4.23.

Scenario 1.2(c)

Scenario 1.2(c) has been introduced to overcome the major difference between the results for the original task of scenario 1.2(b) and those using a linear retention curve by adopting the value from the semi-analytical solution for the right-hand-side saturation as the boundary condition. Since the results do not provide much additional verification for the simulators able to neglect capillary forces, they will be dealt with in the following subsection.

4.2.3.2 Simulation results obtained under modified conditions

The partners working with OpenGeoSys have applied process definitions of their simulators using gas pressure and capillary pressure as the two independent state variables for hydraulic flow. Phase saturations are to be calculated from these pressures via the retention curve $p_{cap}(S_l)$, but for that being possible, S_l has to be uniquely determinable from (p_g and) p_{cap}, which is not the case if $p_{cap}(S_l) \equiv 0\ MPa$. Mathematically speaking, the retention curve $p_{cap}(S_l)$ has to be invertible, i.e., bijective between the real intervals $[0;1] \longrightarrow [0;\infty]\ MPa$. It may be sufficient to use an injective retention curve instead, i.e., one mapping onto a sub-interval of $[0;\infty]\ MPa$ if all values for p_{cap} computed in the simulation lie in that sub-interval.

The constant zero function given for the scenarios 1.2(b-c) therefore had to be replaced for the simulator to work, and it has been decided to use the following linear function:

$$p_{cap}(S_l) = S_g \; MPa \tag{4.1}$$

where

p_{cap} capillary pressure in MPa, $p_{cap} \geq 0 \; MPa$,

S_φ saturation of phase φ, dimensionless.

With this modified setting with independent phase pressures, both fluid phases now obey their own instance of *Darcy*'s law, leading to spatially continuous distributions of $p_\varphi(t; X)$ for both phases and therefore also of $p_{cap}(t; X)$ for each $t > 0 \; a$. Due to the mathematical nature of the capillary-pressure function, it follows that $S_l(t; X)$ also has to be spatially continuous on the whole closed domain interval $[0; 10] \; m$, meaning that a jump at a boundary as observed with the regular scenario 1.2(b) (cf. Fig. 4.29) must not occur.

Scenario 1.2(b)

The following Figs. 4.30 and 4.31 show profiles of pore-gas pressure and liquid saturation, respectively, obtained with the simulator OpenGeoSys under these modified conditions (blue and dark red curves) contrasted to the previously presented results obtained regularly under neglection of capillary forces (bright red and green curves). The pink curves are omitted here for clarity. As long as the fluids penetrate into the model from both sides, the states evolve in very similar ways for all involved simulators. However, as soon as the fluids start to leak out of the right-hand-side boundary (which takes place around $t \approx 650 \; a$ and for the liquid phase considering capillary forces not much later), saturation starts to decrease towards $S_l(X = 10m) \approx 14.3 \; \%$ under neglection of capillary forces as stated in the preceding subsection whereas it remains at its fixed boundary value $S_l(X = 10m) = S_{l;rhs} = 90 \; \%$ during the OpenGeoSys simulations. At steady state, the saturations match each other in a very good manner on the first six metres of the model domain before diverging from each other towards their respective boundary values. The different desaturation behaviour at the boundary has also an impact on the pore-gas pressure again for similar reasons as described in the previous subsection—a reduced gas mobility on the very right of the model forces the gas-pressure gradient to

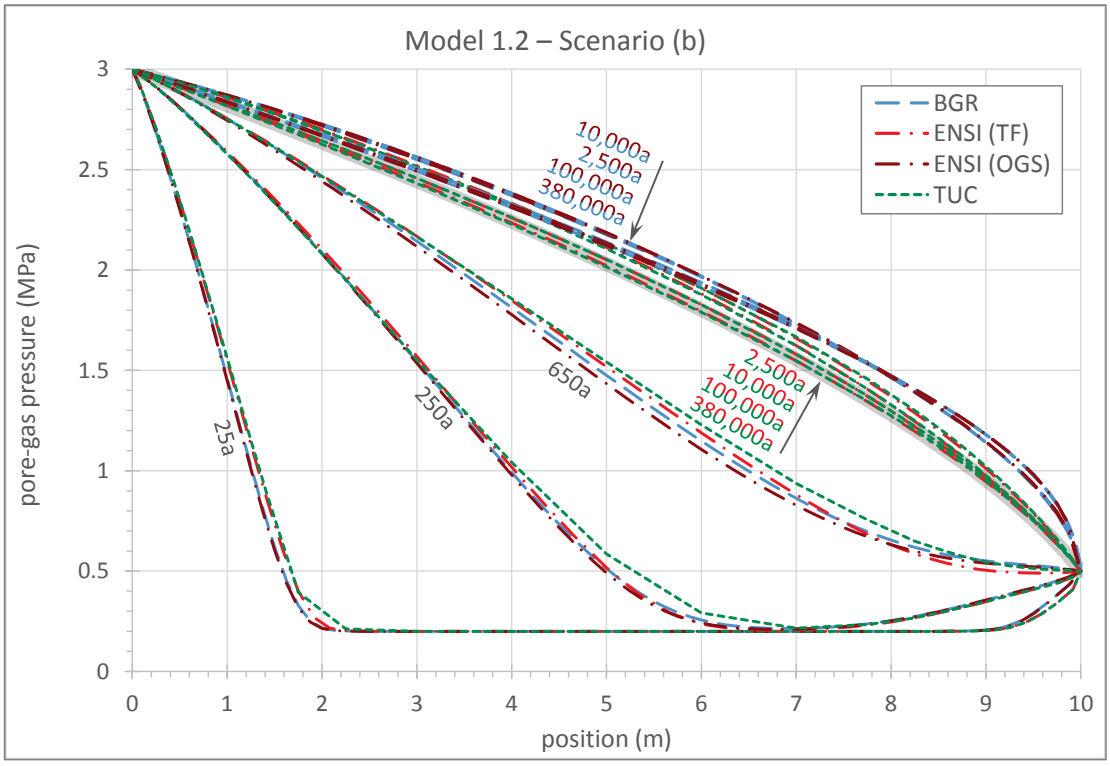


Figure 4.30: Simulation results for scenario 1.2(b) & steady-state semi-analytical solution – spatial distribution of pore-gas pressure

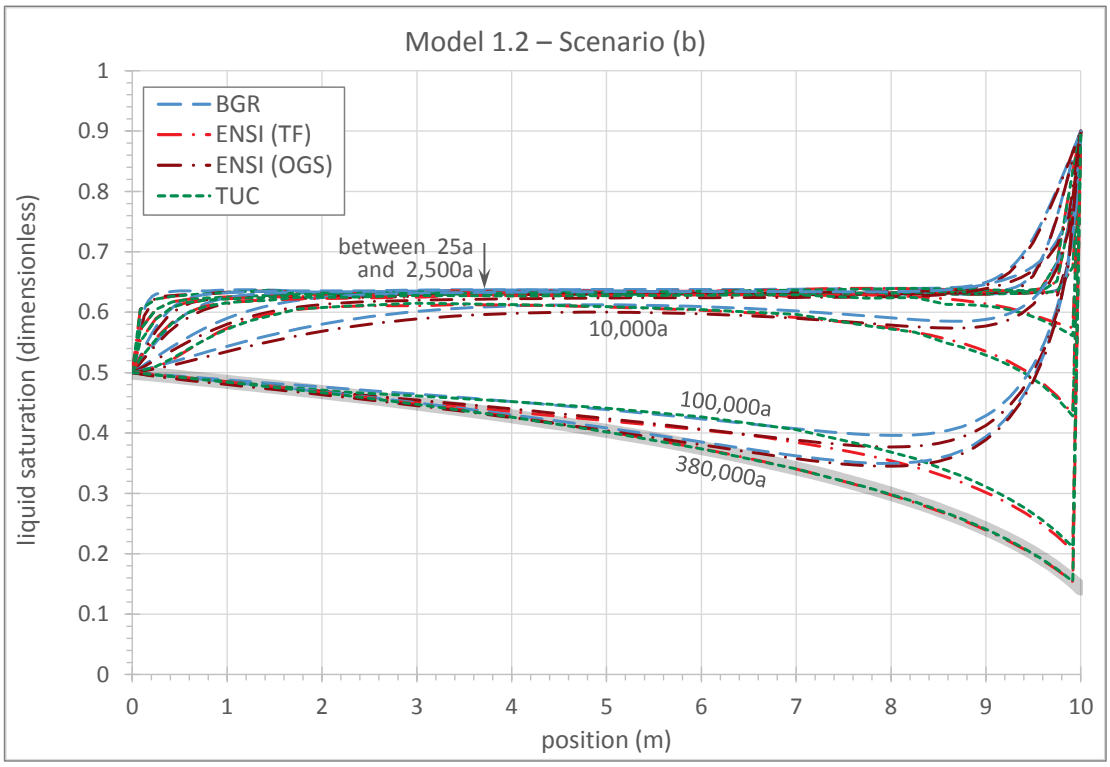


Figure 4.31: Simulation results for scenario 1.2(b) & steady-state semi-analytical solution – spatial distribution of liquid saturation

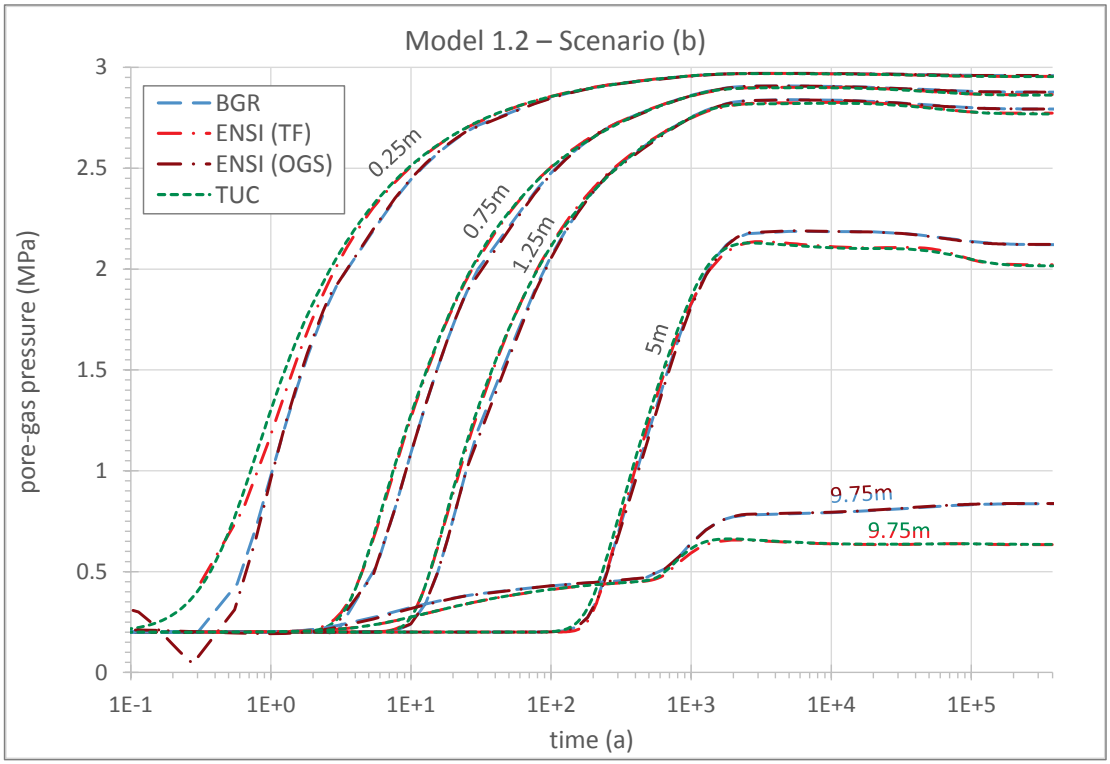


Figure 4.32: Simulation results for scenario 1.2(b) – evolution of pore-gas pressure

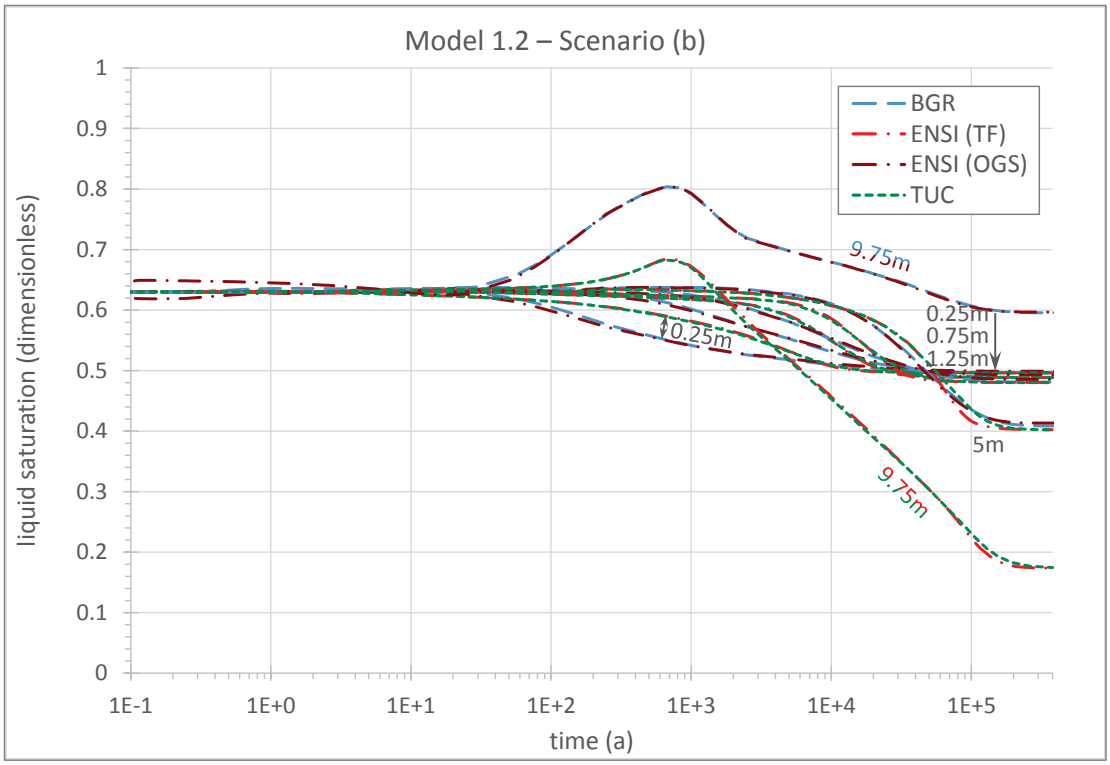


Figure 4.33: Simulation results for scenario 1.2(b) – evolution of liquid saturation

increase significantly in that area, pushing the blue and dark red pressure curves slightly above the others.

In addition to the profiles, Figs. 4.32 and 4.33 depict the corresponding temporal evolutions of the same two quantities. From Figure 4.33, it can be clearly seen that the linear capillary-pressure function already has an influence on liquid flow before its flow direction changes: because of an increased suction inside the model domain compared to the boundary, liquid-flow velocity is slightly higher than when ignoring capillary forces.

In general, aside from few numerical inaccuracies, both OpenGeoSys simulation results agree almost perfectly with one another, which leads to a very clear demonstration of the effect that applying or neglecting capillary forces can have on simulation results.

Scenario 1.2(c)

Upon having compared first results presented for scenario 1.2(b), one further variation was brought into existence in order to improve the conditions for an assessment of simulation results from OpenGeoSys. With scenario 1.2(c), matching the right-hand-side saturation boundary condition to the semi-analytical value was supposed to eliminate the large difference in phase mobilities at least for steady state. Suction is still supposed to have an effect on the flow behaviour, although with $p_{cap} \leq 1\ MPa$, its magnitude is below the changes in pore-gas pressure in most parts of the model, as it was already the case in the preceding scenario 1.2(b).

It can indeed be gathered from Fig. 4.34 that not only the gas pressure's steady state, but also its transient behaviour show only small discrepancies with respect to the application or neglection of capillary forces. The gas flow is only slightly affected by the liquid flow; the latter however still behaves differently during the flow process since the right saturation boundary condition still differs from the initial saturation. An elevated suction at the boundary now moves pore water off the model, leading now to a faster decrease of saturation in its right-hand part according to the development of saturation shown in Fig. 4.35 as opposed to Fig. 4.33. As before, the OpenGeoSys simulation results match each other very well and help visualising a possible impact of capillary forces.

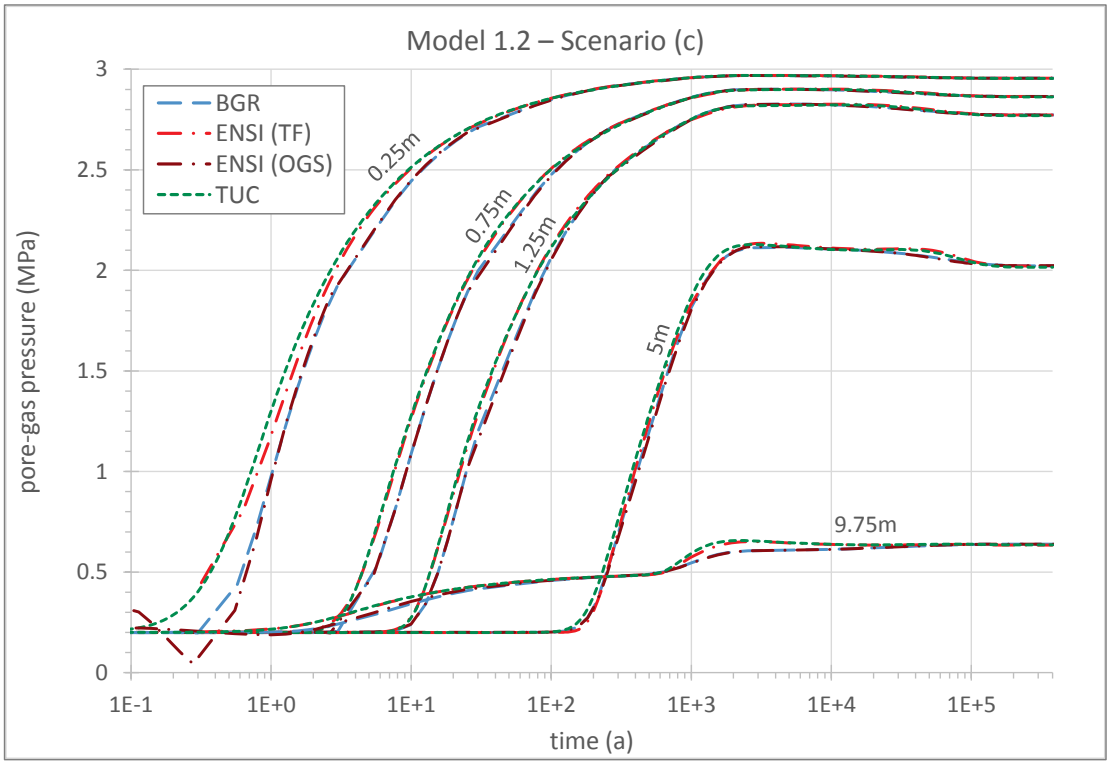


Figure 4.34: Simulation results for scenario 1.2(c) – evolution of pore-gas pressure

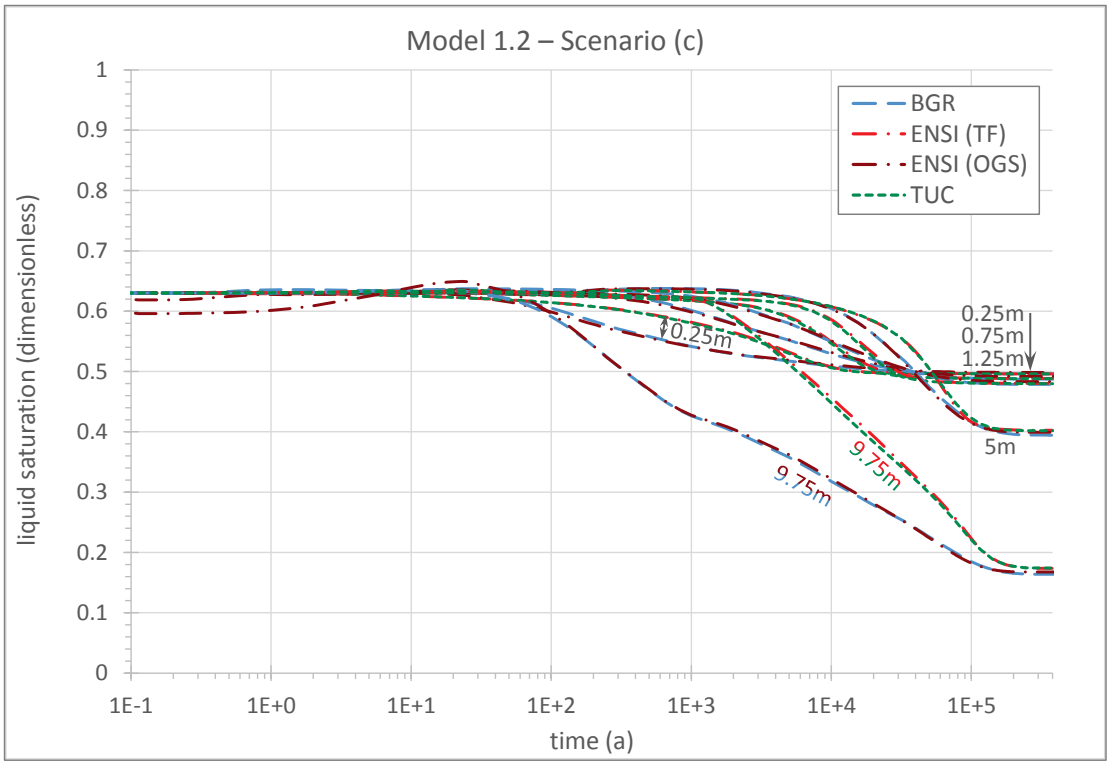


Figure 4.35: Simulation results for scenario 1.2(c) – evolution of liquid saturation

4.3 Model 1.3

4.3.1 Model description

Model 1.3 is an extension of the H^{2}M model 1.2 to a three-sectional model consisting of a drift seal in the centre enclosed by two backfilled drift areas. Both backfilled areas are built from the same material and thus possess the same properties and hydraulic initial states, while those for the drift seal differ from them. The left and right backfilled areas have lengths of 7 and 8 metres, respectively, and the drift seal has a length of 5 metres, for a total of 20 metres. Like with model 1.2, gas generation raises gas pressure and suction on the left model boundary while initial conditions are preserved at its right, mechanically fixed end. Because phase interactions are considered in most scenarios, the fluid-flow directions at steady state are the same as for model 1.2. In contrast to the previous models, new evaluation points were given due to the altered model geometry: they were placed near the boundaries and material transitions at $X = 0.25\ m$, $6.75\ m$, $7.25\ m$, $11.75\ m$, $12.25\ m$, and $19.75\ m$. Figure 4.36 visualises the model's settings. Scenario-dependent points of time were given again for taking state profiles.

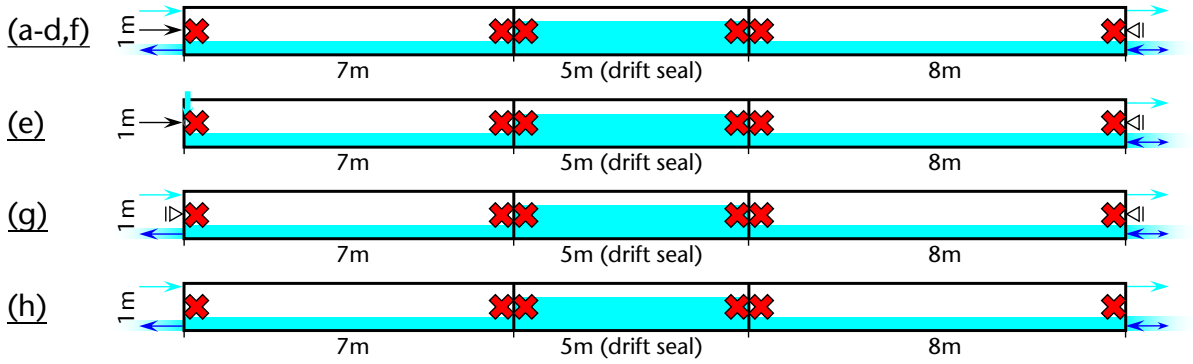


Figure 4.36: Visualisation of model-1.3 geometry and initial and boundary conditions for scenarios 1.3(a-h), and evaluation points

Scenario-dependent points of time for taking profiles are listed in Table 4.5 together with the spatial evaluation points already mentioned. The period of time to be simulated in scenario 1.3(d) amounts to approximately 630 years, meaning that the last two points of time for profiles can be omitted for that scenario.

Evolutions at $X =$	Profiles at $t =$	$\approx$
$0.25\ m$	$2 \cdot 10^6\ s$	$0.063\ a$
$6.75\ m$	$2 \cdot 10^7\ s$	$0.63\ a$
$7.25\ m$	$2 \cdot 10^8\ s$	$6.3\ a$
$11.75\ m$	$2 \cdot 10^9\ s$	$63\ a$
$12.25\ m$	$2 \cdot 10^{10}\ s$	$630\ a$
$19.75\ m$	$5 \cdot 10^{11}\ s$	$16{,}000\ a$
	$2 \cdot 10^{13}\ s$	$630{,}000\ a$
	and for 1.3(e) also at	
	$5 \cdot 10^{13}\ s$	$1{,}600{,}000\ a$

Table 4.5: Positions for temporal evolutions and points of time for profiles concerning model 1.3

For the base scenario 1.3(a), Table 4.6 lists the material parameters for both materials, the fluid parameters for liquid and gas phase, and the initial and boundary conditions. The retention curve and the relative-permeability functions according to *(Mualem/)van Genuchten* are used. The other scenarios of model 1.3 are all based on scenario 1.3(a) with smaller modifications of material parameters, boundary conditions, and the models for phase interactions and effective-stress calculation. Table 4.7 gives an overview of these scenarios and the corresponding alterations in comparison to the base scenario.

4.3.2 Individual approaches for simulation

4.3.2.1 BGR (simulator OGS)

To perform the simulation of model 1.3 in OGS-5, the same numerical implementations and solver settings were used as in model 1.2: The `MULTI_PHASE_FLOW` implementation was used to simulate the hydraulic immiscible two-phase process and `DEFORMATION` was used to simulate the mechanical behaviour. Three material groups were defined and the quadrangular *Taylor/Hood* elements were refined towards the material boundaries as well as towards the left and right domain boundaries.

Material parameters				
		backfill	*drift seal*	
Young's modulus (drained)	E	200	600	MPa
Poisson's ratio	ν	0	0	
porosity	ϕ	0.4	0.3	
intrinsic permeability	K	$2 \cdot 10^{-19}$	10^{-20}	m^2
Biot coefficient	α	1	1	
residual liquid saturation	S_{lr}	0.01	0.05	
residual gas saturation	S_{gr}	0	0	
van Genuchten parameter	m	0.47	0.37	
van Genuchten pre-factor	p_{cap0}	15	35	MPa
pore connectivity parameters	$\tilde{\varepsilon}; \tilde{\gamma}$	0.5	0.5	

Hydraulic initial conditions				
		backfill	*drift seal*	
pore-gas pressure	p_{g0}	0.25	0.2	MPa
liquid saturation	S_{l0}	0.25	0.7	

Fluid parameters		
liquid bulk modulus	$\breve{K}_l$	2,220 MPa
liquid density	ρ_l	1,000 kg/m^3
liquid viscosity	η_l	10^{-9} $MPa \cdot s$
gas viscosity	η_g	$1.8 \cdot 10^{-11}$ $MPa \cdot s$

Mechanical initial condition		
total stress	σ_{tot0}	5 MPa

Boundary conditions		
lhs total stress	$\sigma_{tot;lhs}$	5 MPa
lhs pore pressure	$p_{g;lhs}$	4 MPa
lhs liquid saturation	$S_{l;lhs}$	0.2
rhs displacement	u_{rhs}	0 m
rhs pore pressure	$p_{g;rhs}$	0.25 MPa
rhs liquid saturation	$S_{l;rhs}$	0.25

Table 4.6: Parameters and initial and boundary conditions for scenario 1.3(a)

Scenario 1.3(b)	compressible grains in the drift seal			
	Biot coefficient	α	*drift seal*	0.85
Scenario 1.3(c)	uniform phase-interaction parameters (those of the backfill)			
	residual liquid saturation	S_{lr}	*drift seal*	0.01
	van Genuchten parameter	m	*drift seal*	0.47
	van Genuchten pre-factor	p_{cap0}	*drift seal*	15 MPa
Scenario 1.3(d)	phases are perfectly mobile: $K_{rel;\varphi}(S_l) \equiv 1$			
Scenario 1.3(e)	constant gas source in leftmost zone replaces lhs boundary conditions			
	gas mass generation rate	$\hat{Q}_{m;g;lhs}$		$6 \cdot 10^{-9}$ kg/s
	lhs pore-gas pressure	$p_{g;lhs}$		–
	lhs liquid saturation	$S_{l;lhs}$		–
Scenario 1.3(f)	no suction-induced compaction: $p_{eq} = p_g \implies \sigma_{eff} = \sigma_{tot} - \alpha p_g$			
Scenario 1.3(g)	stress build-up due to mechanically fixed model boundaries			
	lhs total stress	$\sigma_{tot;lhs}$		–
	lhs displacement	u_{lhs}		0 m
Scenario 1.3(h)	neglection of mechanical processes (only H^2 processes)			

Table 4.7: Updated framework conditions, parameters, and boundary conditions for scenarios 1.3(b-h) in relation to the base scenario

4.3.2.2 ENSI (simulator TOUGH-FLAC)

As before, module `EOS3` has been used by ENSI for the TOUGH3 simulation parts.

No issues are reported, and the simulator worked smoothly for all cases.

4.3.2.3 ENSI (simulator OGS)

For these cases, the same approach was used as for model 1.2. Due to the different materials, a refined 2D mesh with 160 cells along x-axis and 8 cells along z-axis was used. The simulation

time of 630,000 years was split into 335 time steps and the one for 1.3(e) of 1,600,000 years into 635 time steps. The parameters, initial and boundary conditions were assigned as described in the task.

4.3.2.4 GRS BS (simulator CODE_BRIGHT)

The modelling approach follows the previous ones. For scenario 1.3(b) dealing with a *Biot* coefficient $\alpha < 1$, GRS BS have not submitted any results since the *Biot* theory is not included in its entireness in CODE_BRIGHT. More information about the modelling procedure can be found in *Czaikowski & Friedenberg (2020)*.

4.3.2.5 TUC (FTK simulator)

As before, module `EOS3` has been used by TUC for the TOUGH2 simulation parts.

4.3.3 Results and conclusions

The geometry of model 1.3 consists of three sections which closely resemble that of the preceding model and can therefore be seen as an extension of model 1.2. Nevertheless, its variations are not limited to neglecting capillary pressure or altering boundary values, but instead, several parameters and physical or technical assumptions have been changed to observe their effects on the multiphysical system. In addition, the quantities to record are augmented with mass flow rates for both hydraulic phases. The three sections represent three different parts of a closed emplacement drift consisting of two different materials, arranged in the order backfill, drift seal, backfill. Discretisations according to Fig. 4.37 have been applied based on those of the preceding models, and they mostly show refinements not only near the model boundaries, but also around the material transitions. Major issues from the preceding model had been solved before or while dealing with model 1.3, but new ones came up. Besides a few numerical oscillations and perturbations, new coupling errors arose, determination of permeability at material transitions came into view, and a coarse temporal discretisation led to a different initial gas flow behaviour. Mass flow rates seem not to be standard output quantities of the participating simulators, as they could be collected only from up to three parties. Furthermore, different treatments of solubility equilibria of phase-extraneous components, in particular of water vapour in the gas

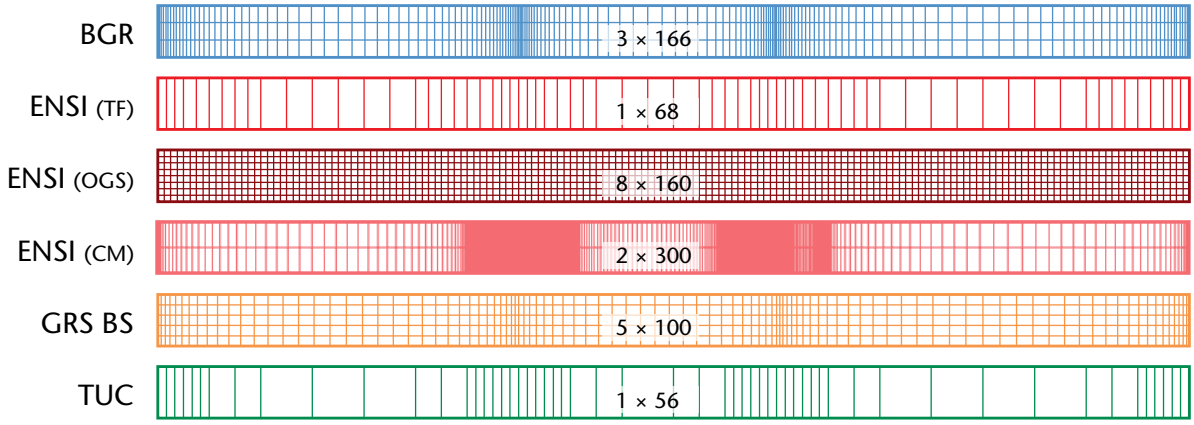


Figure 4.37: Discretisations applied by the partners for simulations of scenario 1.3

phase, led to little differences in steady-state curves. For the scenario 1.3(d) with simplified relative-permeability functions $K_{rel;\varphi}(S_l) \equiv 1$, a semi-analytical solution for the steady state has been derived based on the solution for model 1.2 with appropriate adaptions and extensions. This way, a possibility for verification could also been offered for model 1.3 which proved to be a real benefit—the steady-state simulation results collected from the partners show the best mutual matches from all model-1.3 scenarios. However, although some of the stated issues could be overcome in the course of the project, others remained.

Many curves for any particular quantity show strong similarities across different scenarios. A complete collection of simulation results is provided on the CD-ROM enclosed in this report. Therefore, only some results of every model-1.3 scenario will be presented and discussed in this section.

Scenario 1.3(a)

Scenario 1.3(a) represents the base case and will be discussed first. While the partners' results for pore-gas pressure from the profiles in Fig. 4.38 are in good accordance, first discrepancies can be seen in the saturations in Fig. 4.39 concerning the pink curves. Besides a slightly elevated pressure in the central drift-seal area at steady state, a different short-term behaviour in the left backfill section can be gathered from the figure upon a closer look onto the horizontal lines' levels. Corresponding observations can also be made for the profiles for pore-water pressure shown in Fig. 4.40 where the nature of *van Genuchten*'s retention curve visually enlarges these discrepancies. The elevated steady-state liquid pressure in the drift seal is supposed to originate

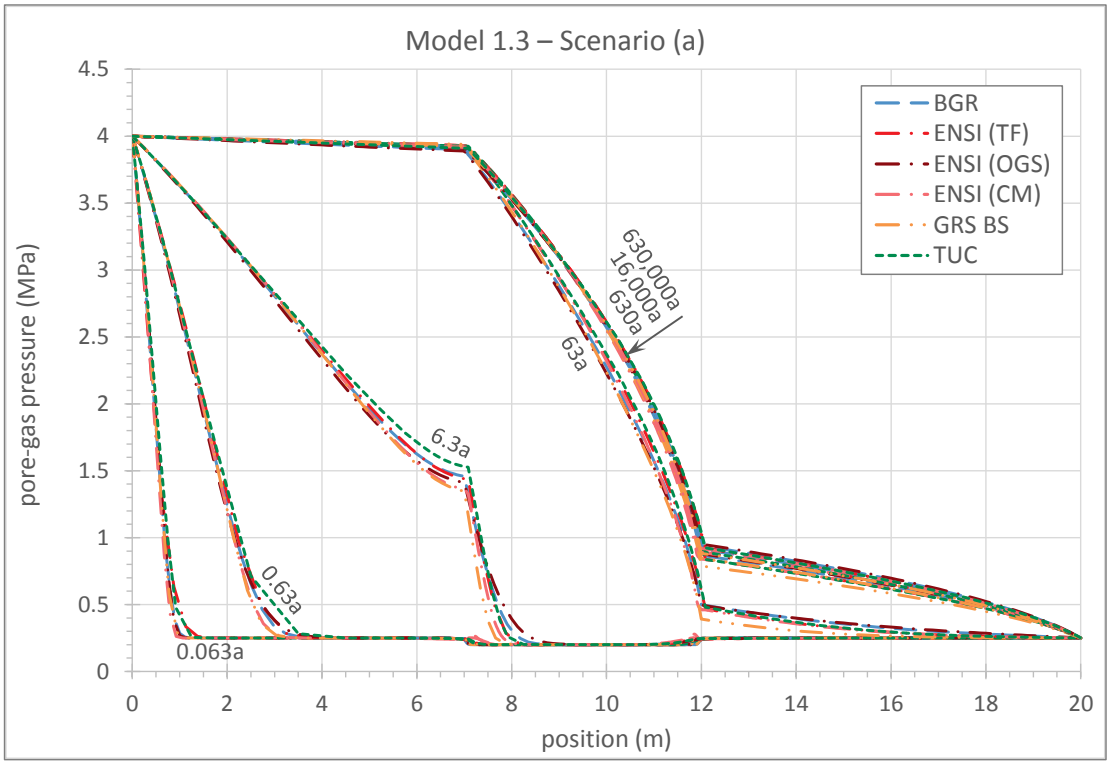


Figure 4.38: Simulation results for scenario 1.3(a) – spatial distribution of pore-gas pressure

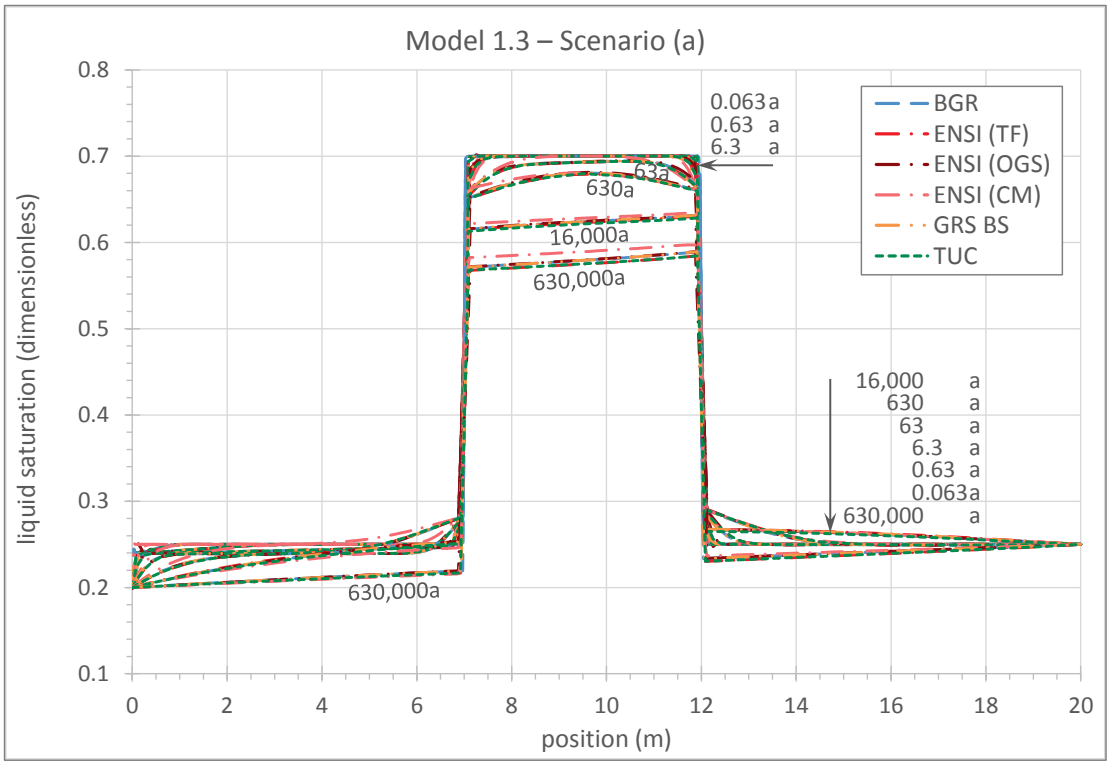


Figure 4.39: Simulation results for scenario 1.3(a) – spatial distribution of liquid saturation

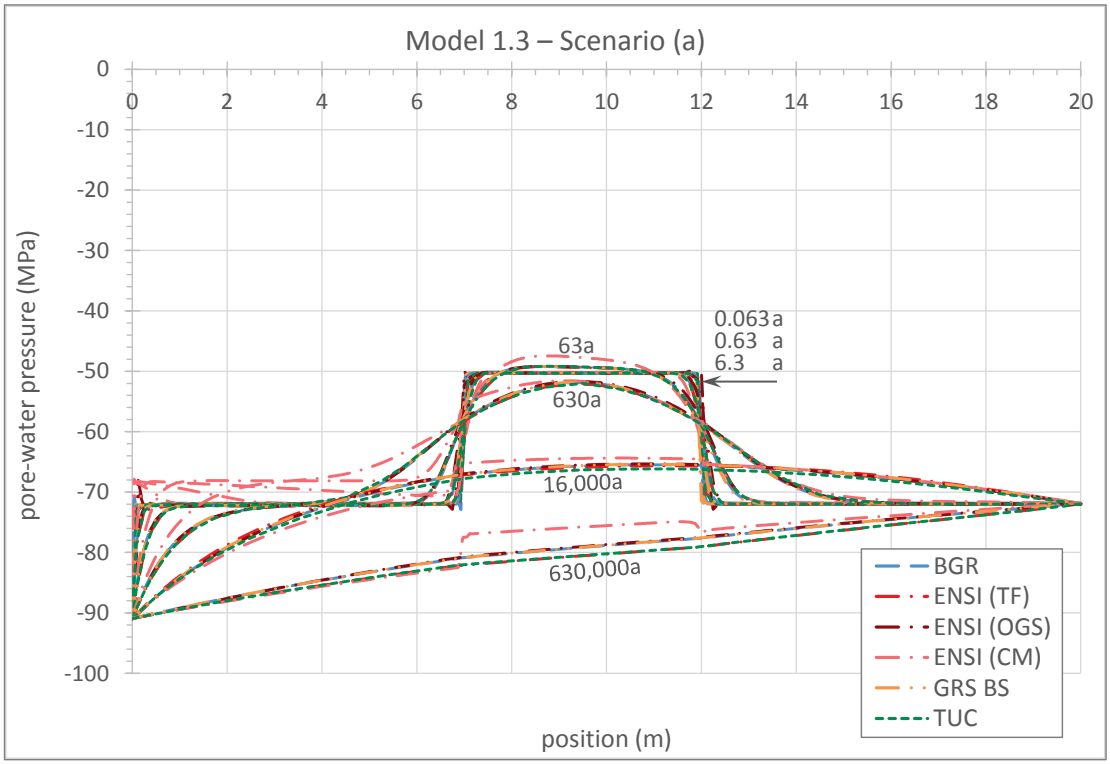


Figure 4.40: Simulation results for scenario 1.3(a) – spatial distribution of pore-water pressure

from an erroneous setting for the HM coupling, as the corresponding results from the uncoupled scenario 1.3(h) lack this elevation (cf. Fig. 4.53), and as the state is physically contradictory because of the liquid pressure's gradient pointing in the wrong direction in the absence of sources in the seal and sinks to its immediate right in the backfill.

The differences in saturations on the left-hand side stem from a missing HM-coupling relation. Gas penetrates into the model raising the gas pressure in the area it has got to so far, while the pore water almost does not move inside that area (the drainage of water to the boundary is significantly slower). If no coupling to mechanical processes were considered, saturation would remain unchanged where water has not drained off yet (cf. Fig. 4.53), and pore-water pressure would increase in the same way as the pore-gas pressure does. When coupled to mechanical processes though, the gas-pressure increase affects the material by expanding it, therefore expanding its pore space, and the different fluid compressibilities allow the gas volume to enlarge while keeping the only very little compressible liquid at its previous volume. The ratio of volumes consequently changes in favour of the gas phase, thus reducing the saturation in Fig. 4.39 and raising the suction. Coincidentally, the effects of gas-pressure increase and desaturation-induced rise in suction cancel almost out, leaving the pore-water pressure in Fig. 4.40 nearly at

its initial level of $p_l \approx -72\,MPa$. While most simulators reproduce this coupling behaviour appropriately, the pink curves seem to lack it and show an uncoupled evolution of liquid pressure and saturation instead.

Another observation that can be made in Fig. 4.40 is a difference in steady-state curves for pore-water pressure. The curves obtained with both simulators involving TOUGH lie a little below those from the other simulators. A hint to the process responsible for the discrepancy was found within scenario 1.3(e), and since that scenario is more suitable to explain this process, an explanation also for the present scenario will be given in the respective paragraph further below.

Figure 4.41 shows the displacement profiles developing with time. The pink steady-state curve lies a little below the others due to the elevated pore-water pressure in the drift seal discussed above. The elevated pressure, which corresponds to a decreased suction, reduces the compaction the seal undergoes. Consequently, the drift seal expands towards the free boundary at the left and pushes the left backfill section in front of it.

Finally, profiles for the mass flow rate for gas and liquid phases can be seen in Figs. 4.42 and 4.43, respectively. Concerning gas flow, discrepancies among the partners' results are larger than before, and especially the flow rate at steady state shows deviations of about 15 %. The partners' results for the liquid-flow rate on the other hand are in very good agreement.

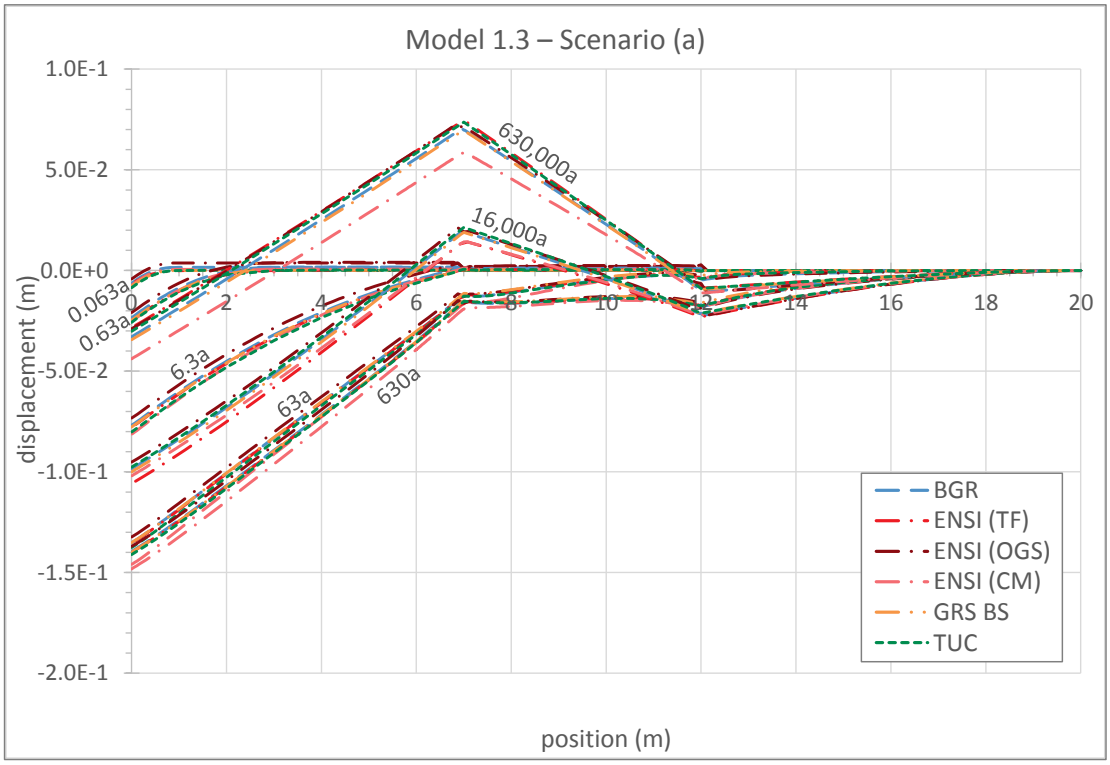


Figure 4.41: Simulation results for scenario 1.3(a) – spatial distribution of displacement

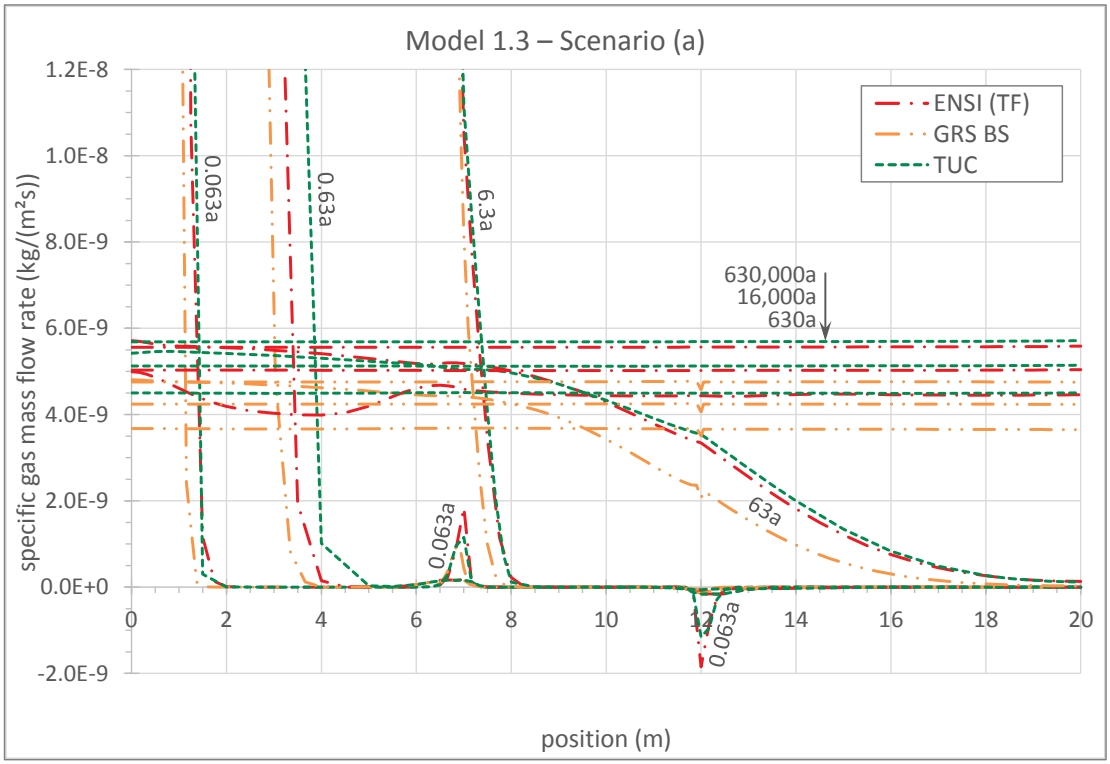


Figure 4.42: Simulation results for scenario 1.3(a) – spatial distribution of gas-flow rate

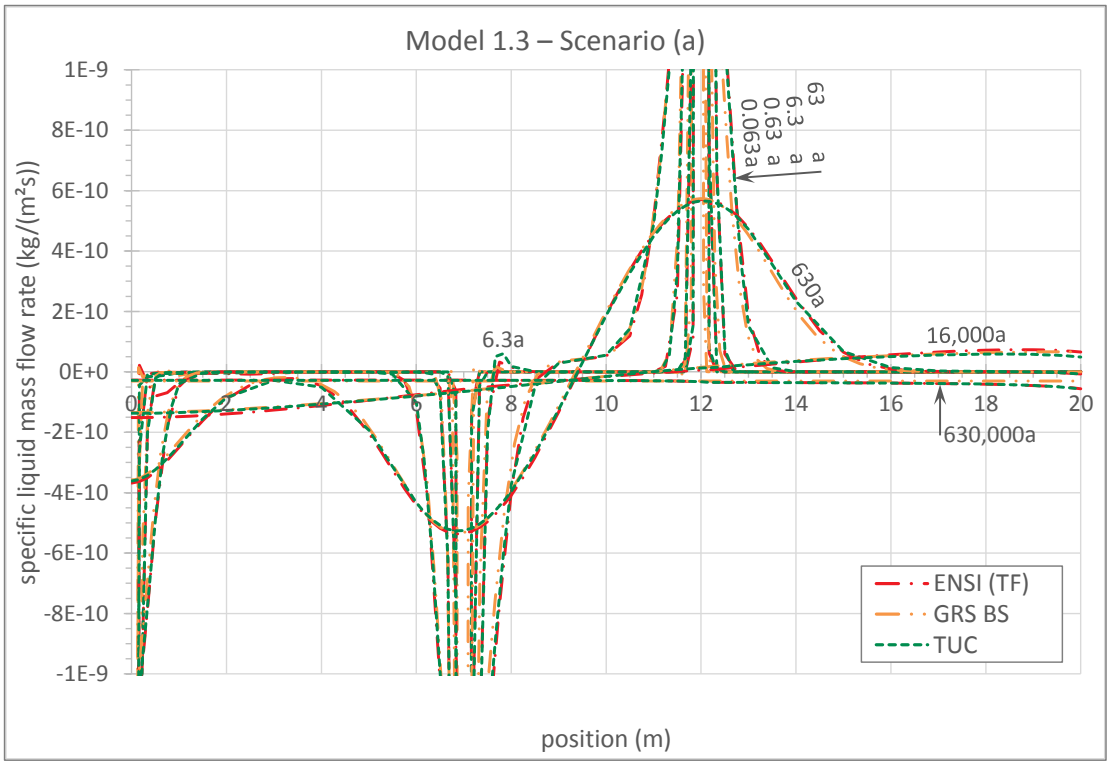


Figure 4.43: Simulation results for scenario 1.3(a) – spatial distribution of liquid-flow rate

Scenarios 1.3(b-d)

Scenario 1.3(b) deals with an increased compressibility of the solid grains in the drift-seal section. The simulation results turned out to be very similar to those of scenario 1.3(a), with slight changes only in displacements which can be seen comparing Fig. 4.44 to Fig. 4.41. As seen with scenario 1.1(b), an increased grain compressibility attenuates the effect of pore-pressure changes on mechanical deformations. The desaturation process in the drift seal that led to a higher suction and to a compaction of the material in scenario 1.3(a) has a reduced effect in scenario 1.3(b) leading to a slightly larger expansion of the seal and thus of the overall model.

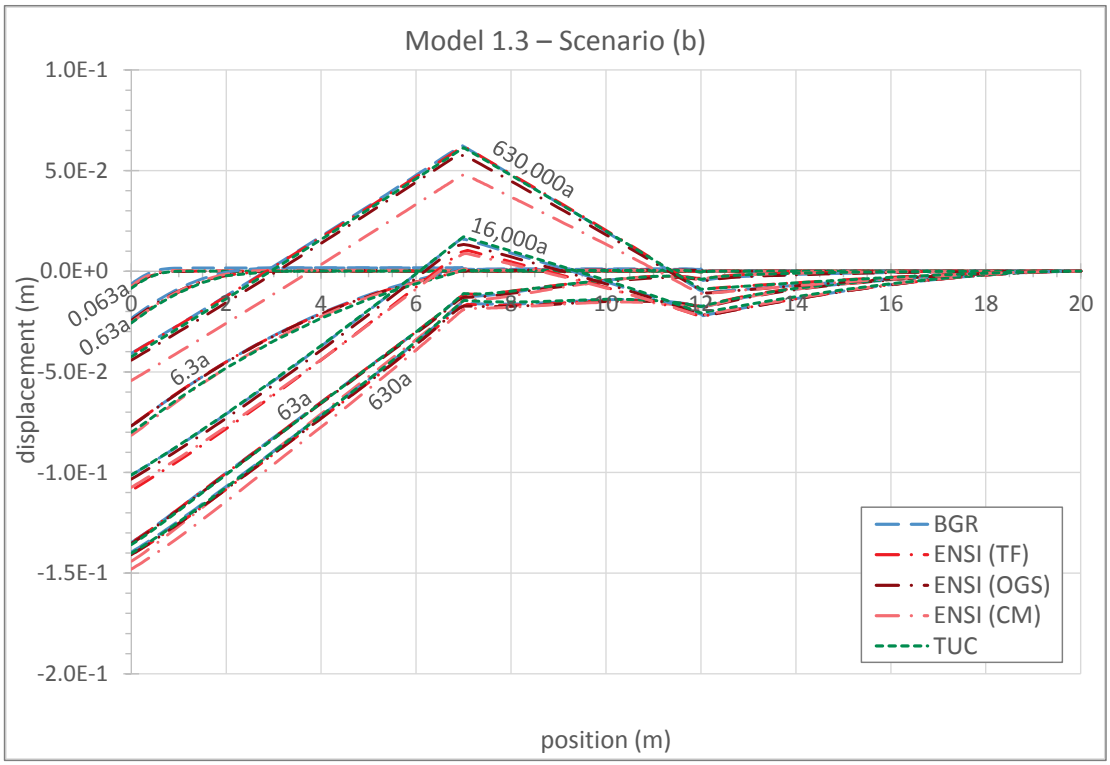


Figure 4.44: Simulation results for scenario 1.3(b) – spatial distribution of displacement

In scenario 1.3(c), the parameters of the *(Mualem/)van Genuchten* models for relative permeabilities and capillary pressure have been unified to the parameter set for the backfill. While the *Darcy* flow balances out the spatial phase-pressure distributions and turns the initial discontinuities at the material transitions into smooth curves in both scenarios 1.3(a,c) (cf. Figs. 4.38 and 4.40), the saturation retains its jumps for the base scenario because of its fixed connection to the retention curves which differ between backfill and drift seal by means of their different parameter sets. When the retention parameters become unified in scenario 1.3(c), the continuity

of the phase-pressure curves implies the continuity of the saturation curve. Figure 4.45 illustrates the stated causal relationship by the corresponding simulation results which are generally in good mutual agreement aside from the apparently missing HM coupling elaborated on above.

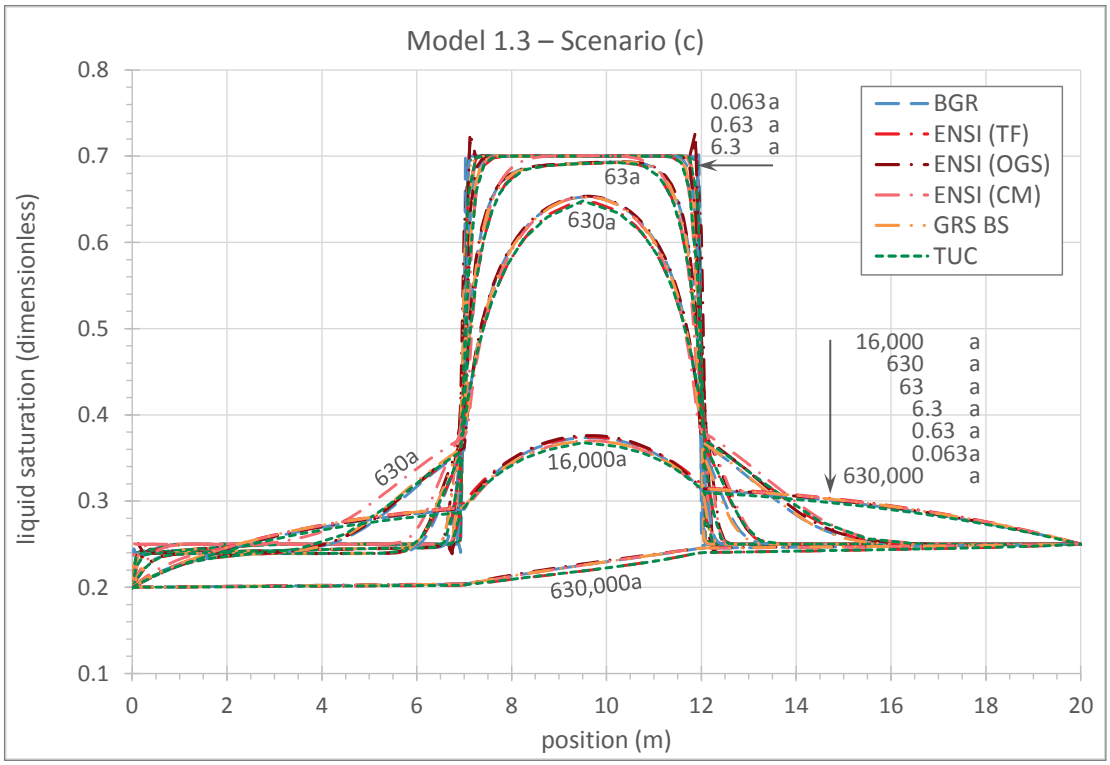


Figure 4.45: Simulation results for scenario 1.3(c) – spatial distribution of liquid saturation

Proceeding with scenario 1.3(d), Figs. 4.46 and 4.47 contain the profiles for both phase pressures. In this scenario, both phases adopt the materials' intrinsic permeabilities without taking into account their saturations, i.e., $K_{rel;\varphi}(S_l) \equiv 1$. This approach allows a simple computation of a fully analytical solution for the steady state if the capillary pressure function is invertible, like the one by *van Genuchten* used here. In the figures, the analytical solution is depicted as a wide, light grey curve lying underneath the numerical results, and it can be gathered from them that the steady-state simulation results match the solution in a very good manner, as already indicated above. Also, the transient coupled flow behaviour is well reproduced, especially of the pore water. The by means of $K_{rel;\varphi}(S_l) \equiv 1$ more sharply elevated gas mobility in the drift seal and the more sharply elevated liquid mobility in the backfill sections—in both cases in relation to the elevated mobility of the respective other phase—lead to lower pressure slopes in relation to the slopes of the respective other section or sections when comparing the profiles with those of scenario 1.3(a) (cf. Figs. 4.38 and 4.40).

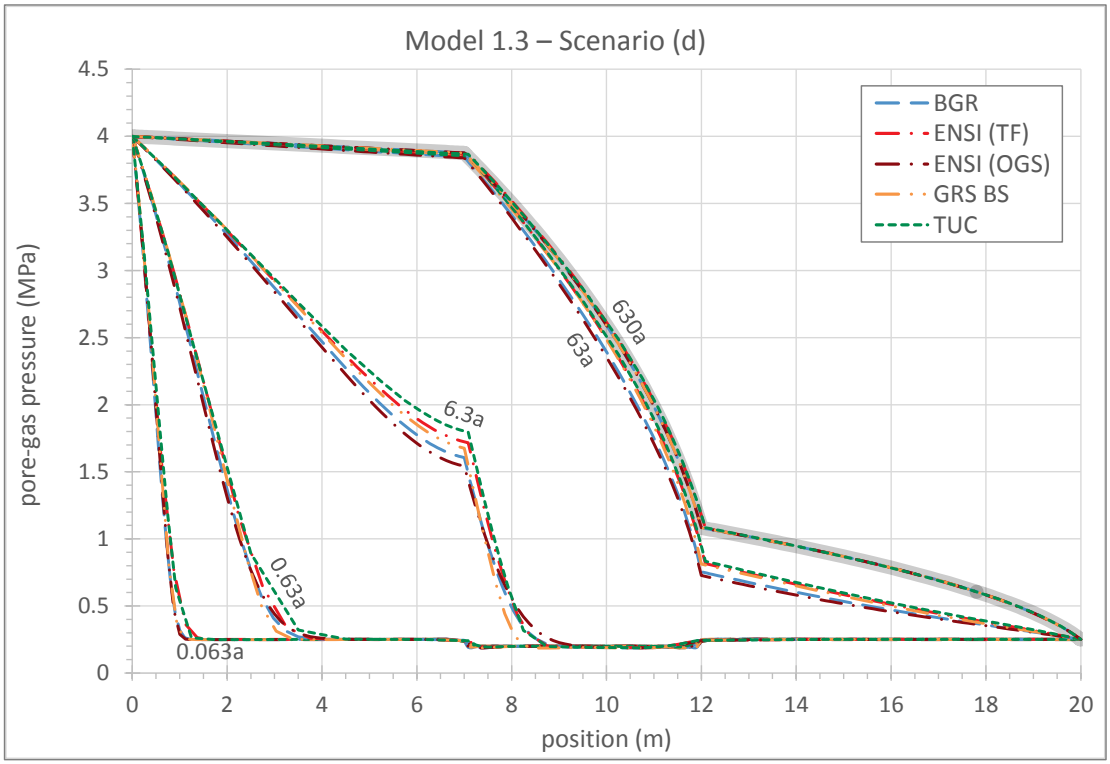


Figure 4.46: Simulation results for scenario 1.3(d) with steady-state analytical solution – spatial distribution of pore-gas pressure

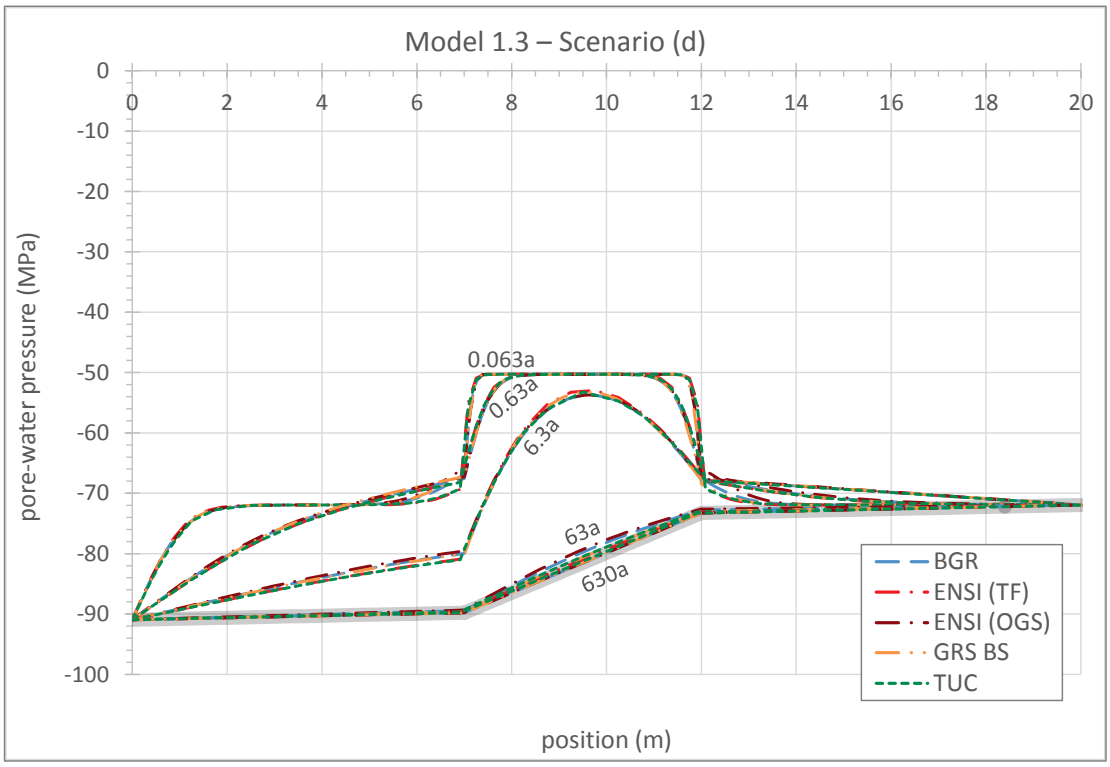


Figure 4.47: Simulation results for scenario 1.3(d) with steady-state analytical solution – spatial distribution of pore-water pressure

One observation from Fig. 4.47 that particularly catches the eye is the good agreement of all submitted liquid-pressure curves in the drift seal at steady state. While for all other model-1.3 scenarios, differences between the results obtained with TOUGH and those from other hydraulic simulators can be seen which are attributed to water evaporation (cf. the explanation further below for scenario 1.3(e)), taking saturation dependencies out of the relative permeabilities removes these differences, although evaporation takes place in this scenario 1.3(d) as well.

Scenario 1.3(e)

The left-hand-side boundary condition has been replaced by a constant gas source for scenario 1.3(e). Since the effects of phase-extraneous components, i.e., water vapour and dissolved air, had been considered small by the project coordinator, the composition of this gas source has not been defined any more closely. In the TOUGH simulators for example, sources applied to model zones relate to fluid components instead of phases, so dry air is passed into the model with the source term given in scenario 1.3(e). In the OGS-5 implementation, the two fluids are not miscible at all.

Figure 4.48 illustrates the evolution of gas-pressure profiles in the course of time, and it can clearly be seen that removal of the fixed left-hand-side value for pore-gas pressure opens up space for variation of this pressure not only near the left model boundary, but also in large parts of the model. The fixed gas generation rate also leads to a slower increase of gas pressures compared to all other model-1.3 scenarios where an initially high but later significantly decreasing pressure gradient induces an initially fast supply of gas which is slowing down with time. Figure 4.49 shows liquid-pressure profiles evolving with time. Leaving aside the pink curve, it can be seen from the figure that pore-water pressure appears to become constant at steady state. This observation can be explained by the fact that the only source of liquid is given by the right-hand-side boundary condition, a fixed value calculated from the prevailing gas pressure and saturation, which is attained in the whole model after the *Darcy* flow has equilibrated the pore-water pressure.

A closer look however reveals that the green and bright red curves show a pressure drop towards the left-hand side of the model. Consequently, pore water is constantly flowing from the right-hand-side boundary through the whole model towards the left, but this flow never manages to increase the liquid pressure any further, i.e., the amount of water appears to reduce during that

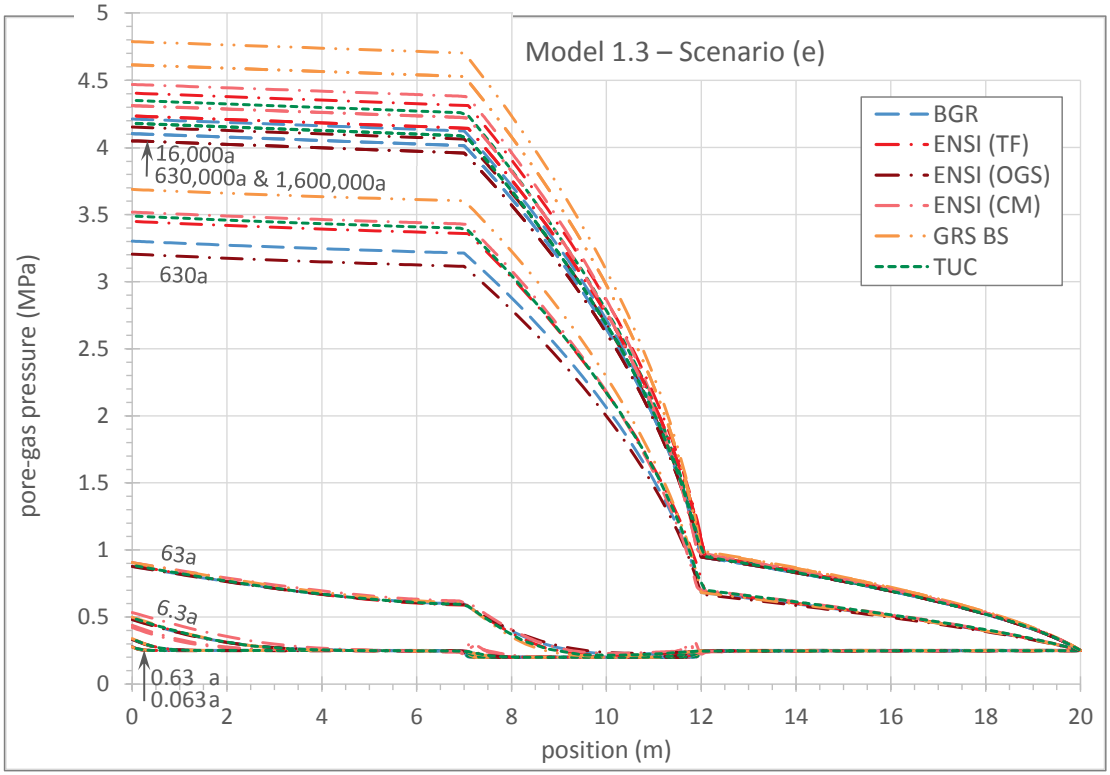


Figure 4.48: Simulation results for scenario 1.3(e) – spatial distribution of pore-gas pressure

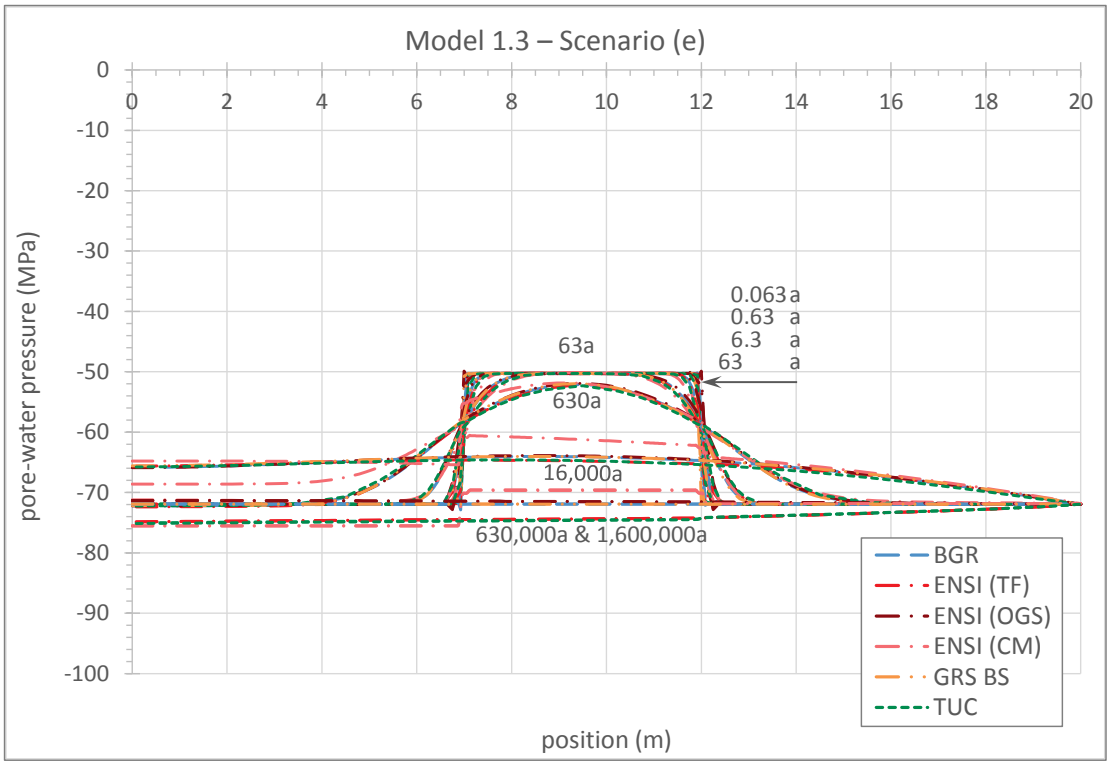


Figure 4.49: Simulation results for scenario 1.3(e) – spatial distribution of pore-water pressure

process. Bearing in mind that both the green and bright red curves are obtained with a TOUGH simulator responsible for the flow processes and recalling the composition of a gas source within this simulator, it can be concluded that the dry air passed into the leftmost model zone dilutes the gas phase, making it capable of taking up some more water vapour from the liquid phase. The more the gas decreases in pressure in the course of its way towards the right-hand side, the more water it is able to take up, and the capacity for water vapour uptake particularly increases significantly when pressures become small, i.e. here, in the right backfill section. This way, the pore water circulates in scenario 1.3(e) from the right model boundary to the left side via the liquid phase and back via the gas phase.

Although in the other scenarios, the left-hand-side boundary condition only provides a source of vapour-saturated air instead of dry air, the vapour-uptake process for decreasing gas-phase pressures still applies to those scenarios when using TOUGH. Since liquid-pressure values are fixed at the model boundaries, water inflow on the right-hand side has to be stronger in order to compensate for the evaporation-induced loss of liquid water within the model. This way, the pore-water-pressure gradient has to be steeper on the right side and flatter on the opposite side in order for the pressure to end up at the same left-hand-side boundary value as for the other simulators which do not consider this vapour-uptake process to a similar extent.

Scenario 1.3(f)

In scenario 1.3(f), the definition of equivalent pore pressure as the saturation-weighted mean of phase pressures is replaced by simply using the pore-gas pressure. Both definitions are linked by Eq. (4.2):

$$\sum_{\varphi} S_{\varphi} p_{\varphi} = p_g - S_l p_{cap} \tag{4.2}$$

where

S_{φ} saturation of phase φ, dimensionless,

p_{φ} pore pressure of phase φ in MPa,

p_{cap} capillary pressure (suction) in MPa, $p_{cap} \geq 0\ MPa$.

According to this equation, using gas pressure as the equivalent pore pressure therefore means to ignore the suction for computation of effective stresses and deformations.

Taking a look at scenario 1.3(a) with Figs. 4.38 to 4.41, it can be gathered that in the drift seal, the large decrease in saturation by a factor of ≈ 0.8 leads to a significant increase in suction by a factor of ≈ 1.6, increasing the initial $S_l p_{cap} \approx 35\ MPa$ to a value of $> 46\ MPa$ as opposed to increasing p_g by only up to $4\ MPa$. The observed compaction of the drift seal therefore occurs as a consequence of the decrease in equivalent pore pressure by more than $7\ MPa$. In both backfill sections, the term $S_l p_{cap}$ remains almost constant, so the equivalent pore pressure rises by almost $4\ MPa$ in the left backfill and by up to $0.7\ MPa$ in the right one, and the materials expand correspondingly.

As suction is neglected in the current scenario 1.3(f), only the change in gas pressure determines the deformation behaviour together with the materials' different *Young* moduli (the total stress remains constant in this scenario as well). And since this pressure increases in the whole model domain in relation to its initial state, the whole model expands, as can be seen in Fig. 4.50. The three sets of simulation results submitted show a good mutual agreement, particularly concerning the displacements displayed in this figure.

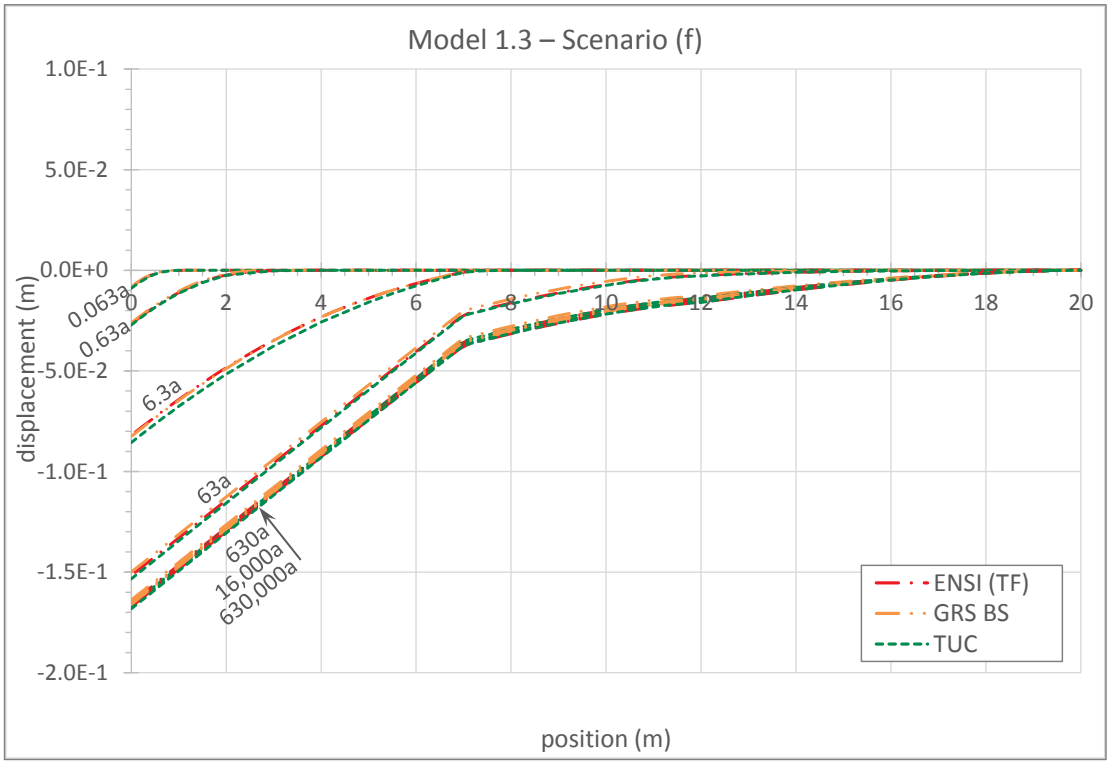


Figure 4.50: Simulation results for scenario 1.3(f) – spatial distribution of displacement

Scenarios 1.3(g-h)

Scenario 1.3(g) differs from the other model-1.3 scenarios in the mechanical boundary conditions. While in the other scenarios, the total stress is kept constant and the model is allowed to change its total length at the left boundary, the model is clamped on both sides for the current scenario which in turn allows the total stress to change. Therefore, an expansion of the overall model like in almost all other model-1.3 scenarios is suppressed and converted into a pressure-induced increase of total stress instead. Figure 4.51 shows the displacement profiles which are in their relations between different sections comparable to the corresponding displacement distributions from scenario 1.3(a) (cf. Fig. 4.41). The simulation results other than the ones in pink match each other in a very good manner.

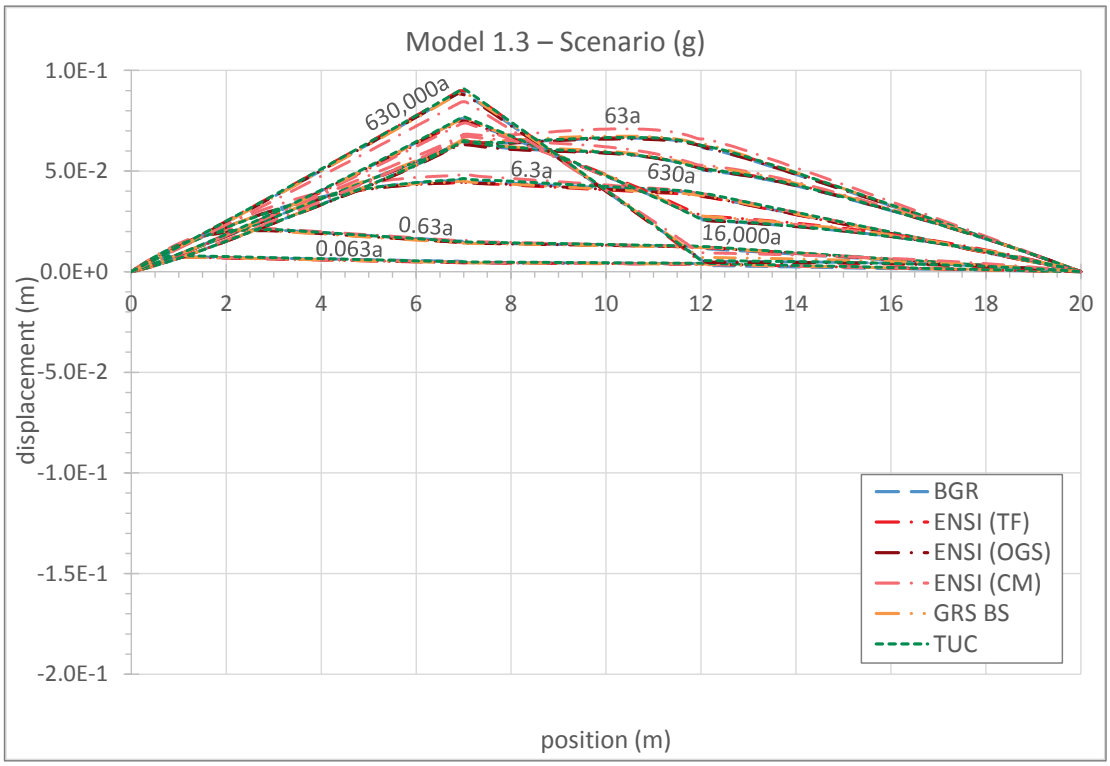


Figure 4.51: Simulation results for scenario 1.3(g) – spatial distribution of displacement

The three total-stress evolutions submitted and depicted in Fig. 4.52 contain minor discrepancies in the longer term but are nevertheless in good accordance with each other. In preceding displacement evolutions like the one in Fig. 4.41, it can be seen that the left boundary of the expanding model first moves far to the left before returning near to its initial position, for scenario 1.3(a) e.g. by about 14 *cm* (0.7 % of the total model length) before returning to a final expansion of approximately 3 *cm* (0.15 % of the total model length, ≈ 21 % of the maximum

expansion). When movement is confined, a hydraulic stress develops to an extent corresponding to the stated expansions: the pressure increase in the left model section lets the backfill material exert a total stress of almost 1.9 MPa while later on, the suction-induced compaction of the drift seal compensates for 1.5 MPa of that stress, reducing it to $\approx$ 21 % of the maximum value.

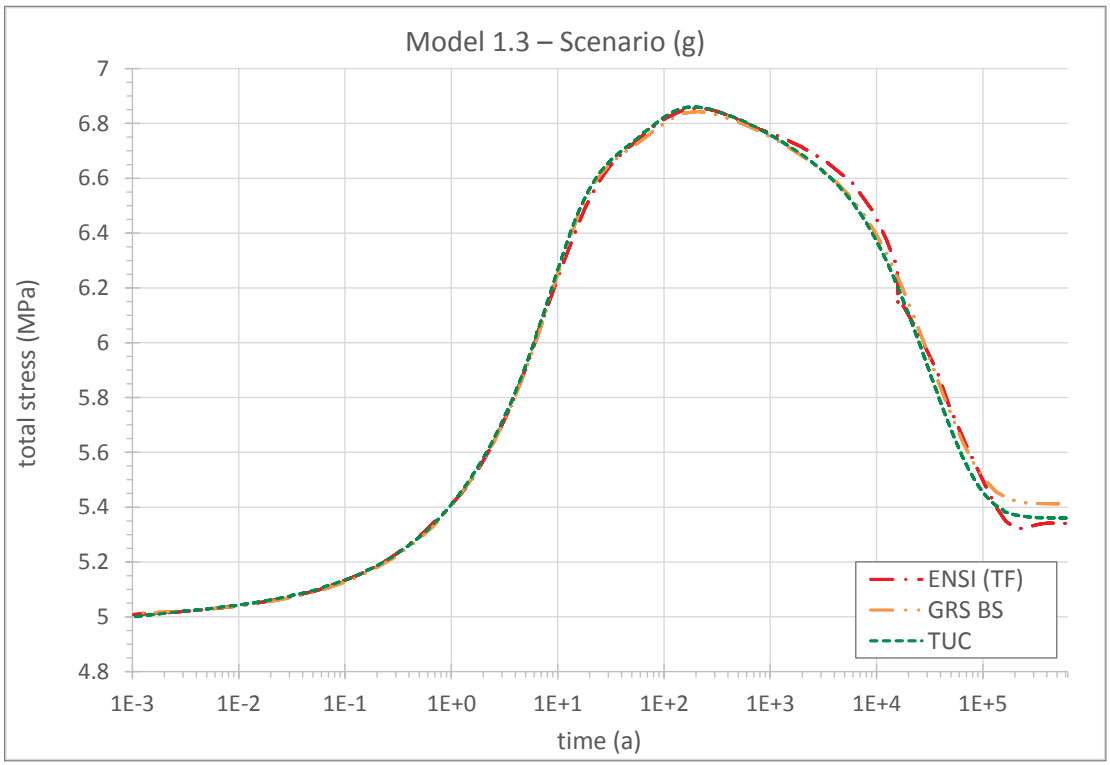


Figure 4.52: Simulation results for scenario 1.3(g) – evolution of total stress

Finally, scenario 1.3(h) is defined by omitting mechanical processes completely. The obtained results show again a very similar behaviour to that from scenario 1.3(a), and the major difference beyond vanishing displacements lies in the lack of the HM-coupling effect described for scenario 1.3(a). The increasing gas pressure does not have any influence on zone lengths any longer, so saturations are unaffected by this pressure increase. The errors in the pink curves visible in the preceding scenarios—one of which came from the aforementioned lack and the other is supposed to have its origin in the mechanical processes or coupling as well—do not appear in the corresponding simulation results. As a consequence, Fig. 4.53 shows generally (aside from numerical oscillations at the left model boundary) very good agreements of the collected simulation results for saturation.

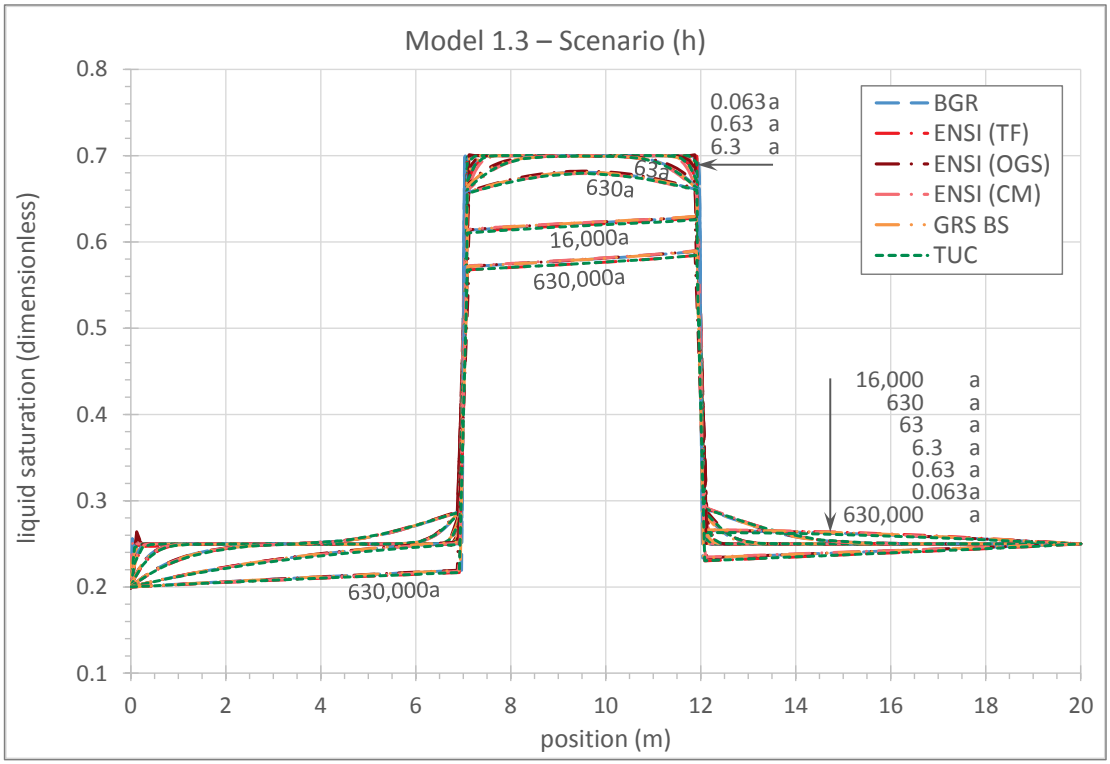


Figure 4.53: Simulation results for scenario 1.3(h) – spatial distribution of liquid saturation

4.4 Model 1.4

4.4.1 Model description

With respect to model 1.3, its successor is of increased complexity in several aspects. Model 1.4 additionally takes thermal processes into account, and its geometry is extended to include besides a drift seal and backfilled areas two sections of backfilled voids around a heat-generating waste canister as well as a section of host rock behind the entrance of the emplacement drift from the (imaginary) access drift. Due to the one-dimensionality of the model, a clear distinction between canisters and surrounding backfill is not possible; instead, "canister sections" are defined with blurred properties calculated from backfill and canister properties. The geometric model is depicted in Fig. 4.54 and consists of a repository and a host-rock section of 20 metres each, with the former being further subdivided into canister sections, drift seal, and backfilled areas filling up the space between the other repository elements. Symmetry conditions—no flow and no deformation—prevail at the left model boundary, while at the outer boundary of the host rock, far-field conditions prevail, i.e., pressures, saturations, and temperature are fixed at their initial state. Heat and, if applicable, gas are produced in both canister sections simultaneously by means of simple, piecewise linear source terms. Table 4.8 summarises the spatial

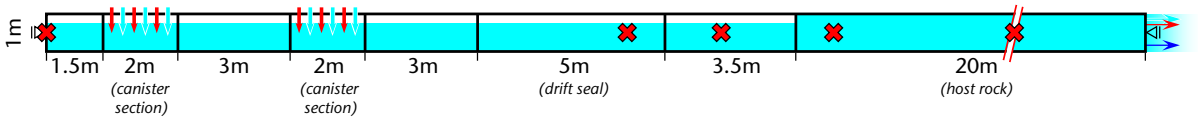


Figure 4.54: Visualisation of model-1.4 geometry and initial and boundary conditions for scenarios 1.4(a-e), and evaluation points; unlabelled sections represent backfill

Evolutions at $X =$	Profiles at $t =$	$\approx$
$0 \quad m$	$5 \cdot 10^{7} \quad s$	$1.6 \quad a$
$15.5 \quad m$	$5 \cdot 10^{9} \quad s$	$160 \quad a$
$21 \quad m$	$1.5 \cdot 10^{11} \quad s$	$4{,}800 \quad a$
$30 \quad m$	$3 \cdot 10^{11} \quad s$	$9{,}500 \quad a$
and for S_l also at	$3.7 \cdot 10^{11} \quad s$	$12{,}000 \quad a$
$18 \quad m$	$1.5 \cdot 10^{12} \quad s$	$48{,}000 \quad a$
	$1.5 \cdot 10^{13} \quad s$	$480{,}000 \quad a$
	and for 1.4(a) also at	
	$5 \cdot 10^{8} \quad s$	$16 \quad a$

Table 4.8: Positions for temporal evolutions and points of time for profiles concerning model 1.4

and temporal points preset for the evaluation of results, with the steady state in scenario 1.4(a) being attained already after 400 years.

If all multiphysical processes and source terms defined for model 1.4 are considered, fluid-flow processes are primarily controlled by the initial differences in liquid pressure due to an increased suction in the drift seal and the initial liquid overpressure in the host rock. Gas sources increase pressures and intensify gas flow and drainage into the host rock. The temperature increase induced by the heat sources, especially at the start of the course of multiphysical processes when they are most active, changes fluid densities and therefore affects saturations. Heat drains away towards the model's right, according to the simulation results in section 4.4.3 practically unaffected by convective transport. Deformations are induced by the changing equivalent pore pressures and by heating.

The main scenario features TH^{2}M-coupled processes, and the simulations were supposed to be contrasted to simulations under omission of gas sources, hydraulic processes, and/or mechanical processes, but otherwise under the same conditions. For reasons around the definition history of model 1.4, the main TH^{2}M scenario has been given the letter (d), with 1.4(d2) being a slight variation of it, and a sketch of the other scenarios and their relations to 1.4(d) is shown in Fig. 4.55. The parameters, initial and boundary conditions, and source terms for gas and heat generation for scenarios 1.4(b,c,d,e)—which also serve as a basis for derivation of parameters for scenarios 1.4(a,d2)—are given by Tables 4.9 and 4.10 and by Fig. 4.56. While most fluid parameters have been defined constant, densities may vary upon changes in pressure and temperature: for the liquid phase by means of Eqs. (2.8) and (2.30) with constant bulk modulus and thermal expansion coefficient, and for the gas phase based on the ideal gas law.

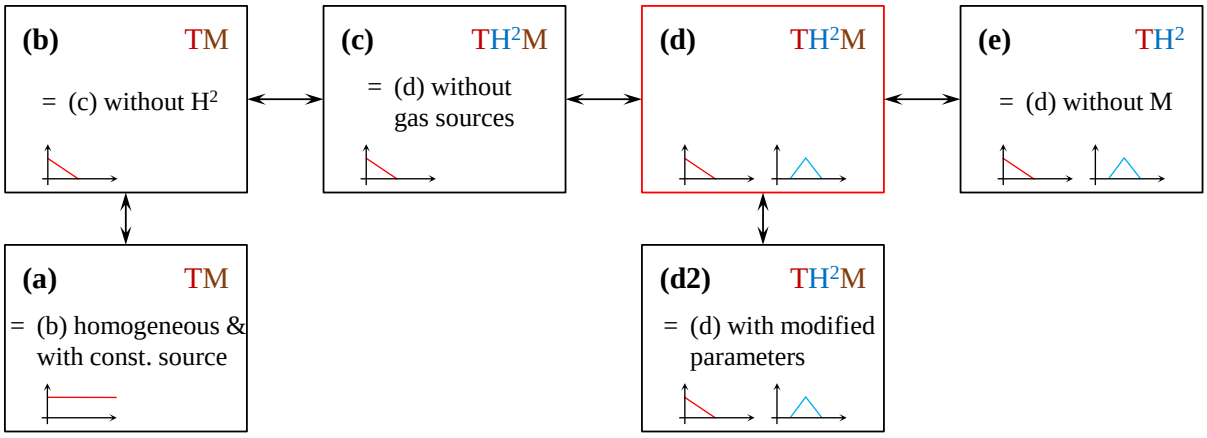


Figure 4.55: Scenarios for model 1.4

Material parameters						
		r e p o s i t o r y			*host rock*	
		backfill	*canister sections*	*drift seal*		
Young's modulus (dr.)	E	45	150	400	8,000	MPa
Poisson's ratio	ν	0	0	0	0	
porosity	ϕ	0.42	0.37	0.35	0.17	
intrinsic permeability	K	$2 \cdot 10^{-18}$	$1.75 \cdot 10^{-18}$	$2 \cdot 10^{-20}$	10^{-20}	m^2
Biot coefficient	α	1	1	1	1	
res. liquid saturation	S_{lr}	0.03	0.03	0.1	0.16	
res. gas saturation	S_{gr}	0.005	0.005	0.005	0.005	
van Gen. parameter	m	0.5	0.5	0.3	0.28	
van Gen. pre-factor	p_{cap0}	12	12	26	30	MPa
pore connectivity par's	$\tilde{\varepsilon}; \tilde{\gamma}$	0.5	0.5	0.5	0.5	
thermal conductivity *(grains)*	λ_s	2.1	11.57	2.1	2.3	$W/(m\,K)$
thermal conductivity *(bulk)*	λ_{dry}	$\Rightarrow$ 1.23	$\Rightarrow$ 7.33	$\Rightarrow$ 1.38	$\Rightarrow$ 1.91	$W/(m\,K)$
thermal conductivity *(bulk)*	λ_{wet}	$\Rightarrow$ 1.49	$\Rightarrow$ 7.55	$\Rightarrow$ 1.59	$\Rightarrow$ 2.02	$W/(m\,K)$
spec. heat capacity *(grains)*	c_s	2,500	3,450	2,500	2,700	$J/(kg\,K)$
density *(grains)*	ρ_s	2,500	3,450	2,500	2,700	kg/m^3
lin. th. exp. coeff. *(skeleton)*	α_s^{th}	$5 \cdot 10^{-6}$	$5.9 \cdot 10^{-6}$	$4 \cdot 10^{-6}$	$3.6 \cdot 10^{-6}$	K^{-1}

Hydraulic initial conditions				
	r e p o s i t o r y		*host rock*	
pore-gas pressure	p_{g0}	0.1013	4	MPa
liquid saturation	S_{l0}	0.75	0.995	

Table 4.9: Basic material parameters and hydraulic initial conditions for model 1.4

Fluid parameters				
		liquid phase	*gas phase*	
viscosity	η_φ	$5 \cdot 10^{-10}$	$2 \cdot 10^{-11}$	$MPa \cdot s$
thermal conductivity	λ_φ	0.65	0.03	$W/(m\,K)$
specific heat capacity	c_φ	4,200	2,000	$J/(kg\,K)$
density	ρ_φ	970	$\frac{M_g}{R} \cdot \frac{p_g}{T}$	kg/m^3
bulk modulus	$\breve{K}_\varphi$	2,100	p_g	MPa
volumetric thermal expansion coefficient	γ_φ^{th}	$6 \cdot 10^{-4}$	$\frac{1}{T}$	K^{-1}

Mechanical and thermal initial conditions		
total stress	σ_{tot0}	8 MPa
temperature	T_0	25 $°C$

Source terms (per canister section)		
peak gas generation rate	$\hat{Q}_{m;g}^{max}$	10^{-9} kg/s
peak heating power	$\hat{Q}_Q^{max}$	1.5 W

Boundary conditions		
lhs displacement	u_{lhs}	0 m
rhs displacement	u_{rhs}	0 m
rhs pore-gas pressure	$p_{g;rhs}$	4 MPa
rhs liquid saturation	$S_{l;rhs}$	0.995
rhs temperature	T_{rhs}	25 $°C$

Table 4.10: Basic fluid parameters, thermal-mechanical initial conditions, and boundary conditions for model 1.4

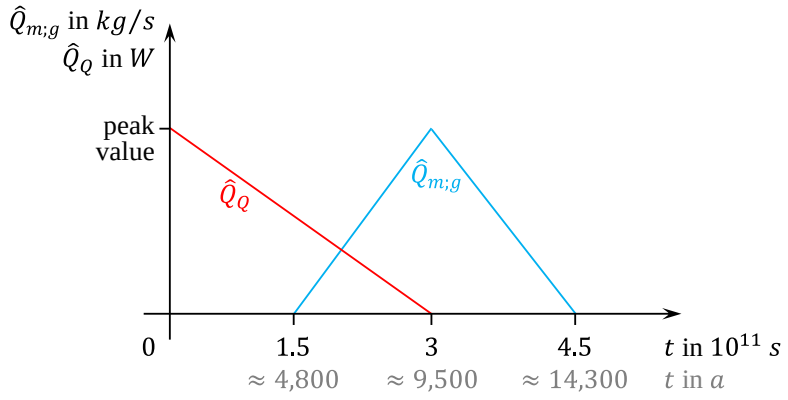


Figure 4.56: Basic heat and gas source terms for model 1.4

Scenario 1.4(a) was conceived as a "pre-study" for model 1.4 in that it features major simplifications for comparability with an analytical solution. The scenario covers only TM processes in a homogeneous material, i.e., a 40-metres-long backfill area, with persistent and constant heat-source terms for both canister sections taking the "peak" value from Table 4.10. A changed set of parameters was chosen for the backfill material based on its parameters from Table 4.9 and fluid parameters from Table 4.10, assuming the initial hydraulic state remains unchanged throughout the course of TM processes. This parameter set is given in Table 4.11 and applies together with the basic TM initial and boundary conditions.

Blurred material parameters					
Young's modulus (dr.)	E	71.6 MPa	thermal conductivity	λ	1.43 $W/(m\,K)$
Poisson's ratio	ν	0	specific heat capacity	c	1,640 $J/(kg\,K)$
porosity	ϕ	0	density	ρ	1,756 kg/m^3
Biot coefficient	α	0	lin. thermal expansion coeff.	α^{th}	$5 \cdot 10^{-6}$ K^{-1}

Table 4.11: Material parameters for scenario 1.4(a) in relation to scenario 1.4(d)

Scenario 1.4(d2) was defined as an alternative to 1.4(d) with a few settings appearing more realistic but not necessarily implementable exactly as defined with every simulator applied in this project. The fluid parameters are not predefined and have to be computed by functions provided by the simulator. The joint simulation of this scenario thus comes close to a conventional benchmarking where every participant has to choose their own approach to simulate the benchmark as appropriate as possible. Table 4.12 shows the modified assumptions for scenario 1.4(d2) in comparison to 1.4(d).

4.4.2 Individual approaches for simulation

4.4.2.1 BGR (simulator OGS)

To simulate the TH^{2}M coupled processes, the implementation `MULTI_PHASE_FLOW`, `HEAT_TRANSPORT`, and `DEFORMATION` were used in a staggered coupling scheme. There was no iterative coupling between these processes due to numerical instabilities. Hence, the processes are coupled but in one way: H^2→T→M. Therefore, a compaction or extension of the

Modified material parameters

		repository			host rock	
		backfill	*canister sections*	*drift seal*		
Biot coefficient	α	1	1	1	0.8	
residual gas saturation	S_{gr}	0	0	0	0	
thermal conductivity *(bulk)*	λ_{dry}	$(1-\phi)\lambda_s + \phi\lambda_g$				$W/(m\,K)$
	λ_{wet}	$(1-\phi)\lambda_s + \phi\lambda_l$				

Modified fluid parameters	Modified boundary condition		
thermodynamical functions of the simulators	rhs liquid saturation	$S_{l;rhs}$	1

Modified hydraulic initial conditions

		repository	host rock
liquid saturation	S_{l0}	0.75	1

Table 4.12: Updated parameters and initial and boundary conditions for scenario 1.4(d2) in relation to scenario 1.4(d)

material will have no influence on the pore-fluid pressures and there is no thermal pressurisation of the fluid phases. Numerical solver settings as well as the meshing with quadrangular *Taylor/ Hood* elements are otherwise similar to the settings in model 1.2 and 1.3.

The initial full saturation of the material on the right side of the model posed a challenge for the numerical solution: In the gas mass balance equation as implemented in OGS-5, several terms are multiplied by the gas saturation which becomes zero in the fully saturated materials. This can make it impossible for the linear solvers to find a solution satisfying the gas mass balance and thus, the solution does not converge. This problem is solved by defining a residual gas saturation of $S_{gr} = 0.1$ in all material groups.

4.4.2.2 ENSI (simulator OGS)

Model 1.4 was simulated with a staggered approach of the thermo-hydro-mechanical processes in OGS-5. The mesh used for this case was a 2D mesh with 440 cells along x-axis and 3 cells along z-axis. A simulation time of 630,000 years was split into 416 time steps when only thermal sources were involved and into 10,235 time steps when also hydraulic sources were involved. The parameters, initial and boundary conditions were assigned as described in the task. During the simulation, the pressure gradient is evolving. The saturation is decreasing within the seal and increasing in backfill left and right of the seal. Due to the stress conditions, an initial compaction of the model occurs. The seal is further compacted due to the increasing effective stress following the desaturation whereas the backfill expands.

4.4.2.3 GRS BS (simulator CODE_BRIGHT)

The modelling approach for the THM-coupled simulation is the same as in the previous models. Due to the project end for GRS BS in April 2020, there could be some differences to the final specification for this model. However, GRS BS performed all THM-coupled simulations following an interim specification. More information about the THM-coupled modelling can be found in Section 7 of *Czaikowski & Friedenberg (2020)*.

4.4.2.4 TUC (FTK simulator)

As before, module `EOS3` has been used by TUC for the TOUGH2 simulation parts. The alternative of using simplified dependencies of thermodynamic properties η_φ, ρ_φ, and u_φ on pressure and temperature by means of constants η_φ, $\breve{K}_\varphi$ and γ_φ^{th}, and c_φ^V, respectively, instead of by TOUGH2-intrinsic functions has not yet been fully implemented by TUC. Therefore, TUC simulated only scenario 1.4(d2) which does not restrict to these simplified relations.

4.4.3 Results and conclusions

With the increased complexity of the model geometry and with the inclusion of thermal processes and their couplings to others, defining a set of model-1.4 scenarios appropriate for a simulator benchmarking under equal conditions (processes, equations, parameters) proved to be significantly more difficult than for the preceding HM-coupled models, *Lux et al. (2021)*.

These difficulties manifested for one thing in more complex process interactions (e.g., variation of fluid densities, boiling of water, vapour content of the gas phase) where simplifications for the sake of comparability of results could not be implemented in the participating simulators, or only with considerable effort for marginal benefit beyond this project. For another thing, many test simulations by the project coordinator with the FTK simulator reached technical limits, and comparative test simulations using FLAC were not able to confirm or comprehensibly refute the FTK test simulation results for the settings where FTK produced apparently reasonable results. Scenarios had to be redefined several times, and the restrictions on assemblies and travelling emerged in the final year of project processing prevented comparisons, further analysis, and joint discussions of the collected simulation results. These results therefore can be regarded as "first results" (cf. Figs. 4.9 and 4.27 from the preceding models) and will be presented further below in this subsection. Discretisations have not been submitted for model 1.4 by the other partners, so Fig. 4.57 contains only the one by the project coordinator.



Figure 4.57: Discretisation applied by one of the partners for simulations of scenario 1.4

Some simulation results collected after having defining the final set of scenarios for model 1.4 are depicted in the Figs. 4.58 to 4.63. An analytical solution could be derived for the TM-coupled model from the first scenario. Because of the missing opportunity for presentation and joint analysis of the simulation results, a discussion of discrepancies in the results of other scenarios beyond the first is kept at a minimum in this report.

Scenario 1.4(a) is considered as a kind of pre-study for model 1.4, and as such it covers only TM processes in a basically homogeneous medium with a simple constant thermal source in both canister sections and an analytical solution for comparison. Figure 4.58 shows the evolution of temperature at the four positions $X = 0\ m$, $15.5\ m$, $21\ m$, and $30\ m$, while Figs. 4.59 and 4.60 show profiles of temperature and displacement for four different points of time, $t \approx 1.6\ a$, $16\ a$, $160\ a$, and $4{,}800\ a$ (exact values in seconds correspond to these approximate values in years). It can be gathered from these three figures that the analytical solution is generally matched by the simulation curves in a good manner. The temperature evolution for position $X = 30\ m$ has erroneously been recorded at position $X = 33\ m$ by two of the partners, and the dark red curves show a delay of processes after $t = 30\ a$, presumably stemming from the sudden coarsening of time-stepping around that time.

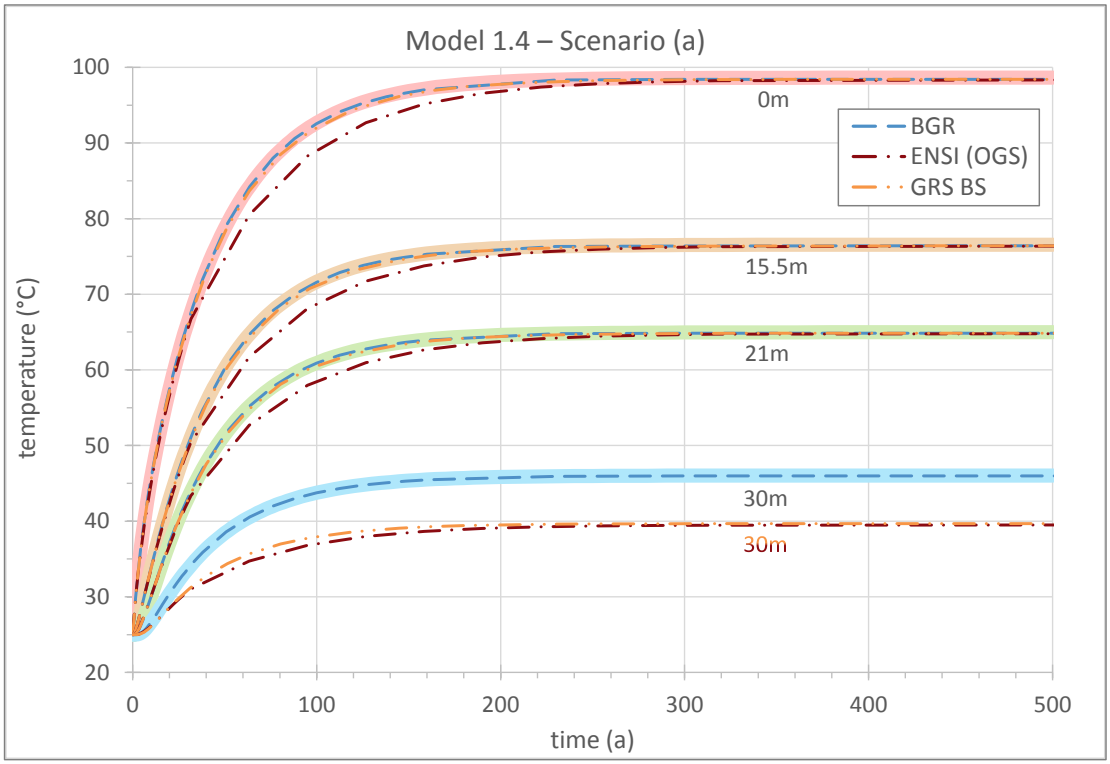


Figure 4.58: Simulation results for scenario 1.4(a) compared to analytical solutions – evolution of temperature

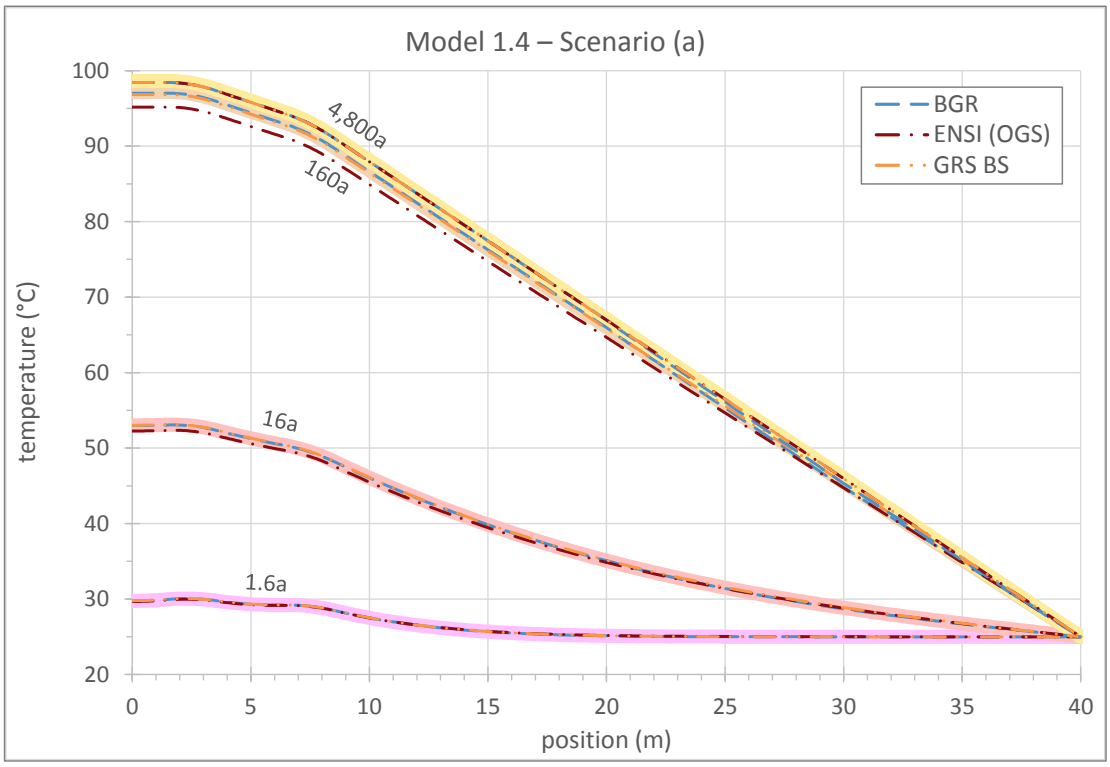


Figure 4.59: Simulation results for scenario 1.4(a) compared to analytical solutions – spatial distribution of temperature

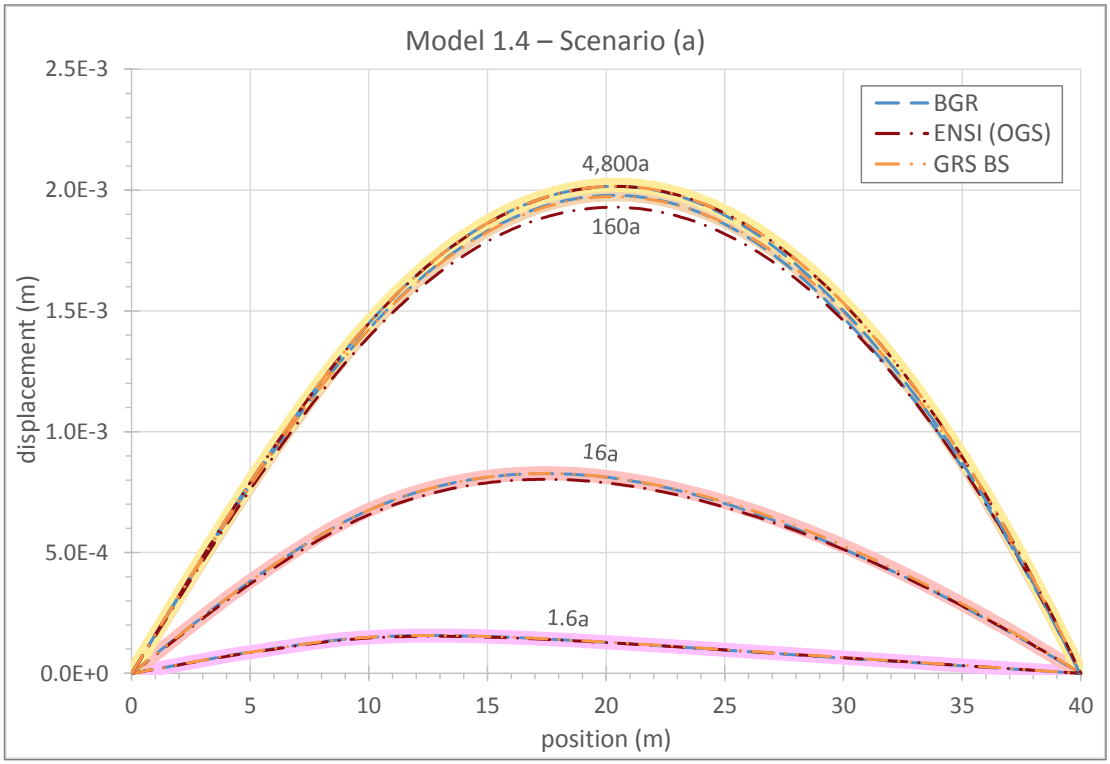


Figure 4.60: Simulation results for scenario 1.4(a) compared to analytical solutions – spatial distribution of displacement

Erroneous positions for recording temporal evolutions appears also to have been the predominant issue for most other discrepancies in temperature curves, the ones for the TH^{2}M scenario 1.4(c) being displayed in Fig. 4.61. Aside from that issue, temperatures evolve very similarly in the partners' simulations, leading to very good agreements of corresponding curves. This observation can also be made with all other scenarios except for 1.4(b). Contrary to this, liquid saturations tend to show significant mismatches, as can be seen in Fig. 4.62 for scenario 1.4(c) and Fig. 4.63 for scenario 1.4(d2).

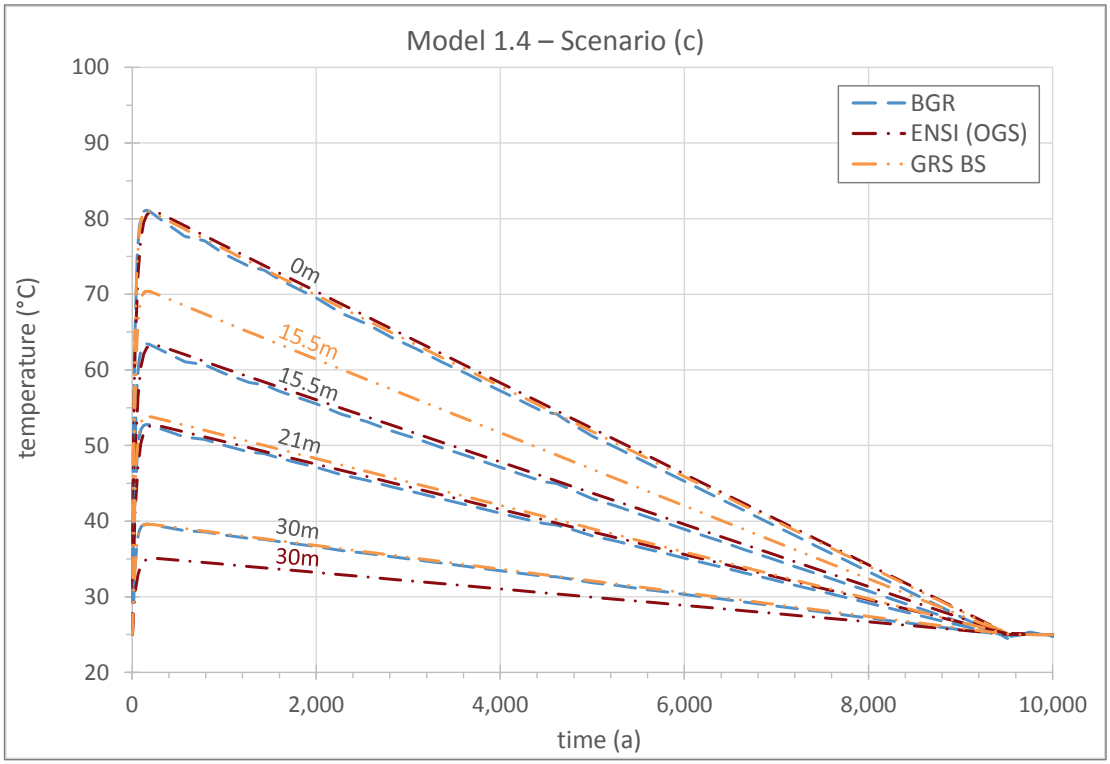


Figure 4.61: Simulation results for scenario 1.4(c) – evolution of temperature

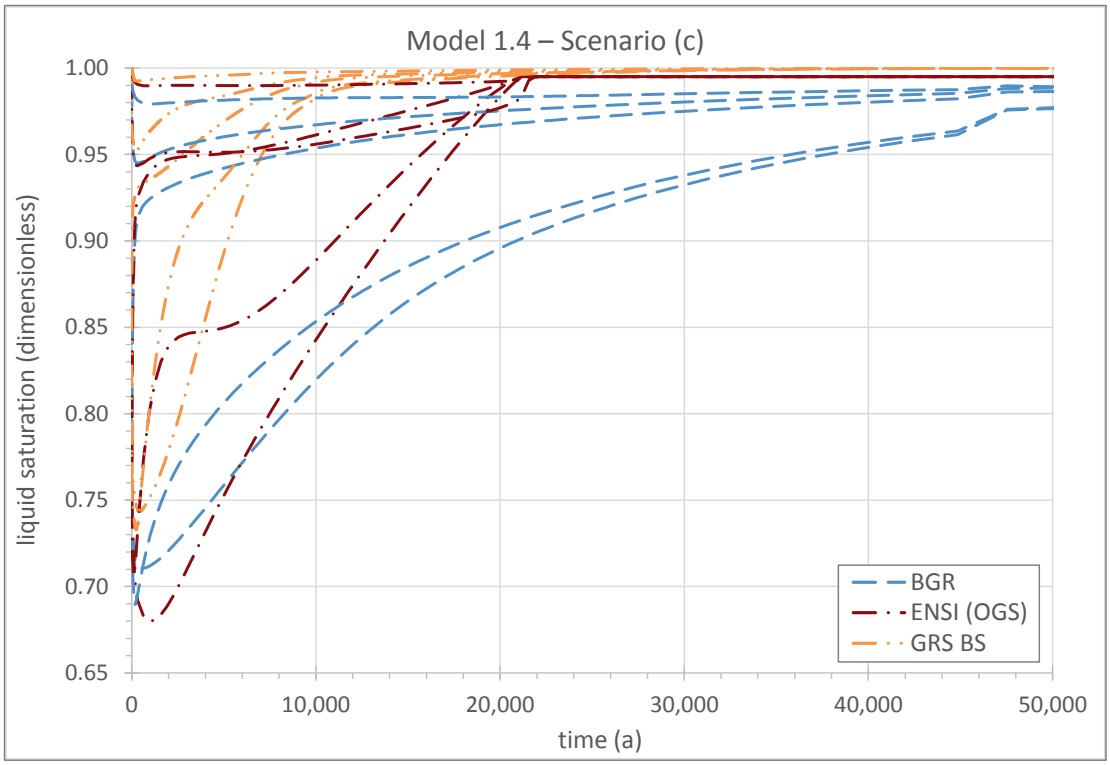


Figure 4.62: Simulation results for scenario 1.4(c) – evolution of liquid saturation

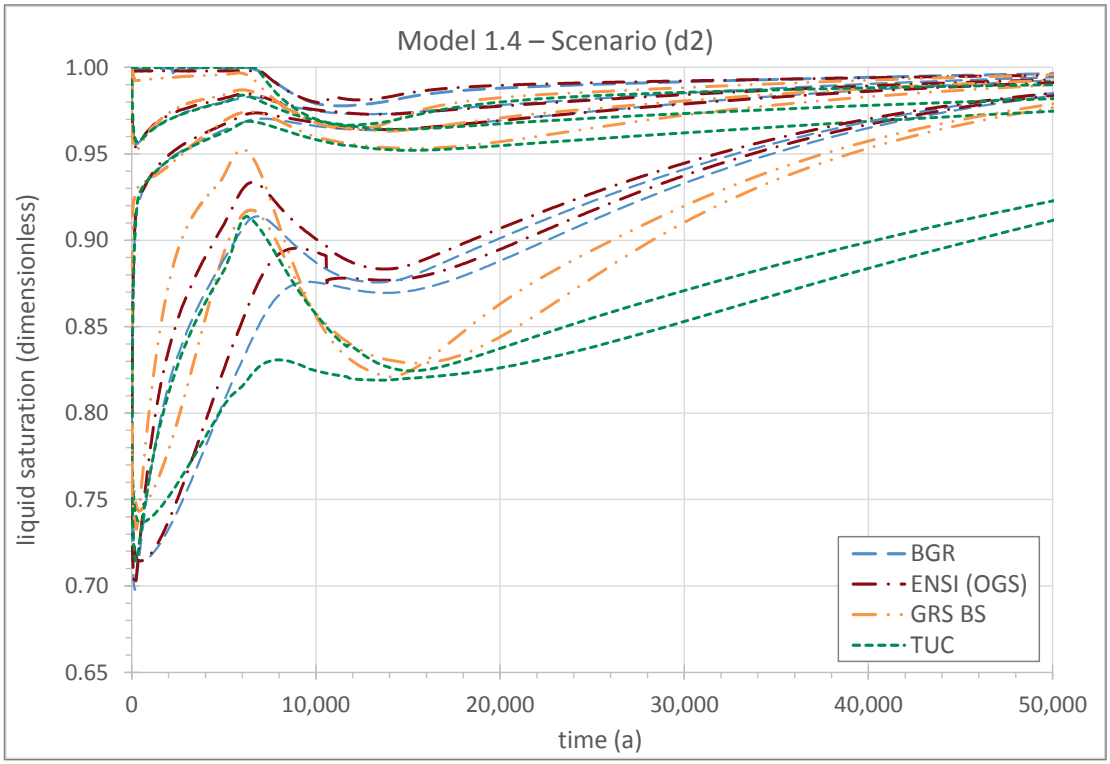


Figure 4.63: Simulation results for scenario 1.4(d2) – evolution of liquid saturation

Upon comparing the "first results" for model 1.4 shown in Figs. 4.61 to 4.63 with some first results for previous models shown before in Figs. 4.9 and 4.27 and how those could be improved in the course of the project, it appears to be probable that more in-deep discussions together with fixing some obvious input errors could have helped to resolve most, or at least some of the discrepancies in the simulation results for model 1.4. However, the experience gathered from these first model-1.4 simulations and from the test simulations shows that a deeper understanding of TH^2M process interactions is required to classify and assess the causes for the discrepancies in the results. In addition, a more profound knowledge of how the process interactions are implemented in the various simulators and of which possible settings and inputs are offered by the simulators is essential for defining model scenarios appropriately for a simulator benchmarking.

5 Discussion and lessons learnt

The BenVaSim project has been initiated to help verifying numerical simulators brought into operation for numerical reproductions and predictions of multiphysically coupled systems (thermal and/or hydraulical and/or mechanical) in the context of underground radioactive waste disposal. Various simplified one-dimensional models and scenarios have been simulated in this first project stage by different project partners with simulators they usually apply in this scientific discipline. The contrasting of their simulation results has proven to be a highly helpful and effective method for verification of the workings of such simulators. Despite quite a number of differences in the project partners' first results, joint discussions and analyses promoted the understanding of the workings and capabilities of the own simulator and thus helped to overcome many of these differences. The final results presented in this report show a great deal of agreement in many cases, including mechanical quantities (displacements and stresses) and also where a comparison to an analytical or semi-analytical solution cannot be offered. Literature research about benchmarkings conducted in the framework of a pre-project revealed that particularly mechanical processes seem to be more likely responsible for discrepancies in simulation results, *Lux & Rutenberg (2018)*, but regarding the results from this project, it appears that the simple linear-elastic isotropic constitutive model by *Hooke* does not pose any severe issues for coupled simulations. Analytical or semi-analytical solutions have been handed out to the partners in advance, and consequently, very good accordance of the corresponding simulation results with these solutions has been attained due to the direct possibility to check one's results. But also many results for scenarios without this possibility agreed well with one another, which founds the success of this project as a verification approach for the participating simulators.

Hence, with respect to the overall objective the BenVaSim project aims to contribute to, many results of the HM-coupled simulations do not give cause for questioning the correctness and basic suitability of the involved simulators for applications in the field of subsurface radioactive waste disposal. Two processes require improvements of the corresponding simulators (proper consideration of grain compressibility, dealing with soft materials under two-phase flow). However, the positive results cannot be regarded as more than an indication of correct simulator workings and suitability because of the models' simplicity and particularly the restriction to the very simple mechanical constitutive model. In addition, several processes relevant for modelling and simulating repository behaviour have been taken into account when defining models

and scenarios, but this choice is not exhaustive. For the coupling to thermal processes, no specific statement can yet be made from the conducted simulations. Further steps to simulator verification would therefore have to be taken.

With respect to further simulator benchmarkings or other measures of simulator verification and quality assurance by third parties, the simulation results of this project form a data base that can help serving these purposes. A comprehensive collection of figures obtained from this project can be found on the CD-ROM enclosed in this report.

Working on the BenVaSim project generated experience not only for the people conducting the numerical simulations, but also for the project coordinator responsible for making up reasonable model suggestions focused on the aim of such a project and taking care of the properties and pitfalls of the involved simulators and of the partners' specific aims. In addition, institutions funding cooperative scientific research and in particular benchmarkings can make decisions for future projects based on experience from the BenVaSim project. Scientific benefits made and lessons that can be learnt by the partners and by other research groups involved in a benchmarking comprise:

- A few flaws of simulators in special cases have been detected, and some of them could be eradicated. Note that remedial action or further development can only be taken if the corresponding parts of the simulator's source code are accessible and modifiable, or if the code's developer contributes to this remedy. In addition, remedial action is desirable if it constitutes a scientific advantage worth the investment in time and funding, which has not always been the case in the course of the BenVaSim project. For example, neglecting capillary pressures as it was required for a few H^2M scenarios of this project is not a key feature simulators need to have for application on subsurface problems, since fluid phases occurring in geomaterials usually possess different wetting properties and therefore develop surface tension and capillary forces at their interfaces. As another example, replacing sophisticated simulator-intrinsic physical laws by simplified equations or constants for no further use can be deemed unnecessary.
- Aside from erroneous inputs into the simulator which are easily detectable or, to a certain extent, identifiable by a rough estimate from the results and therefore correctable, input errors may also lead to simulation results still appearing plausible if no comparative curve is at hand. Likewise, errors in the implementation of process equations may lead to inconspicuous simulation results, which can be the case both with self-developed and with

pre-existing simulator codes. As this project stage shows, these effects may already occur for seemingly simple model geometries, and especially the latter effect confirms the recommendation to perform predictive numerical analyses with more than one verified simulator within a single institution or organisation.

– When reconducting simulations and compiling their results for evaluation, it is advisable to replace the complete tabular data for the resimulated scenarios by the new results instead of only a part of it in order to avoid possible inconsistencies between profiles and/or temporal evolutions.

– For the accuracy of results and especially in order to avoid numerical oscillations, it can be crucial to reconsider spatial and temporal discretisations of the model with respect to areas with elevated thermal or hydraulic activity or jumps in stresses (high gradients) and also with respect to given points of time for profiles.

– The descriptions in the manual of a simulator may be misleading. On the one hand, documentation may be outdated or incomplete and fail to reveal features that are indeed implemented in the code (which affects particularly non-commercial simulators but is not restricted to them). Definitions of the equivalent pore pressure derived from phase pore pressures and saturations may serve as an example. On the other hand, the documentation may give the impression that a certain physical model is featured, but its implementation turns out to be incomplete or interpreted differently for different simulators. An example is the relative-permeability model of *Mualem/van Genuchten*: for the liquid phase ($K_{rel;l}$), it is uniformly implemented based on *van Genuchten (1980)*, whereas for the gas phase, an analogously derived expression found in *Parker et al. (1987)*—either with an adjustable or with a preset pore-connectivity parameter deviating from the stated reference—or the alternative expression $K_{rel;g} = 1 - K_{rel;l}$ are all in use. Lastly, the feature of a "linear" retention curve $p_{cap}(S_l)$ may wrongly arouse the impression of permitting the zero function.

Relevant results and lessons to be learnt from the perspective of a simulator-benchmarking coordinator are:

– Manuals may be misleading concerning the features of simulators (cf. the last bullet of the above list for benchmarking participants).

- At project start, the partners decided to leave the discretisation of each model at their own discretion, with possible incompatibilities of discretisation and simulator in mind but also knowing that model discretisation does have an impact on simulation results. It turned out in the course of the project that everybody subdivided the model geometry into rectangular zones. The question whether to prefer unified or individual model discretisations—or put another way, the question whether discrepancies are more significant when arising from the use of different discretisations or from using a discretisation that is less appropriate for a simulator—has not been pursued systematically.
- Solubility equilibria of phase-extraneous components have been considered of only minor importance in the definition of the benchmarks. But the comparison of simulation results showed that particularly the content of water vapour in the gas phase should be kept in mind when interpreting the results and spotting causes for discrepancies and may become important for the analysis of processes and results. However, a modification of corresponding simulator-intrinsic equations is not always easily possible or desired.
- TH^2M-coupled processes allow a variety of interactions where a variety of physical relations is involved, and compiling a set of scenarios for a TH^2M model becomes significantly more challenging than for a H^2M model when aiming at equal framework conditions for the simulations.
- Taken together, the boundaries between a simulator benchmarking and a conventional one (comparing simulation approaches instead of verifying simulator workings) are blurred: on the one hand, framework conditions are required to be the same, on the other hand, this requirement cannot always be put into practice or is not always desired to be met. The questions that have to be decided are firstly, how complex are models allowed to be without involving too many differently defined processes, and secondly, which differently defined processes can be permitted without having an excessive and distorting impact on simulation results and their discrepancies? Broad experience or extended testing will be necessary in order to answer these two questions for defining appropriate benchmarks.

Furthermore, institutions funding cooperative scientific research may gain from this project:

- For the basic scenario 1.1(a), results could be collected from all project partners, whereas the number of submissions in subsequent scenarios and models decreased gradually to a final core of four parties out of eight. It has been part of the idea and definition of this

project to leave the intensity of computational contributions at the partners' discretion, depending e.g. upon the capacities in personnel and time they are able to invest in it or upon the scenarios they consider worth simulating, since only two partners have been funded by PTKA. In case it should be intended to avoid such losses and to bind cooperation partners to a commissioned project, it is suggested to fund as many expedient institutions/organisations as desired and reasonable to form the core if framework conditions permit doing so.

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The references listed below make use of, a.o., the following abbreviations:

BMU Federal Ministry for the Environment, Nature Conservation and Nuclear Safety (Bundesministerium für Umwelt, Naturschutz und nukleare Sicherheit), in between times BMUB

BMWi Federal Ministry for Economic Affairs and Energy (Bundesministerium für Wirtschaft und Energie)

GRS Gesellschaft für Anlagen- und Reaktorsicherheit gGmbH

LBNL Lawrence Berkeley National Laboratory, formerly Lawrence Berkeley Laboratory (LBL)

TUC Clausthal University of Technology (Technische Universität Clausthal)

UPC Polytechnic University of Catalonia (Universitat Politècnica de Catalunya)

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